

高雄醫學大學眼科 實習醫學生手冊

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Topics

Community Eye Health. 2012; 25(79-80): 66-70.

References

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Abbreviation

為了提高眼科實習醫學生的學習效果，避免因眼科儀器、病歷記載、用藥之特殊性而陷入摸索學習之障礙中，特製作本學習指引手冊。此由歷屆本科醫師撰寫的手冊涵蓋大部分實習醫學生在眼科學習時需要的知識，包括病歷記載、基本檢查及儀器操作、常用縮寫、常見術式、眼科用藥以及眼科急症的處理原則，希望同學能善加利用，俾使短暫的眼科實習生涯能夠愉快且充實。若內容有窒礙難懂或需補充之處，也期望各位同學提出改善意見，以嘉惠學弟妹。

除了歷代的前輩醫師們，一併感謝協力作者群：賴昱宏、徐旭亮、張祐誠、黃靖雅、紀立中、鄭雯馨、李怡萱諸位醫師，此外為了方便同學們攜帶與閱讀，特別情商黃靖雅醫師重新校對排版並晉升為數位版本，自2020年初版起歷經多次校對及增添內容，今年已邁入第七版，在此衷心致謝。

祝各位同學

知識增長 學習愉快

眼科部主任 許永明

眼科實習醫學生導師 曾漢儀

2022.10

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Abbreviation

初診表格書寫

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Abbreviation

眼科初診表的填寫，有其特殊的格式。各檢查詳見「眼科基本檢查」章節。

病 歷 No.		姓 名		年 月 日	
診治醫師	診 斷 各項檢查完成後所下的tentative diagnosis				
視 力	右	遠視力	視網膜鏡 角膜弧度儀		
	左		+ L + L		
近 視 力	右	近視力	眼外肌運動情形		
	左		X X		
眼 壓	R	眼 壓	凸眼計		眼 位
	L		> <		
Pupil & L-R	Direct	R(mm) L(mm)	PG	右	ortho正視 各種斜視眼位紀錄
	Indirect				
		瞳孔大小 / 光反射		present glasses 目前眼鏡的度數	

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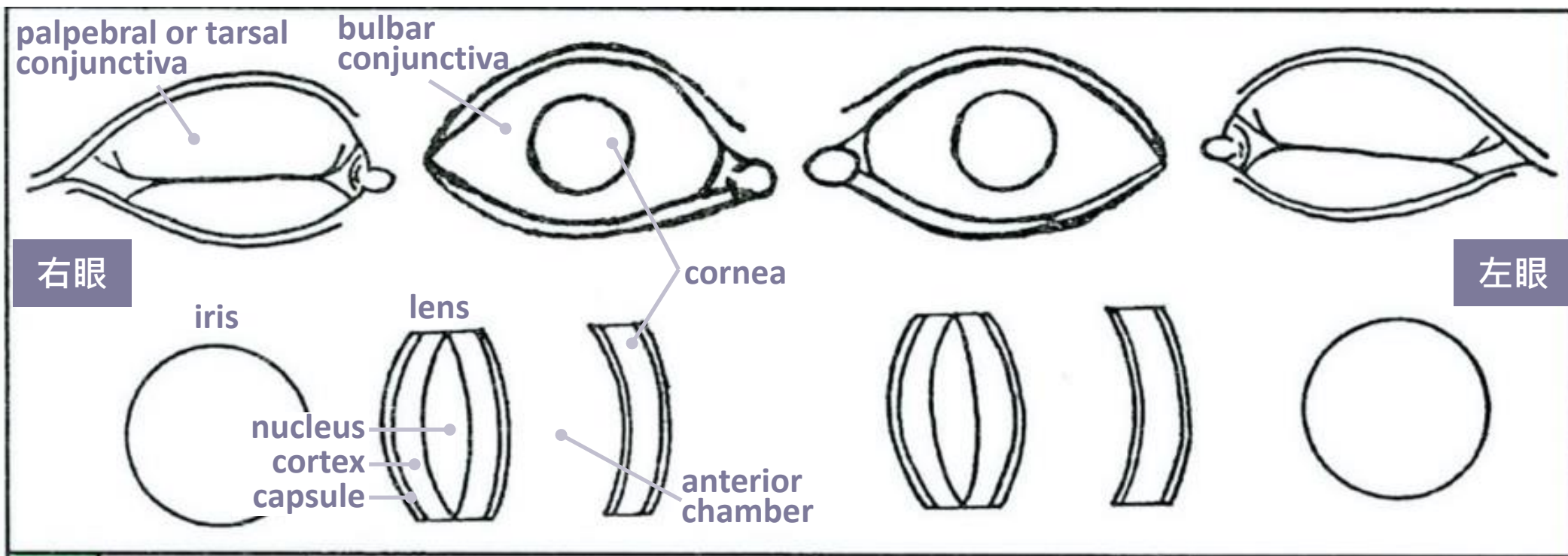


Emergency



Drug

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Abbreviation

	Previous Ocular Surgery or Tx：過去在眼科曾做過的處置 / 雷射 / 手術	
A-R 黏貼電腦驗光儀所列 印出來的左右眼屈光 狀態	Eyedrops：長期使用的眼藥水	
	填寫CC、PI、PH等等	

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Abbreviation

右眼

左眼

眼底檢查

C/D

Disc

Vessels

Macula

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Surgery



Emergency



Drug

Abbreviation

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眼科基本檢查

視力測量

Visual Acuity Determination



細隙燈

Slit Lamp Examination



淚液試驗

Schirmer Test



光反射

Light Reflex



眼壓測量

Tonometry



隅角鏡檢查

Gonioscopy



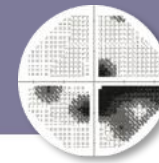
眼底鏡檢查

Ophthalmoscopy



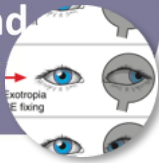
視野檢查

Perimetry



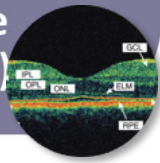
眼球運動及眼位檢查

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Abbreviation

視力測量Visual Acuity Determination

- 視力是指視覺的能力，視力檢查是評估眼睛功能最基本也是最重要之項目。
- 視力測量分為遠視力 (far vision) 及近視力 (near vision) 測量。



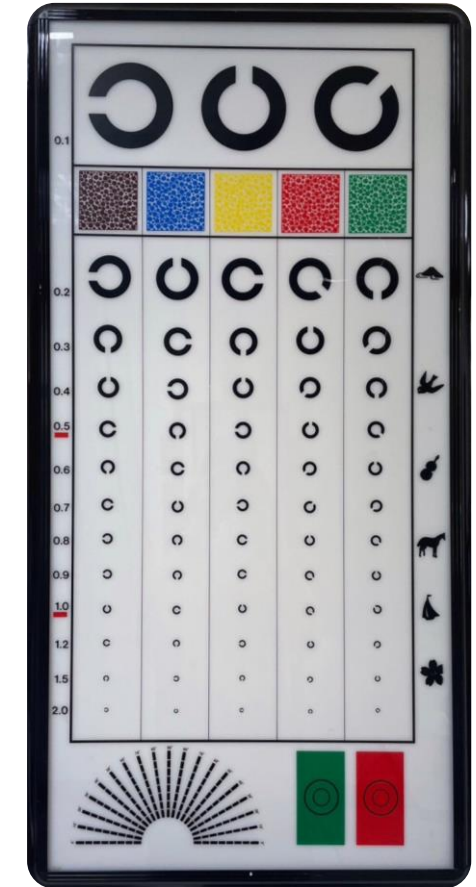
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Abbreviation

遠視力測量 Far visual acuity determination

- 一般常用之視力表，有Snellen chart 及Landolt chart。Snellen chart 量測之距離為6公尺 (20英尺)，看E字母之缺口，測量出的視力值表示為20/20、20/40、20/100.....；Landolt chart 則是距離5公尺，看C字母之缺口方向，測量值表示為0.1、0.2、0.5、1.0、1.2.....，兩種量測值之絕對值，可通用比較。
- 本院目前採用Landolt chart以及auto chart projector (視力表投影儀)。為節省空間，利用鏡像反射之原理，將Landolt chart置於受測者之頭頂位置，並在受測者前方2.5公尺處設置一面鏡子，如此受測者之視覺路徑就等於5公尺。最佳之測量高度為字母與眼睛等高，若是採用鏡射之方式，須注意鏡子是否正確定位，以及受測者坐姿時，可完整見到視力表。故鏡子之折角或椅子之高度，需機動視受測者之姿勢與身高做調整。視力表投影儀則是由投影機、投射螢幕和遙控器組成，有機動性強、視標多元而且亮度可以隨環境調整的優點。在檢測空間不足時，可以彈性縮短投影機與投射螢幕的距離，再藉由調整投影焦距以得到清晰的視標影像；此清晰的視標影像視角與 6公尺視標相同，故測得之視力值不變。



Landolt chart

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Abbreviation

以下是視力測量步驟：

- 一般先量右眼，再量左眼，非受測眼須確實遮蓋。
- 先量裸視，看受測者能看到的最小字母行。例如：0.6, 0.9, 1.0, 1.2..... 若是最小字母行並非每一個皆能辨識，可加以標明，例如：0.9⁻²，即表示0.9那一行雖大部分可辨識，但有兩個字母無法辨識。
- 若是受測者有自己的眼鏡，則戴上眼鏡受測。表達之方式為最小字母行加註 " x PG "。例如：1.0 x PG。PG 表示present glasses。若是戴隱形眼鏡，則表示為 " x CL " (contact lens)。
- 若是受測者沒有眼鏡且視力不佳，可以pinhole測試；若有改善表示眼睛有屈光之缺陷，若無改善則是其他原因。
- 若是受測者之視力，差到連第一排最大之字母皆無法辨識，則請他往視力表走近，直到可看清第一排之字母停止。若是走到距離視力表3公尺處才能看見第一排，則表示為 0.1 x 3/5。
- 若有屈光上的缺陷，須以試鏡片加以矯正，以獲得最佳矯正視力。可以電腦驗光值作為參考，加以逐步矯正。首先矯正球面屈光部分，若為近視放置minus，遠視則放置plus 之鏡片。矯正近視部分時，須由略低度數之鏡片逐漸增加，不可立刻照驗光值就加上該度數之鏡片。遠視部份亦類似，只是遠視度數過頭時，會有變模糊之現象，此時上一個度數是最佳之矯正選擇。不管遠視或近視之矯正，在接近屈光度數時，皆須以最小間隔之25度鏡片逐步更替，以達成最佳矯正視力。

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Abbreviation

- 另外，尚須矯正柱狀屈光，亦即散光部分。散光亦有plus及minus，試鏡片不同於球面鏡，會有一軸度線之刻痕，試鏡時須根據驗光值放置於合適之角度。最好能稍稍轉動，以確定最佳之散光鏡放置角度。此時，影像應最清晰以及散光線 (astigmatism line) 應最均勻。
- 若是無法以眼鏡矯正，則須測量是否能在近距離辨別施測者之手指數，並須標明距離。例如在眼前30公分可以辨識手指數，則表達為ND/30 cm (ND：number of digits) or CF/30 cm (CF：count fingers)。
- 若是視力再更差，則測量是否可辨手動，例如：HM/30 cm (HM：hand motion)。
- 視力最差，則以光源在受測者眼前照射，測試是否有光感，SL (+) or SL (-) (SL：sense of light)。
- 視力檢查結果的表示方式，若患者經電腦驗光輸出如右圖，測得VOD：0.8 (1.0 x -0.25 x cyl -0.75 x 55°)，表示右眼裸視力為0.8，經過最佳矯正視力為1.0，其矯正方式為以球面鏡 (sphere) -0.25 D，柱狀鏡 (cylinder) -0.75 D，軸度55度。同度數的另一個表示法可以將散光軸度轉90度，此時球面鏡的度數需與散光度數相加，並將散光度數前的正負號互換，意即VOD：0.8 (1.0 x -1.0 x cyl +0.75 x 145°)。

2020_06_29 PM 06:55			
REF. DATA NO. 2410			
CYL: MIX			
<R>	S	C	A
	+ 0.25	- 1.50	50
	+ 0.00	- 0.75	55
	- 0.25	- 0.75	55
	- 0.25	- 0.75	55
	- 0.25	- 0.75	55
	S. E.	- 0.75	
<L>	S	C	A
	- 1.50	- 0.75	145
	- 1.50	- 0.75	140
	- 1.25	- 0.75	140
	- 1.50	- 0.75	140
	S. E.	- 2.00	
PD: 63			

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Abbreviation

近視力測量 Near visual acuity determination

- 對於一些臥床病人、行動不便者、或注意力不夠集中的小朋友，可施測近視力。有各種近視力表可供使用，在燈光充足下，置於受測者眼睛前方14英吋 (35公分) 處，量測步驟同上述之遠視力量測。



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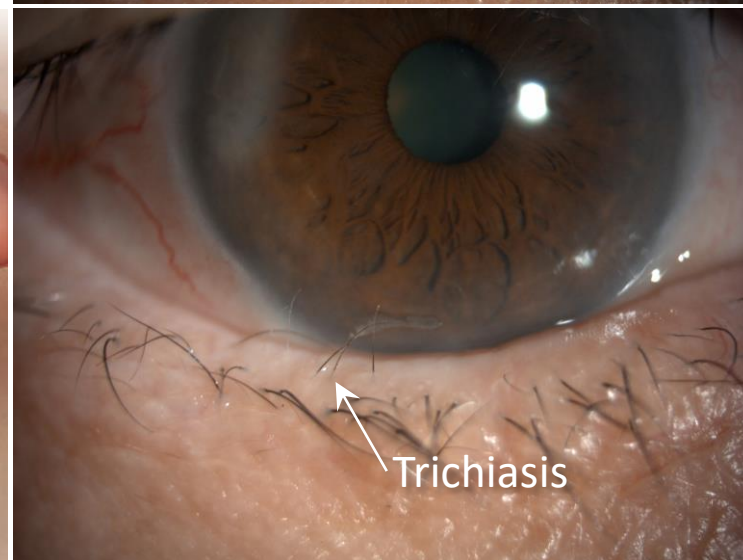
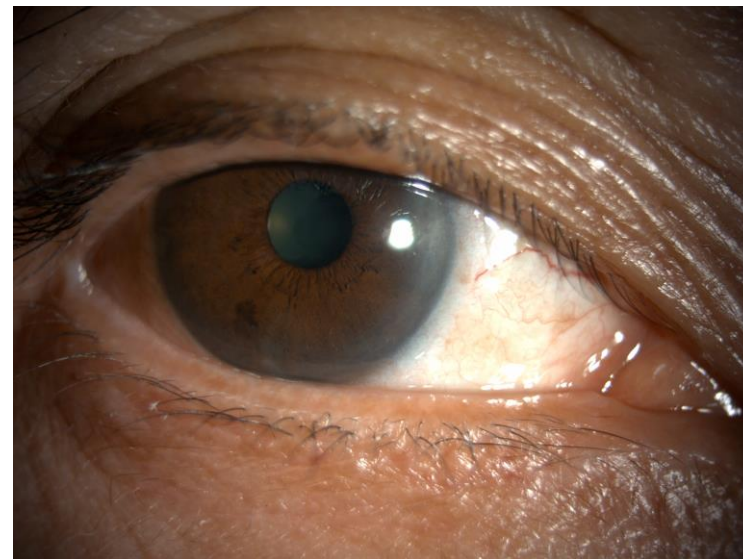
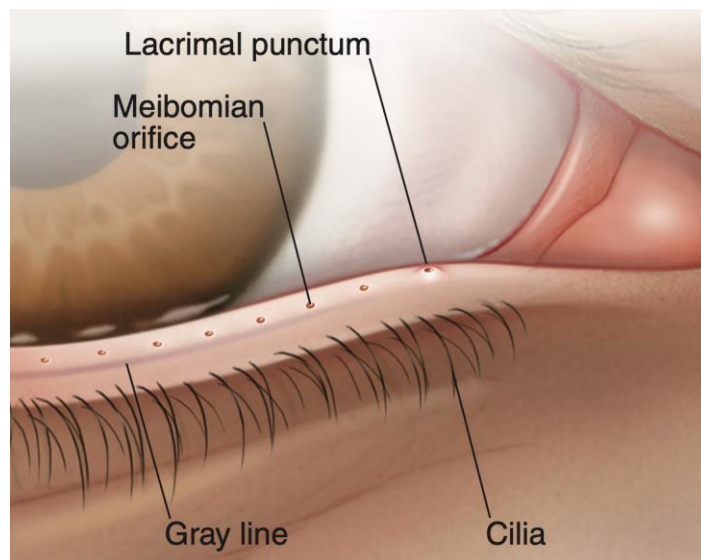


Abbreviation

細隙燈 Slit Lamp Examination

細隙燈是檢查眼前段 (anterior segment) 最主要的儀器，也是眼科門診裡使用頻率最高的儀器。初學者應該熟習一套固定的檢查順序，以免遺漏。以下依序是建議的檢查步驟：

- **眼瞼及眼睫毛 Eyelids and lashes**：眼瞼有無發炎或腫瘤，眼瞼和眼球的相對位置，眼睫毛的方向，淚孔 (puncta) 的狀態，淚液的量 (lacrimal lake) 等等。

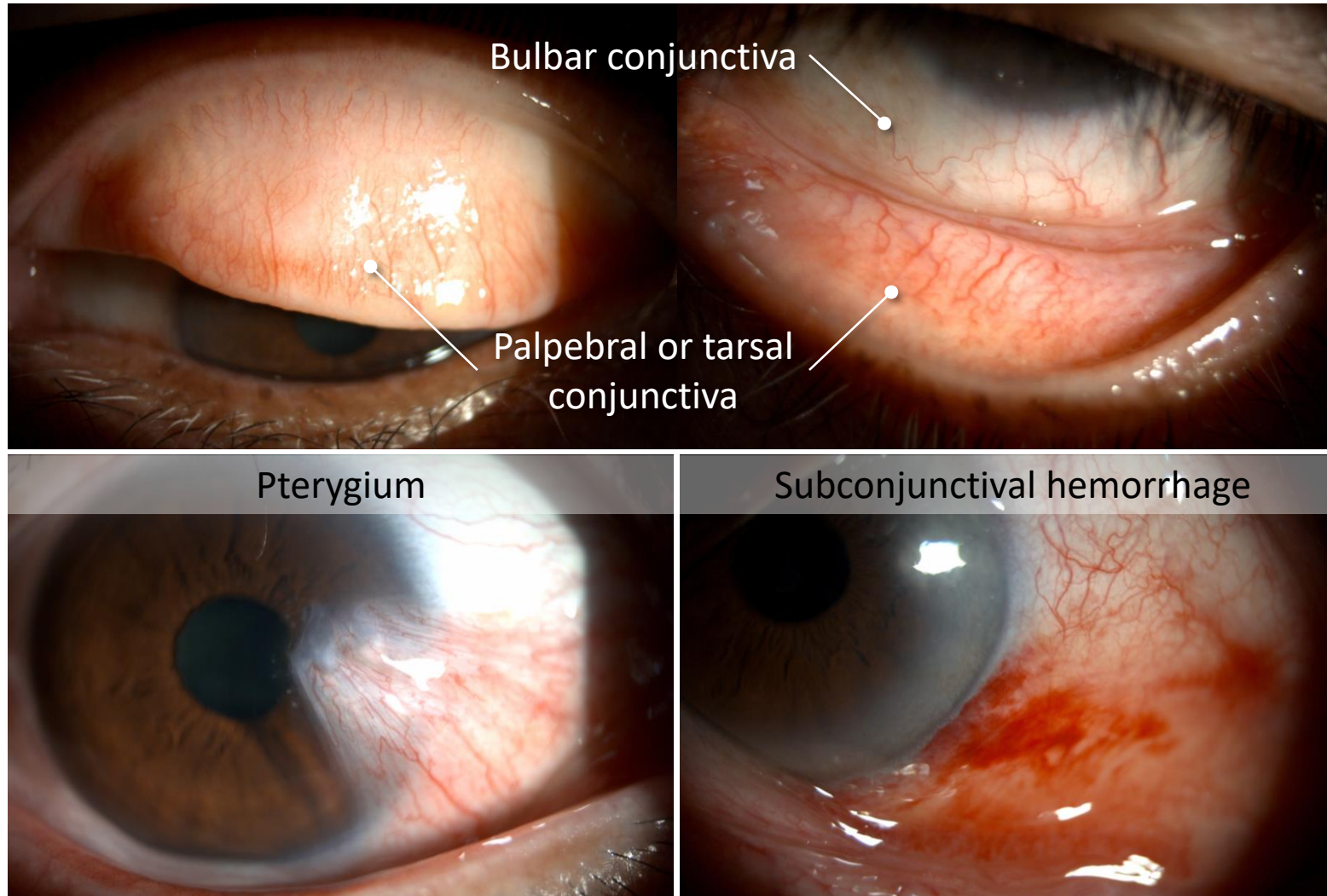


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Abbreviation

- **結膜 Conjunctiva**：要觀察球結膜 (bulbar conjunctiva) 和瞼結膜 (palpebral conjunctiva) 有無濾泡 (follicle，一種lymphoid hyperplasia)、乳突 (papilla，一種vascular hyperplasia)、水腫 (chemosis)、腫瘤、結石 (concretion或lithiasis)、瞼黃斑 (pinguecula)、眼翳 (pterygium)、nevi等等。

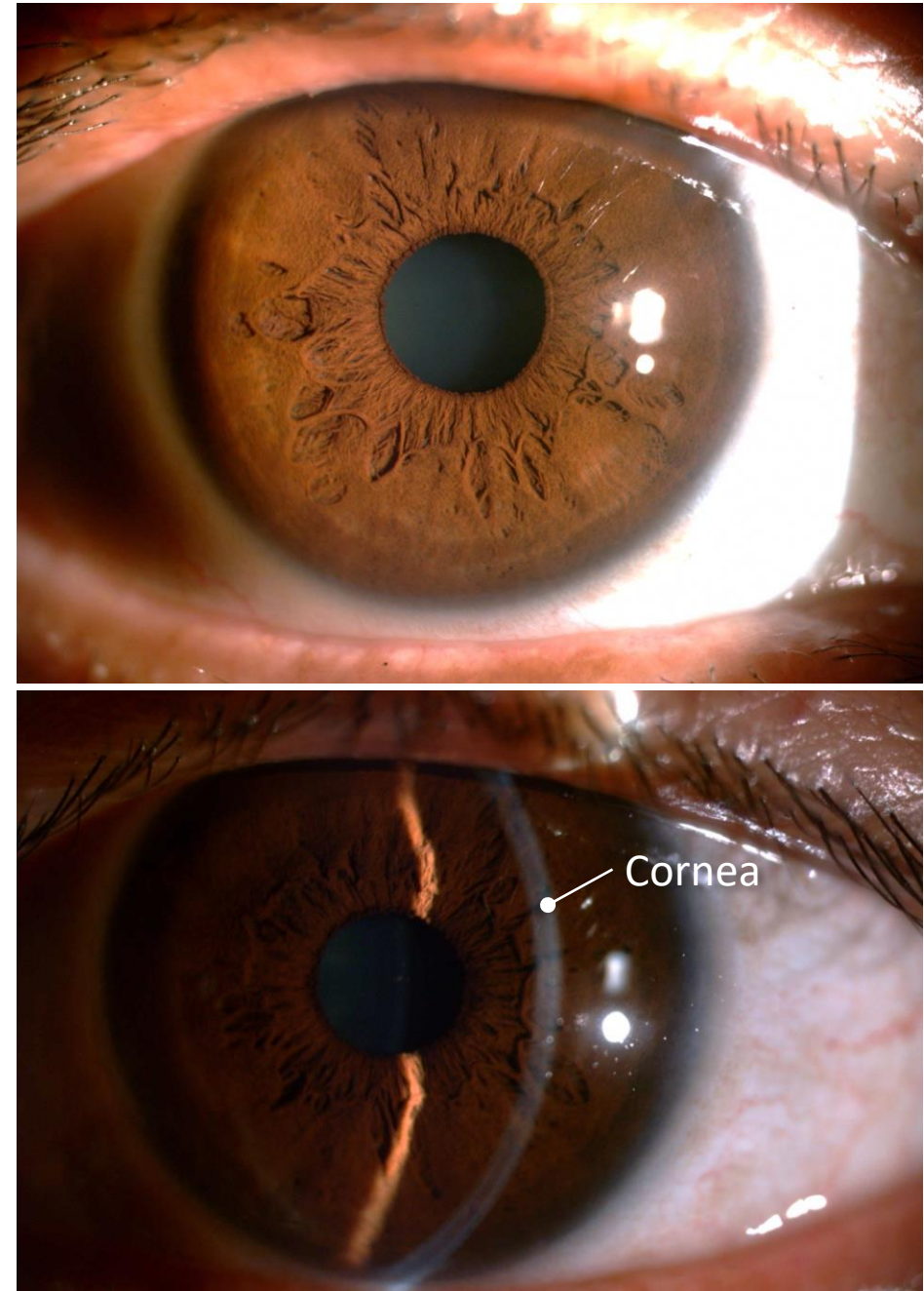


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Abbreviation

- **鞏膜 Sclera**：顏色是否正常，有無發炎、充血、潰瘍等等。
- **角膜 Cornea**：正常角膜為清澈透明，任何混濁皆為不正常。觀察是否有表皮缺損 (epithelial defect，用fluorescein試紙染色於藍光下會呈現綠色螢光)，基質水腫 (常以Descemet's membrane fold呈現)，或有炎症反應 (infiltration，角膜中可見發炎細胞浸潤，呈白色粉末狀)，嚴重時會呈現黃白色的膿瘍 (abscess)；或者有以前發炎或者受傷之後遺留下來的白色癍痕 (corneal leukoma or nebula)，有時候會伴隨血管入侵 (stromal vascularization or pannus)。

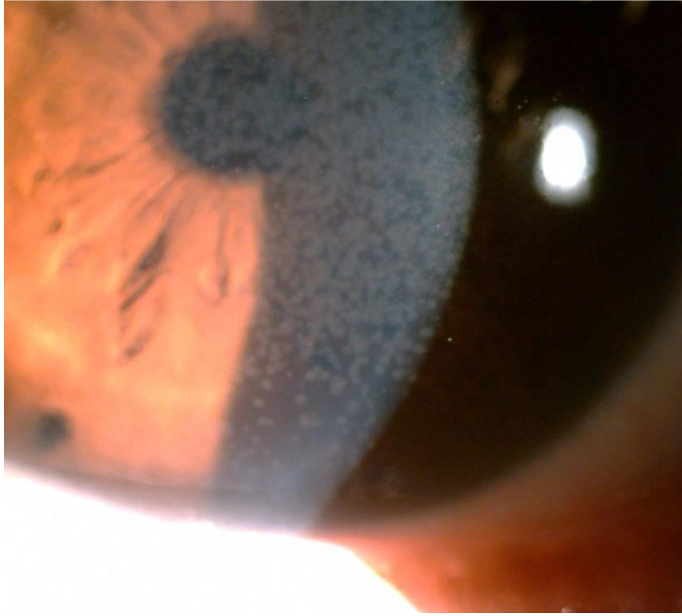


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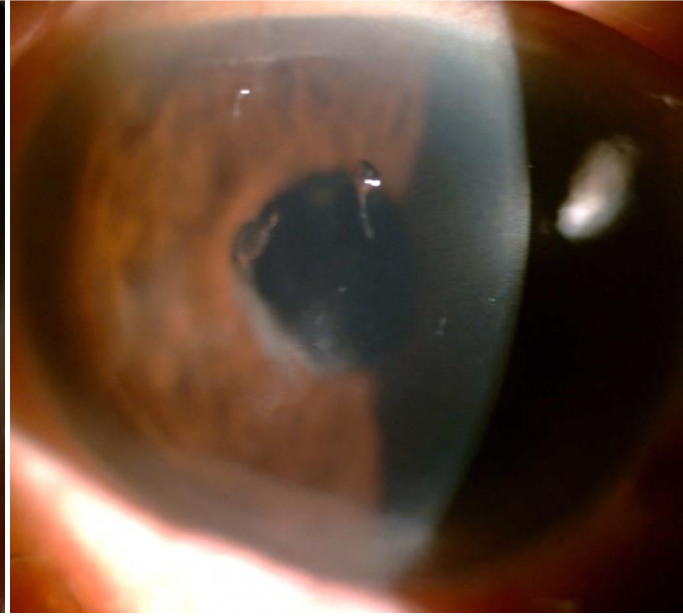


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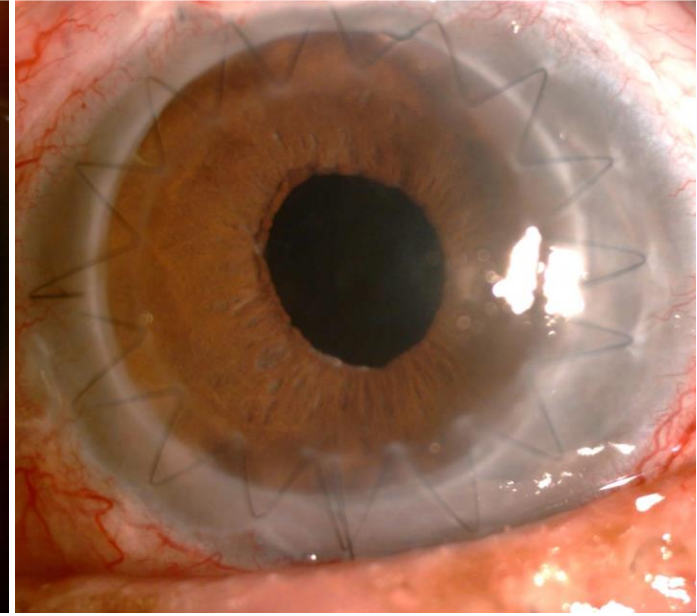
Punctate epithelial erosion



Filamentary keratitis



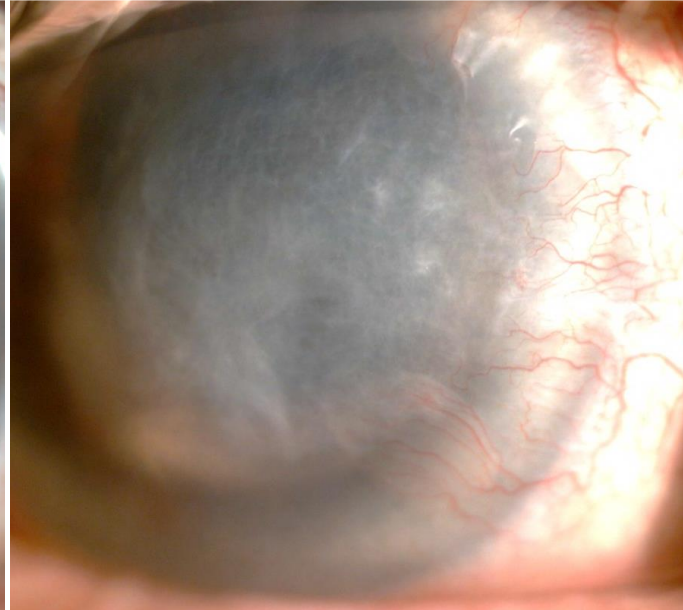
Corneal graft



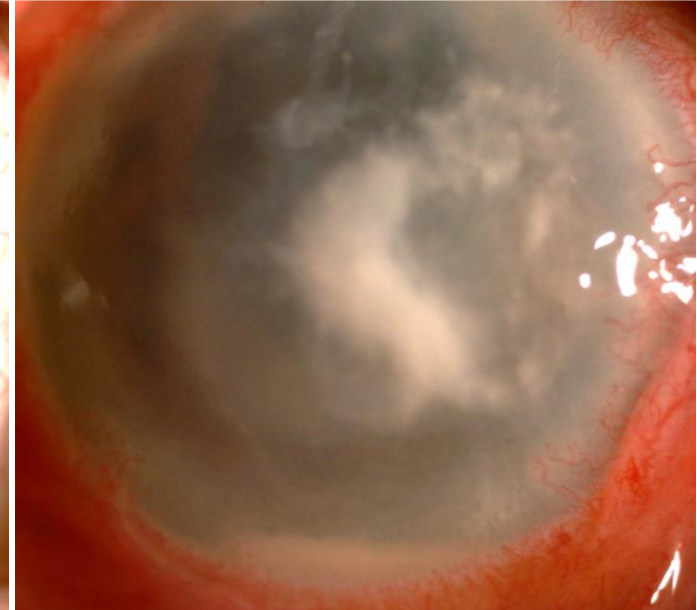
Dendritic keratitis



Corneal leukoma with vascularization



Infectious ulcer with stromal infiltration



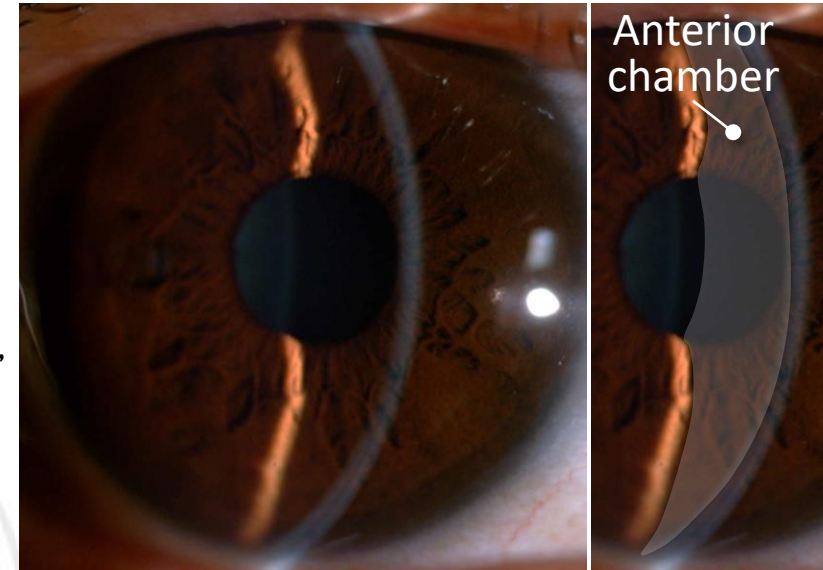
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Abbreviation

- **前房 Anterior chamber**：把光開到最強，背景光降到最低較利于前房的檢查。正常在燈光下應呈現黑色 (也就是空無一物)。

1. 根據在1 mm × 1 mm的範圍內觀察到的發炎細胞數目 (cells)，以及光線通過因發炎反應使蛋白質上升的房水所產生的Tyndall effect (flare)，可以將發炎分為以下幾個等級：



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Abbreviation

Table. The SUN* Working Group Grading Scheme for Anterior Chamber Cells

Grade	Cells in field†
0	<1
0.5+	1-5
1+	6-15
2+	16-25
3+	26-50
4+	>50

* SUN = Standardization of uveitis nomenclature.

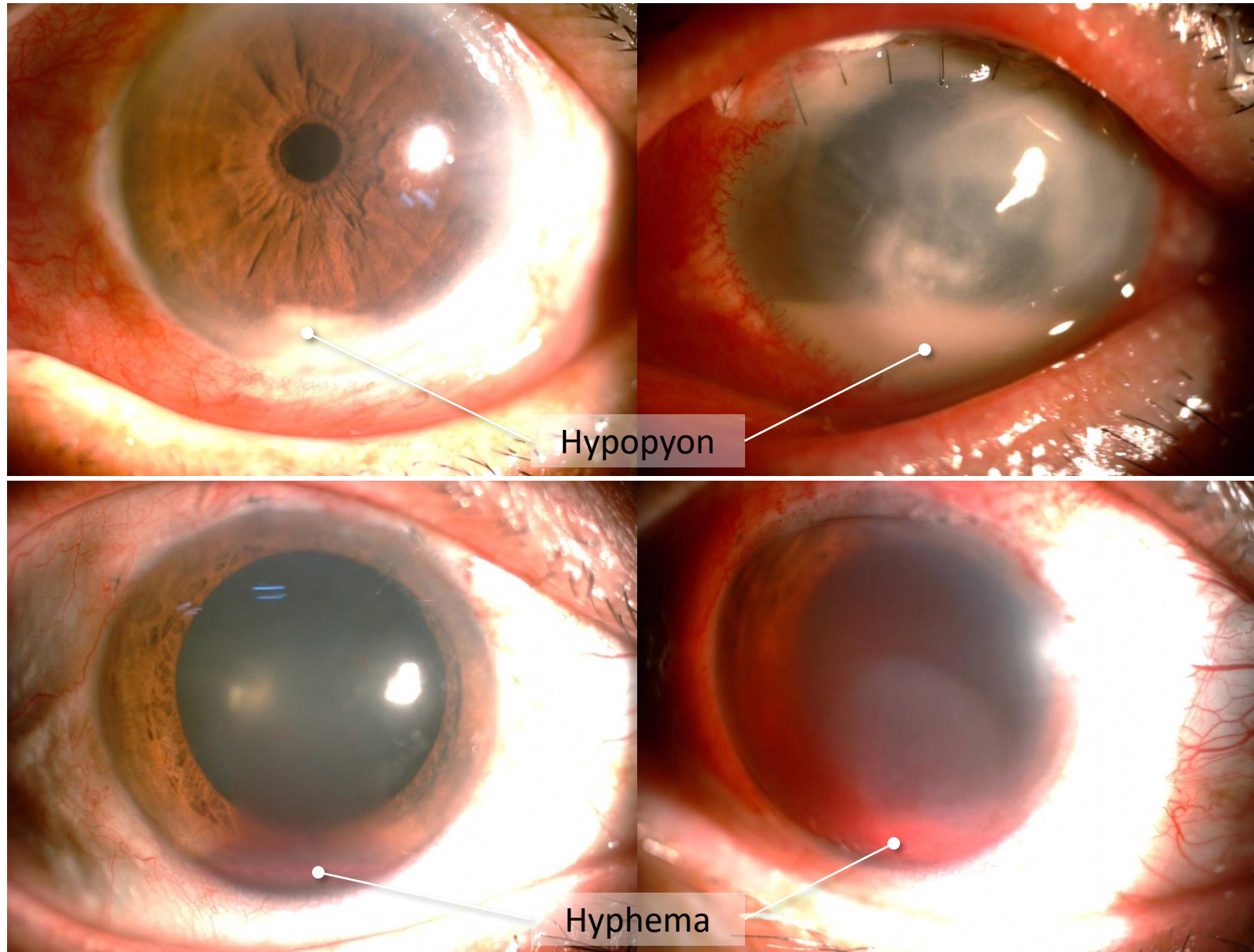
† Field size is a 1 mm by 1 mm slit beam.

Table. The SUN* Working Group Grading Scheme for Anterior Chamber Flare

Grade	Description
0	None
1+	Faint
2+	Moderate (iris and lens details clear)
3+	Marked (iris and lens details hazy)
4+	Intense (fibrin or plastic aqueous)

* SUN = Standardization of uveitis nomenclature.

2. 發炎嚴重時會有前房蓄膿 (hypopyon) 的產生。若以RBC為主叫做前房出血 (hyphema)。



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Abbreviation

3. Van Herick method：評估前房深度的方法，在光線呈60度角下觀察limbus位置前房 (A) 和角膜 (C) 的厚度比 (A/C) (即下圖 Figure 1.6 中的 b/a)。

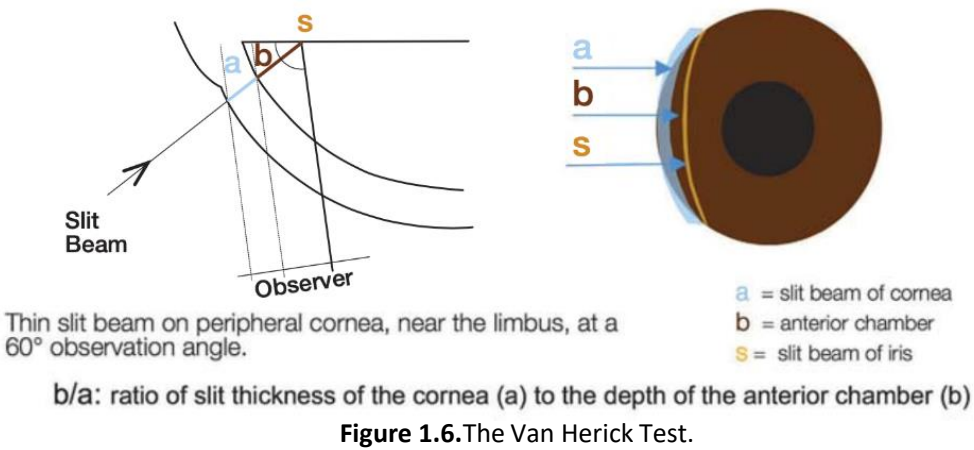
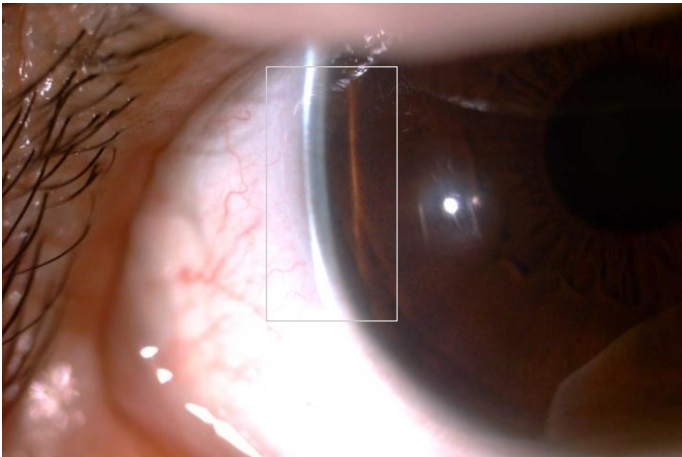


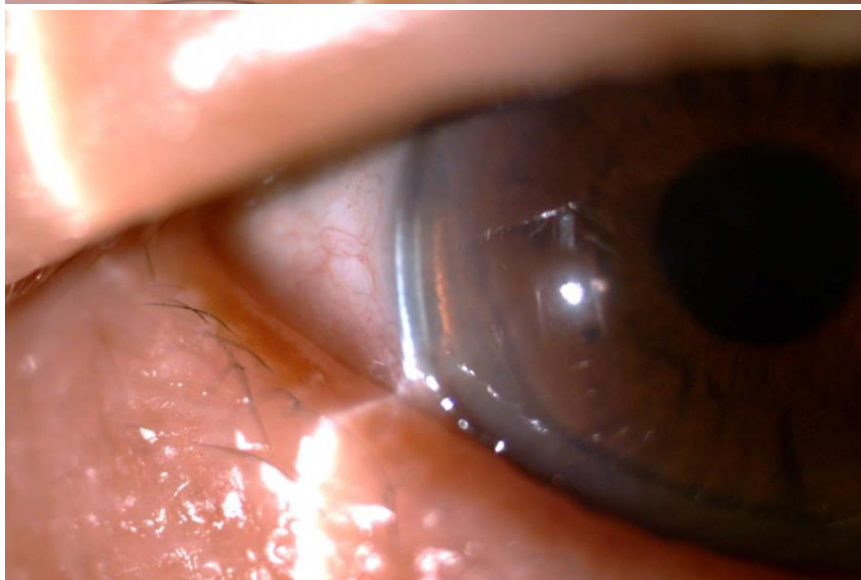
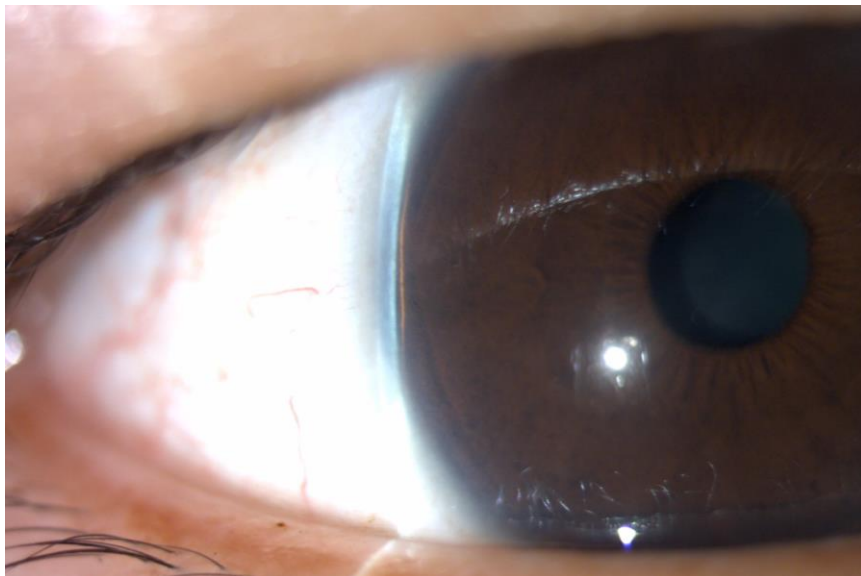
Table. Anterior Chamber Angle Assessment

Anterior chamber depth (A) as a proportion of corneal thickness (C)	Van Herick method		Clinical description (Used in KMHU)
	Grade	Angle status	
A/C = 0	0	Closed	Flat
A/C < 1/4	1	Extremely narrow	Slit
A/C = 1/4	2	Narrow	Shallow
1/4 < A/C ≤ 1/2	3	Open	Moderate
1/2 < A/C < 1	-	-	Deep
1 ≤ A/C	4	Wide open	Very deep

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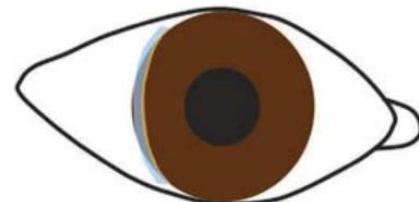


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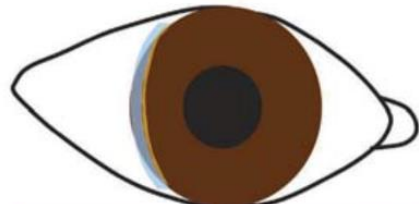


b/a:

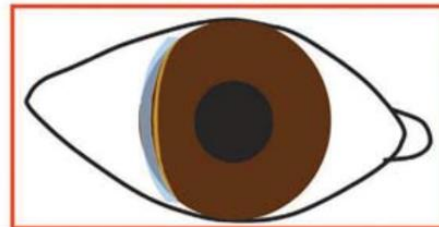
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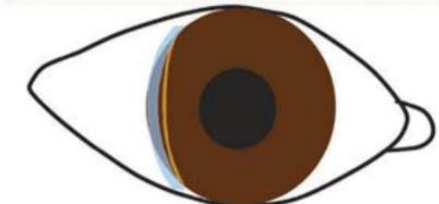
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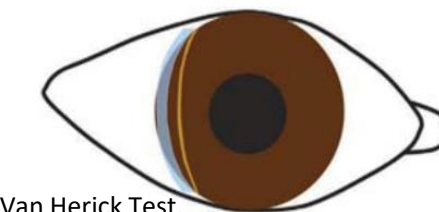
1/4



1/2



1



Grade:

0 Angle closed

1 Angle closure likely (10°)

2 Angle closure possible (angle 20°)

3 Angle closure unlikely

4 Angle closure very unlikely

Figure 1.6. The Van Herick Test.

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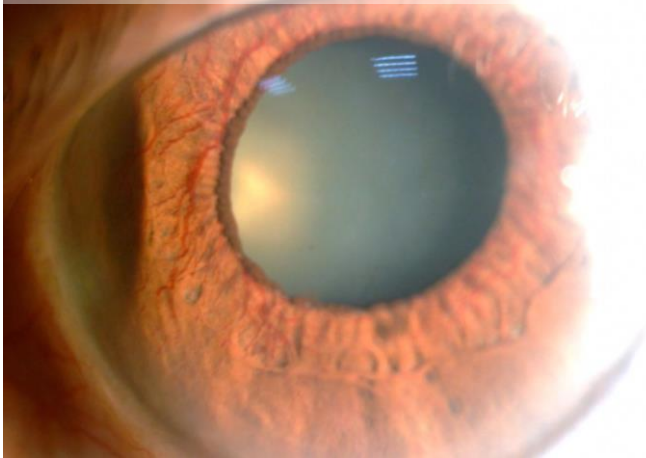
TOC



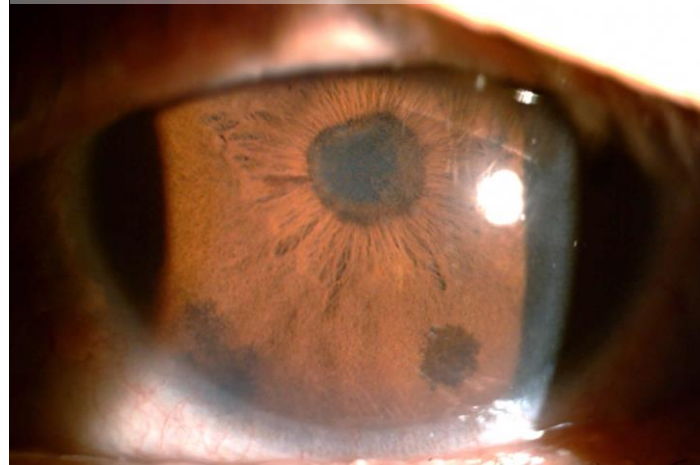
Abbreviation

- **虹膜 Iris**：有無nodule， tumor， 色素的變化、萎縮 (atrophy)、新生血管 (rubeosis， 通常代表有嚴重的retinal ischemic disease， 常會併發neovascular glaucoma)、沾黏 (synechiae， 向後黏到水晶體前囊叫 posterior synechiae， 向前黏到角膜叫 anterior synechiae)。

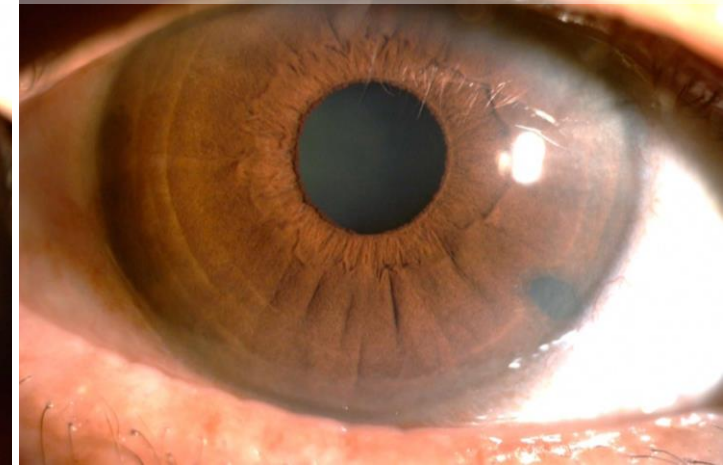
Iris neovascularization



Iris posterior synechiae



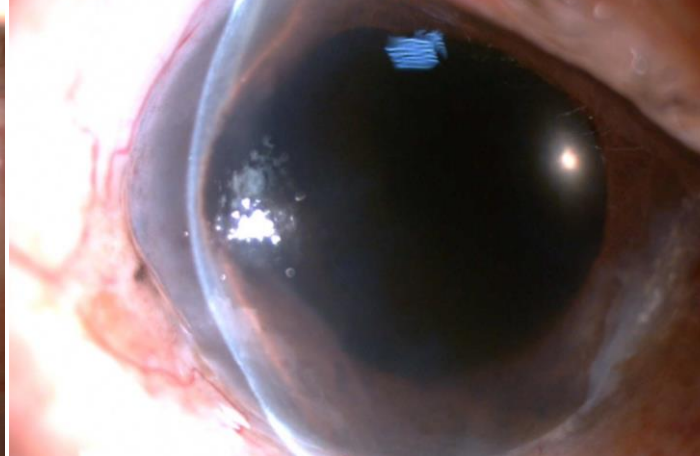
Laser peripheral iridotomy



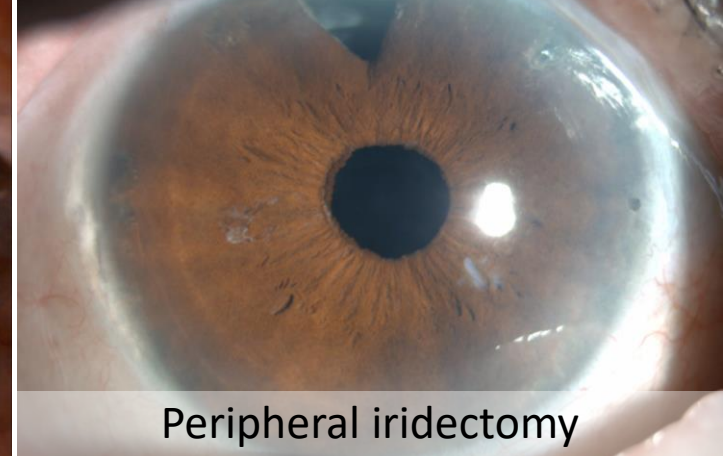
Iris neovascularization



Iris anterior synechiae



Peripheral iridectomy



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Abbreviation

- **水晶體 Lens**：主要觀察是否有混濁現象，即所謂的白內障 (cataract)。根據混濁位置的不同，較常見的描述有三種：nuclear cataract、cortical cataract、posterior subcapsular cataract (PSC)。

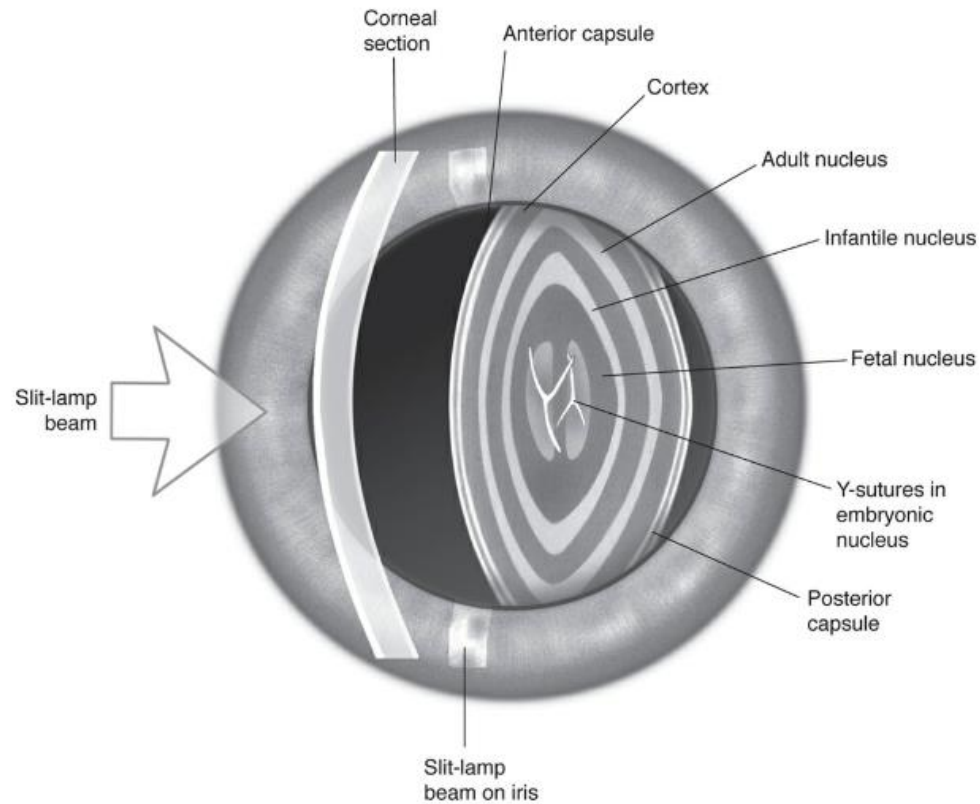
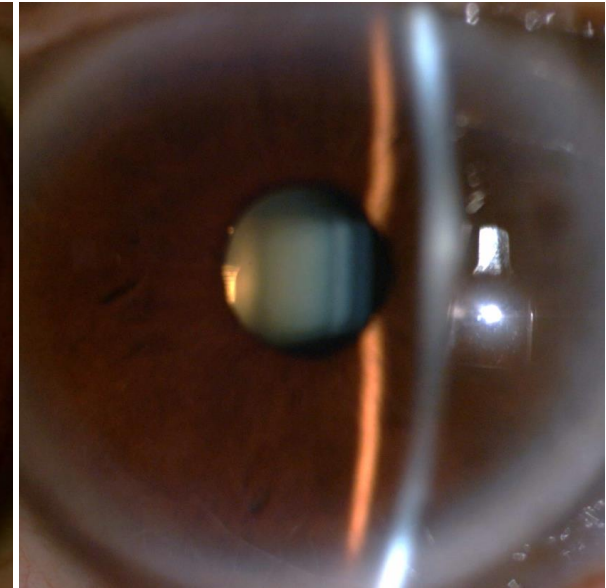
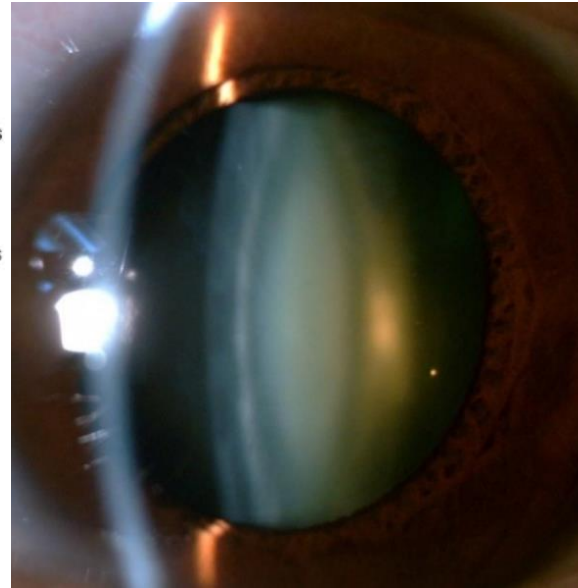


Figure 4-2 Y-shaped sutures, formed during embryogenesis, are visible within the adult lens with the use of the slit lamp. (Illustration by Christine Galapp.)

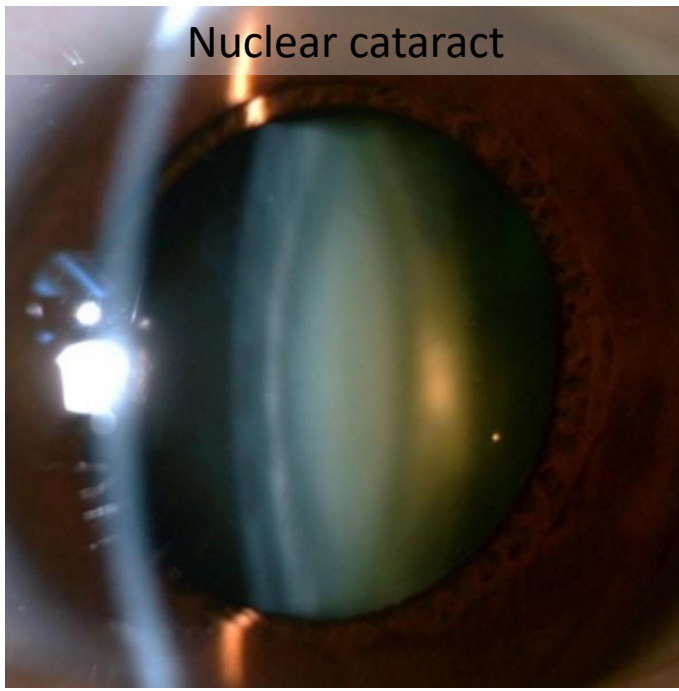


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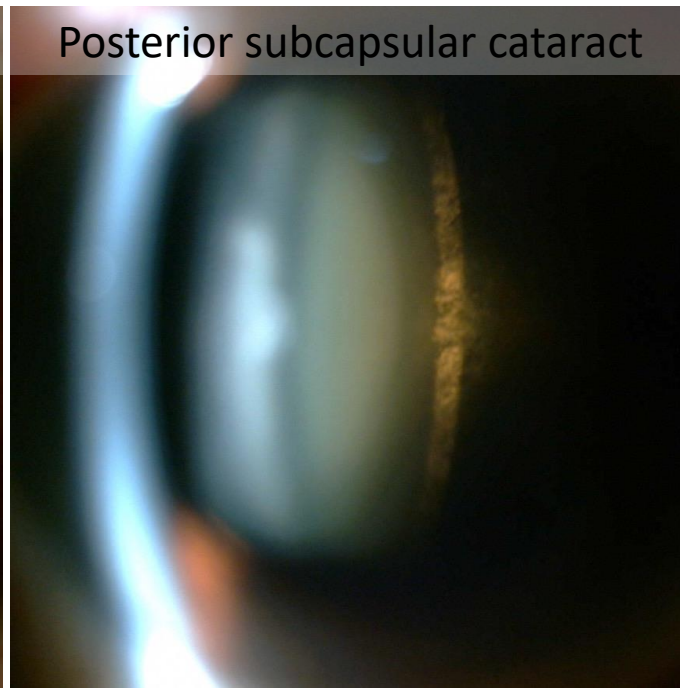


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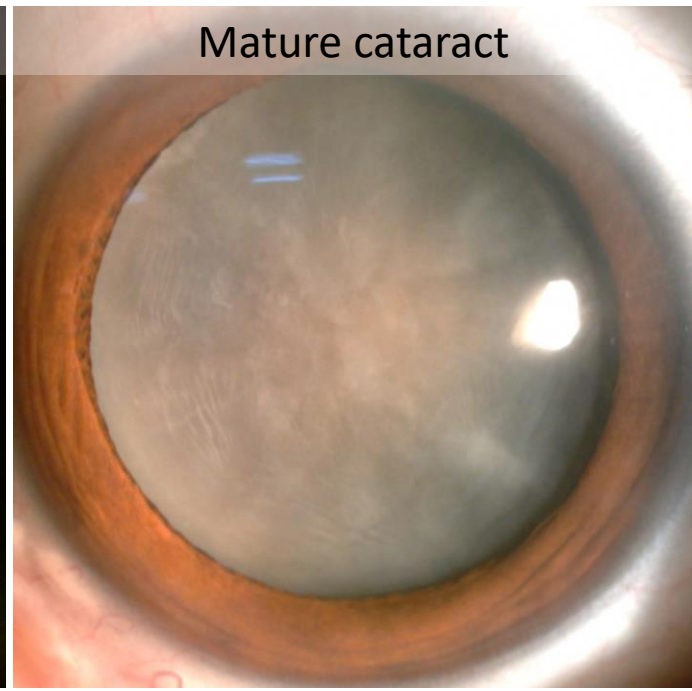
Nuclear cataract



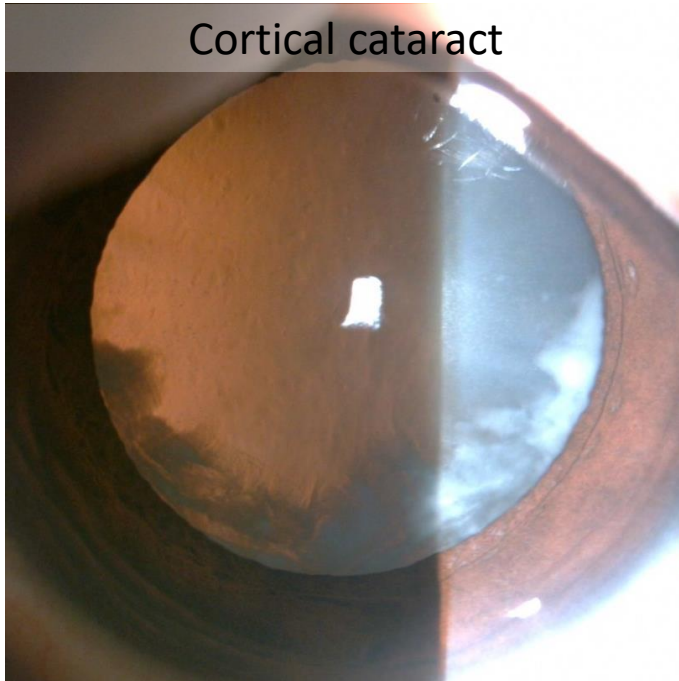
Posterior subcapsular cataract



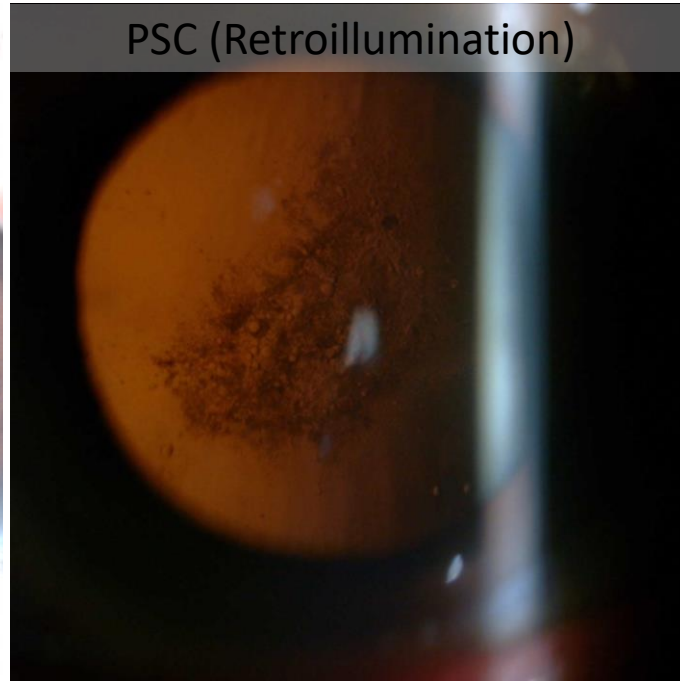
Mature cataract



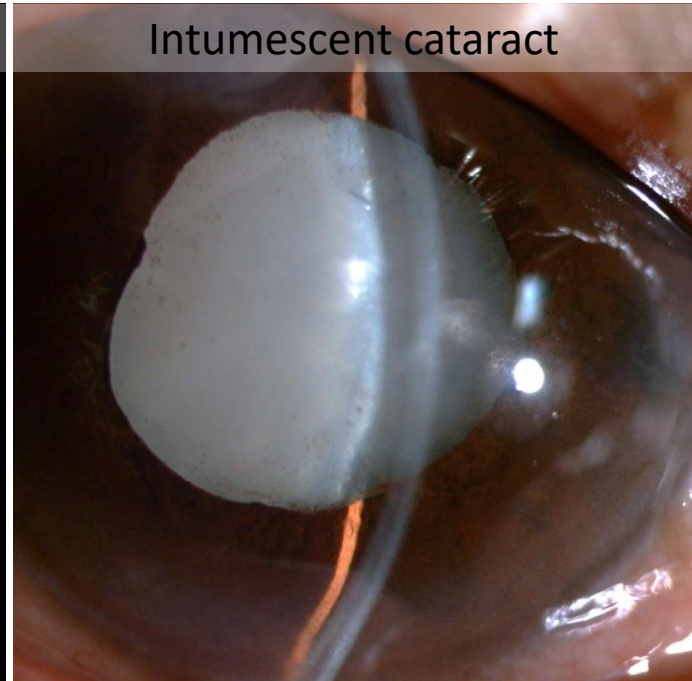
Cortical cataract



PSC (Retroillumination)



Intumescent cataract



TOC



Exam



Surgery

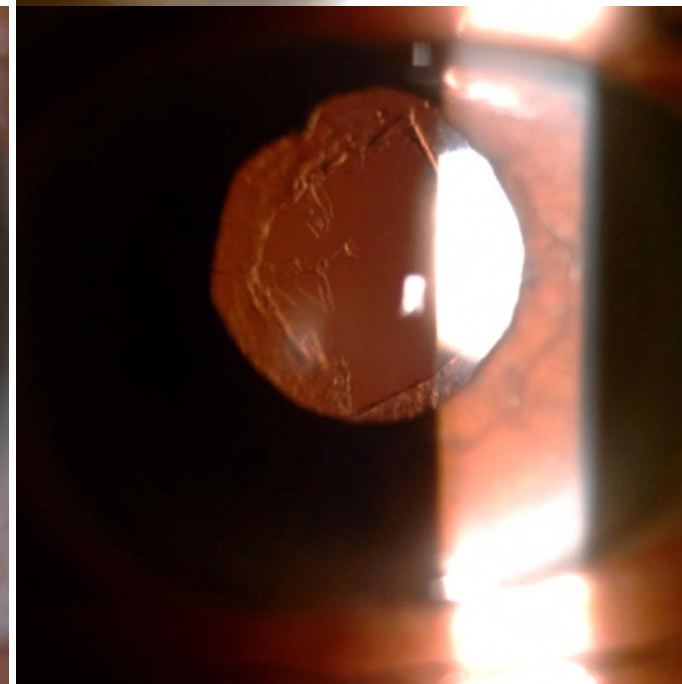
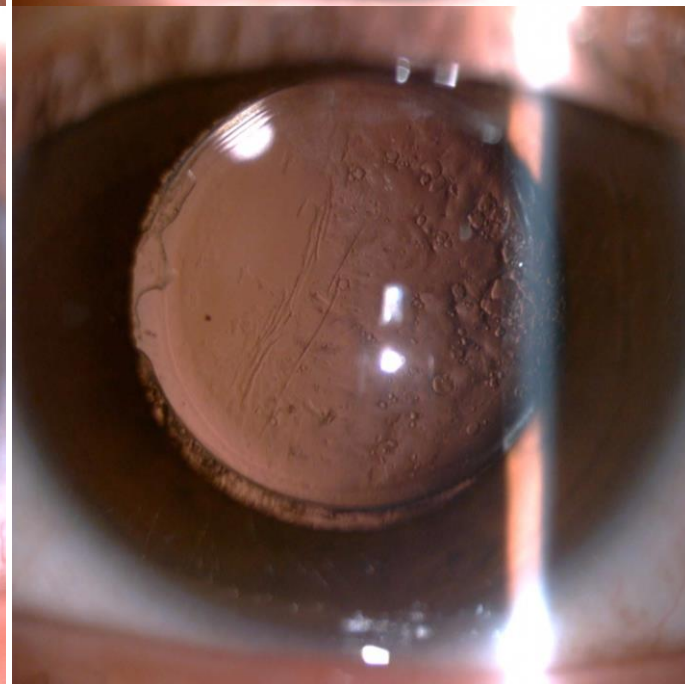
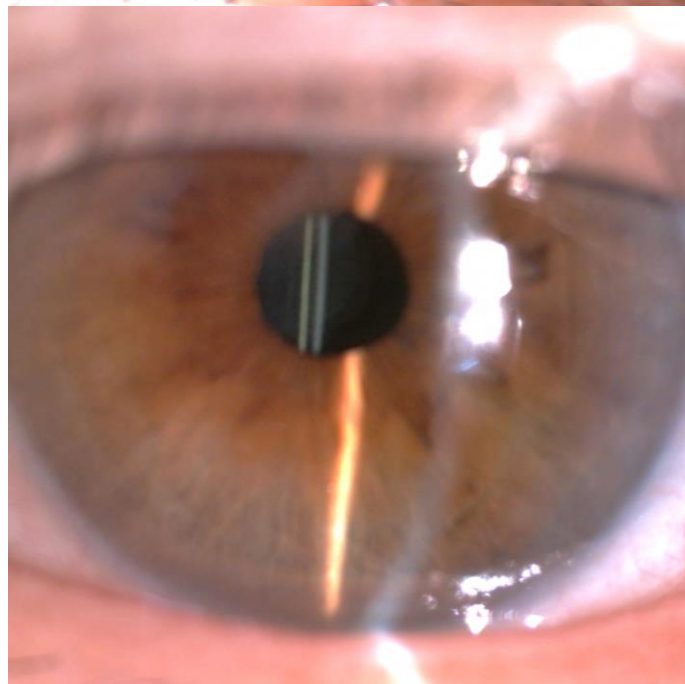
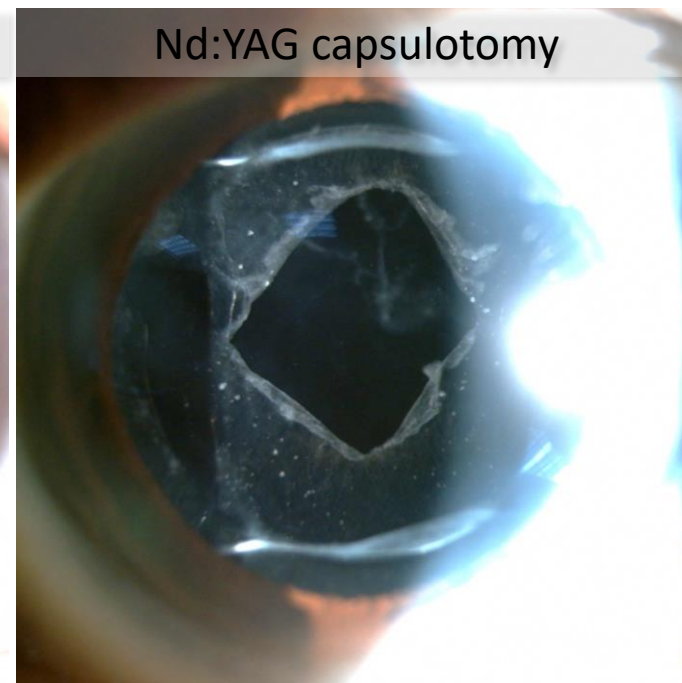
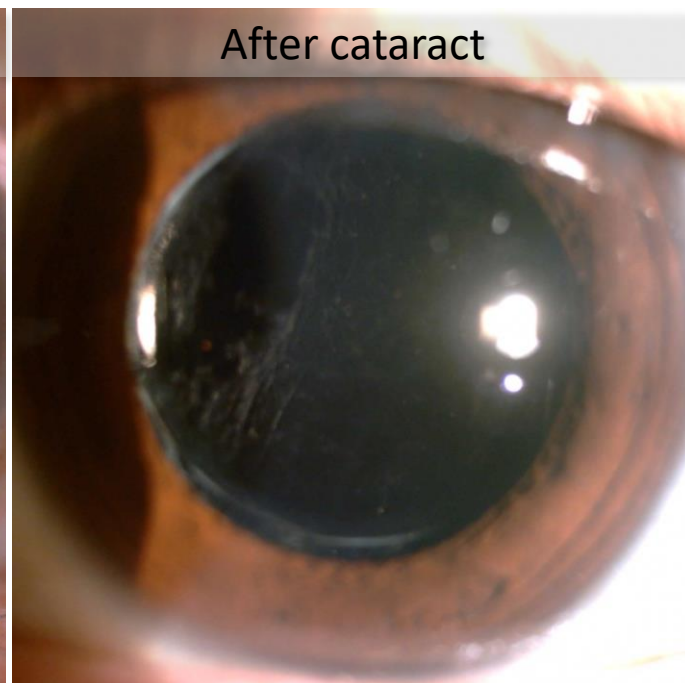
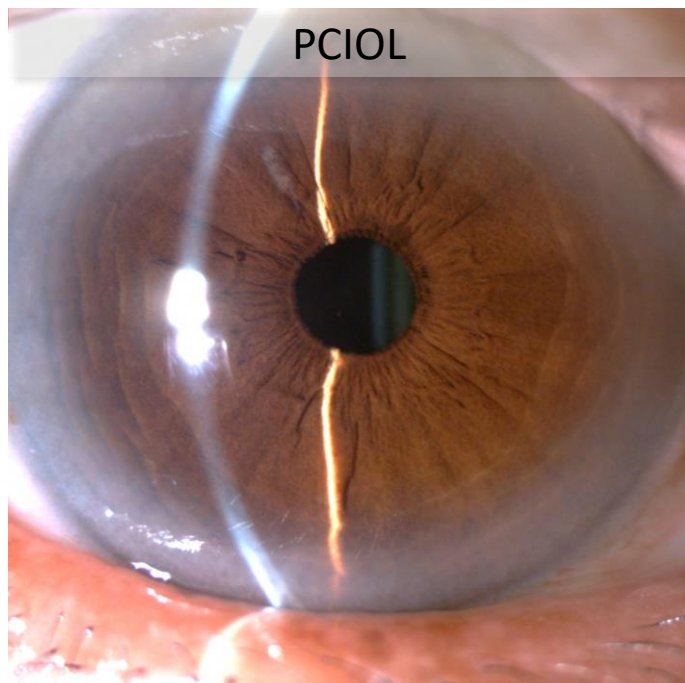


Emergency



Drug

Abbreviation



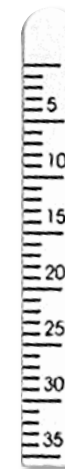
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Abbreviation

淚液試驗 Schirmer Test

- 測量淚液的aqueous layer分泌量，可分為basic secretion和reflex secretion兩種。
- 本科常用的方法是先用麻藥點眼以抑制reflex tearing，亦即只測量basic secretion，因此又叫做basic secretion test。麻醉眼球表面後於下眼瞼外側1/3處放入專用的paper strip五分鐘，判讀紙條變色 (被淚液浸濕) 的長度。正常人應該 $\geq 10\text{mm}$ ；若 $< 5\text{mm}$ 則可判定為異常。



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Abbreviation

光反射 Light Reflex

- 利用手電筒、筆燈或者直接眼底鏡的光源打側光 (ambient light) 觀察瞳孔的直接和間接光反射，正常人應該兩眼瞳孔縮小速度和大小皆對稱。
- 將光源連續交替照在兩眼瞳孔 (swinging flashlight test) 可以檢查出雙眼瞳孔收縮程度有無細微的差別，若呈現paradoxical dilatation (即光線照射時瞳孔散大)，代表該眼視神經功能有問題 (relative afferent pupillary defect, RAPD, = Marcus Gunn pupil)。

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Abbreviation

眼壓測量 Tonometry

主要用在青光眼患者的評估，此外在眼球外傷 (眼球破裂或穿通傷除外)、炎症、視網膜剝離以及眼內部手術後都經常需要量測眼壓。正常人的眼壓值為10~22mmHg。

常用的測量方法有：

- **Digital tonometry**：眼球發炎、角膜不平整或有瘢痕，或者手邊沒有眼壓計時使用。用兩隻手指頭置於病人上眼皮輕輕按壓，正常情形應該要稍有彈性 (sponginess) 的感覺。
- **Schiøtz indentation tonometry**：易貯藏又好攜帶，缺點是不夠精確且讀數必須再經過換算，現已少用。病人要先點眼局部麻醉並且仰躺。紀錄讀數時要同時紀錄使用的砝碼重量，例如 7/5.5 代表使用5.5克的砝碼時讀數是7 (約相當於12mmHg)，需對照專用的換算表才能得知眼壓數值。讀數越小，眼壓越高。
- **Goldmann applanation tonometry**：是公認測量眼壓的gold standard，但是必須先點眼局部麻醉後使用螢光染色，再配合細隙燈操作才能測量，難度比較高。因為是接觸性眼壓計，為避免傳染疾病，在使用過後必須徹底消毒。

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Abbreviation

- **Noncontact tonometer (air-puff)**：門診最常使用的眼壓計，利用氣流 (puff) 將中央角膜壓平來測量眼壓，若角膜不平整時可能就無法測量。優點是不直接接觸眼球所以無需麻醉也不會傳染疾病，測量速度快，適合大量篩檢。
- **Tono-Pen tonometer**：電子式的眼壓筆，比air puff測量來的精確，而且可用於癍痕或不平整的角膜。因為是接觸式眼壓計故同樣得先點眼麻醉才能施行。
- **iCare tonometer**：為反彈式 (rebound) 眼壓計，使用拋棄式的一次性探針，無須局部麻醉，坐立或者平躺皆可測量。

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Abbreviation

隅角鏡檢查 Gonioscopy

配合細隙燈利用mirrored lens觀察隅角構造。

本科常用的是modified Shaffer grading system，根據隅角構造 (Schwalbe's line → trabecular meshwork → scleral spur → ciliary body band) 的可見程度分為grade 0-4。

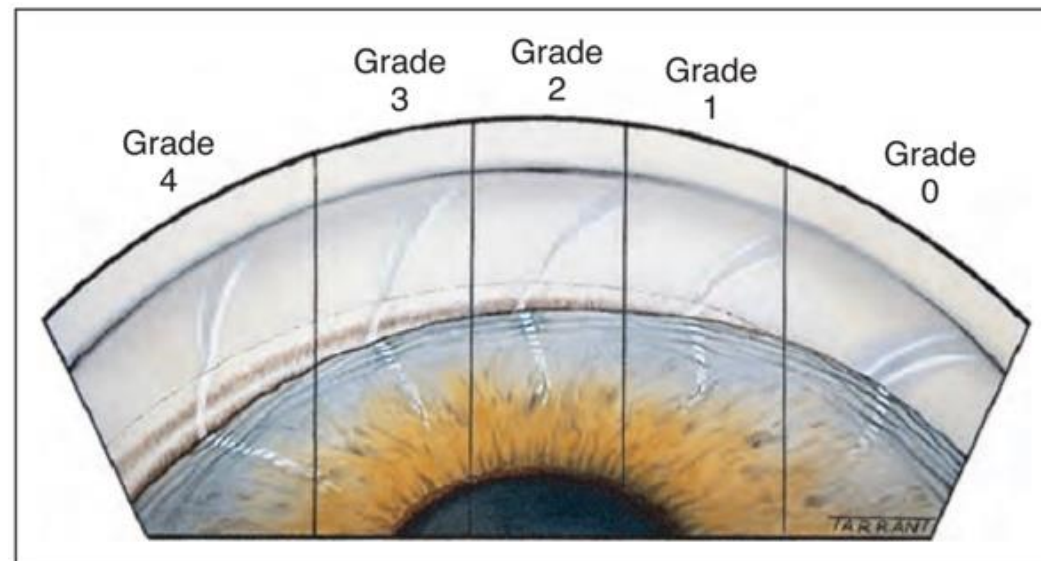
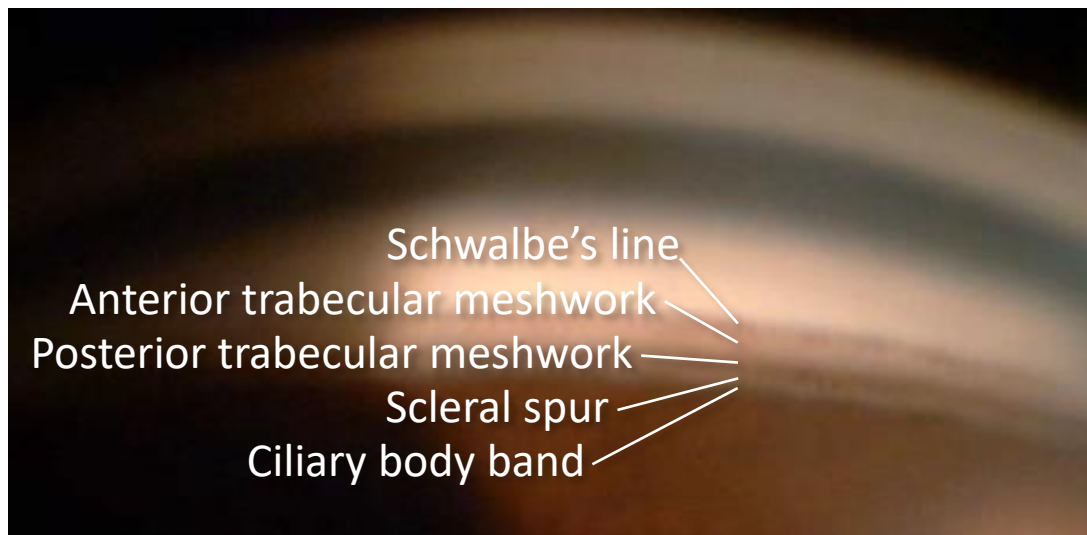


Fig. 11.31 Grading of angle width according to number of visible structures

- Grade 0 : no structures visible (closed angle)
- Grade 1 : Schwalbe's line visible (closure possible)
- Grade 2 : TM visible, but scleral spur invisible (narrow)
- Grade 3 : scleral spur visible (closure impossible)
- Grade 4 : ciliary body band visible (closure impossible)

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Abbreviation

眼底鏡檢查 Ophthalmoscopy

主要用於檢查眼後段 (posterior segment) 的構造，包括玻璃體、視網膜、視神經盤以及脈絡膜。有以下兩種儀器：

- **直接眼底鏡Direct ophthalmoscope**：可提供放大15倍的正立影像。
- **間接眼底鏡Indirect ophthalmoscope**：提供放大倍率比較小 (若採用20D lens可以放大3倍) 的立體倒立影像。

此外尚有 fundus contact lens 可以配合細隙燈使用來檢查眼底，例如常見的 Goldmann three-mirror contact lens 有三面角度不同的反射鏡，以利於做周邊視網膜的檢查。

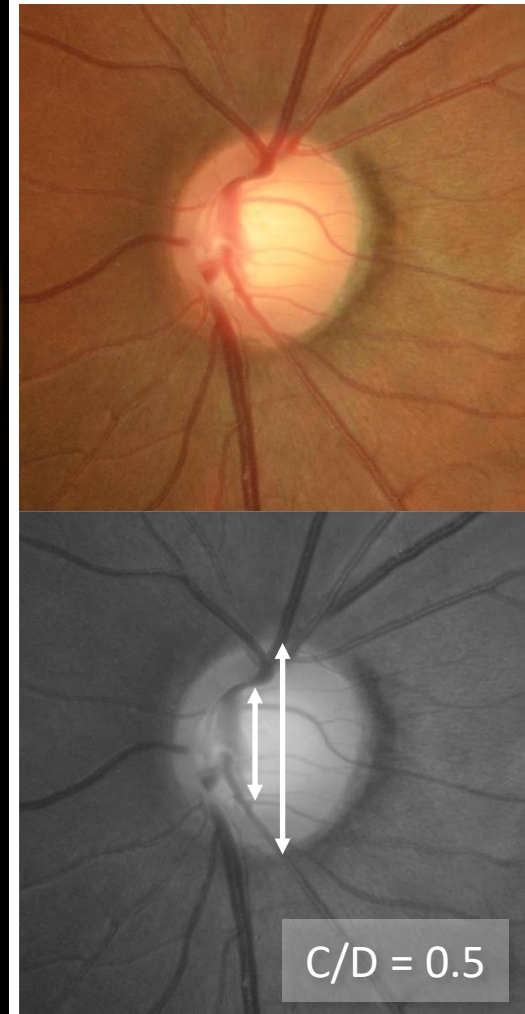
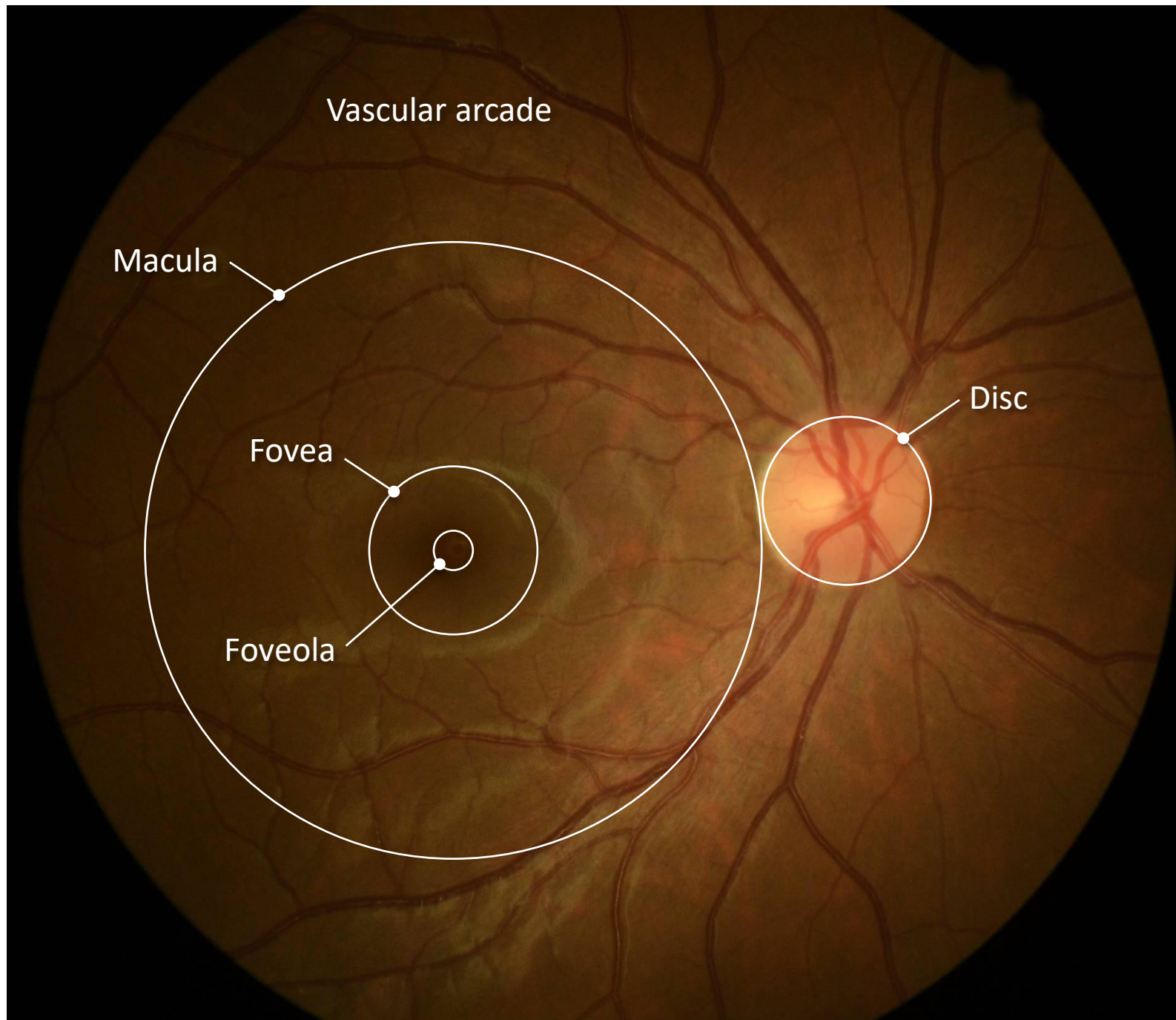
眼底檢查紀錄的重點：

- **視神經盤Disc**：C/D (cup-disc ratio)、color (pale / congestion)、margin (clear / blurred)、high myopia 的人常會有temporal conus。
- **血管Vessel**：A/V (artery-vein ratio)、vessel 的reflex是否正常、有沒有阻塞、周圍有沒有hemorrhage、exudates、edema等等。
- **黃斑部Macula**：健康的fovea會有foveal reflex，不正常的黃斑部會有RPE (視網膜色素細胞) 的變化或者edema、drusen、scar、hole、epiretinal membrane等等。

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Abbreviation



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Abbreviation

直接眼底鏡檢查方法

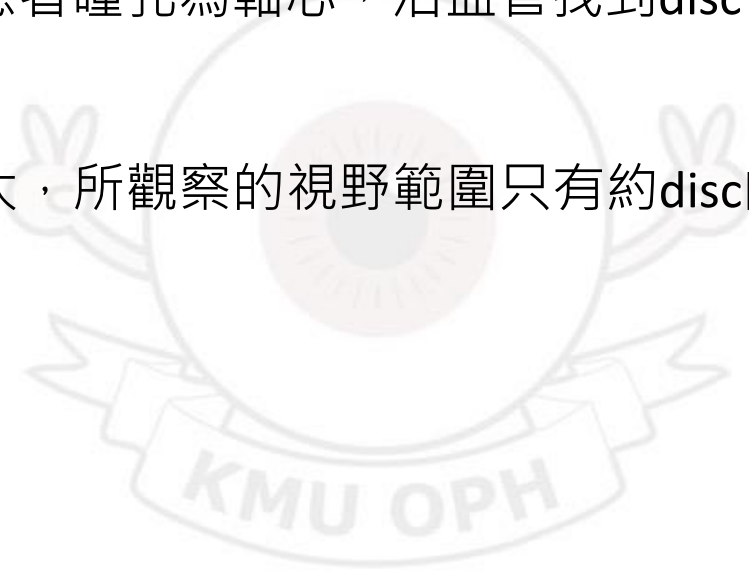
- 室內燈光調暗有助於瞳孔放大以便於觀察眼底，尤其當患者處於未散瞳狀態時。
- 眼底鏡的檢查動作跟病患較為親近，最好有旁人在場。
- 檢查者及患者都需取下眼鏡，請患者直視正前方定點。
- 右手拿眼底鏡以右眼檢查患者的右眼，左手拿眼底鏡以左眼檢查患者的左眼。
- 打開光源檢查燈泡亮度並將度數轉盤歸零，或者也可以事先調整至你自己未戴鏡時的度數(如果你戴近視300度的眼鏡就調到紅色3的位置)。
- 眼底鏡的光源不需調得太亮，以免瞳孔縮小且角膜反光太強產生干擾。
- 一般使用中間的濾鏡白光，及小光圈 (small aperture) 開始檢查。在患者直視角度外側約15度，距離約15公分處，透過觀察窗看患者瞳孔紅反射 (red reflex)，同時食指置於度數轉盤，保持同一視角一邊向患者靠近，一邊調整轉盤對焦，最後以中指固定於患者臉頰，以維持穩定度並保持眼底鏡至眼球的適當距離。轉盤度數約為檢查者與患者的度數總和。

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Abbreviation

- 若患者配合度佳的話，有機會直接觀察到disc，若視野中只看到血管時，在保持透過觀察窗到患者瞳孔呈直線狀態下，以患者瞳孔為軸心，沿血管找到disc，接著再往temporal side移動，找到macula。
- 由於直接眼底鏡的放大倍率較大，所觀察的視野範圍只有約disc的大小，或稍大一些，必須整合不同角度看到的影像。



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Abbreviation



開關 / 調光鈕



FILTERS

-  **Cobalt Blue Filter:** Blue filter used with fluorescein dye permits easy viewing of small lesions, abrasions, and foreign objects.
-  **Red-Free Filter:** Excludes red rays from examination field for easy identification of veins, arteries, and nerve fibers.

APERTURES

-  **Small Aperture:** Provides easy view of the fundus through an undilated pupil. Always start the examination with this aperture and proceed to micro aperture if pupil is particularly small and/or sensitive to light.
-  **Large Aperture:** Standard aperture for dilated pupil and general examination of the eye.
-  **Micro Spot Aperture:** Allows easy entry into very small, undilated pupils.
-  **Slit Aperture:** Helpful in determining various elevations of lesions, particularly tumors and edematous discs.
-  **Fixation Aperture (Coaxial Model only):** The pattern of an open center and thin lines permits easy observation of eccentric fixation without masking the macula.

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Abbreviation

視野檢查 Perimetry

- 正常視野大小應為：superior / nasal / inferior / temporal = 50° / 60° / 70° / 90°。
- 檢查儀器有automated perimetry (Auto-P，自動視野檢查儀)、Goldmann perimeter tangent screen等；因Auto-P受人為因素干擾較少，再現性、敏感性、特異性較高，臨床上較普及。
- Auto-P受檢者採坐姿方式受檢，受檢者必須注視正前方一固定目標 (fixation target)；藉由電腦程式設計，儀器會發出不等強度的光點及聲音，再由受測者以按壓方式表示對光點之反應，待完成檢查後，將光點的亮度轉換呈灰階 (gray scale) 來表示患者視野的狀態。臨床上常用的為檢測三十度視野，電腦輸出視野圖上應在中央偏顳側 (temporal side) 15度處有一生理性盲點 (physiologic blind spot)。

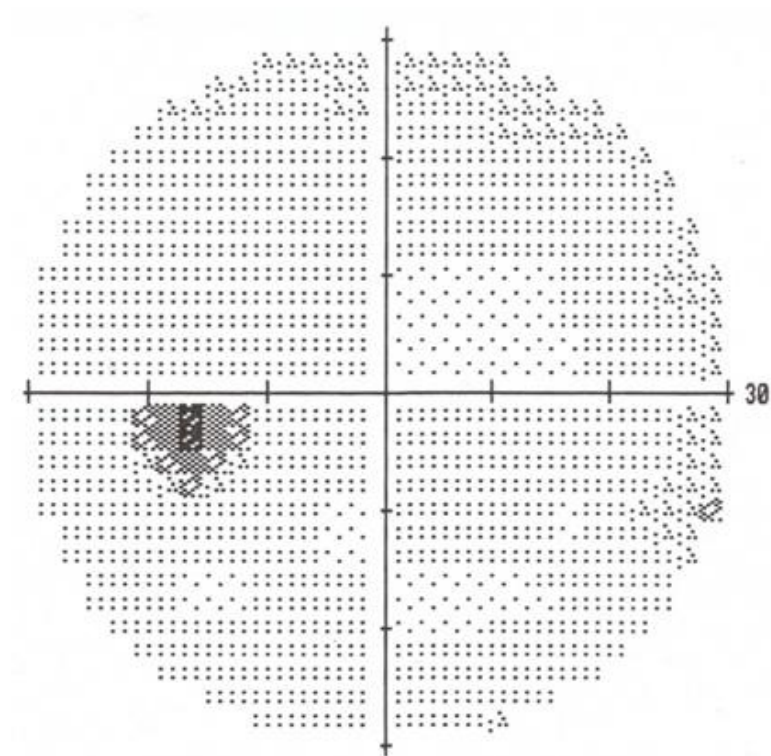
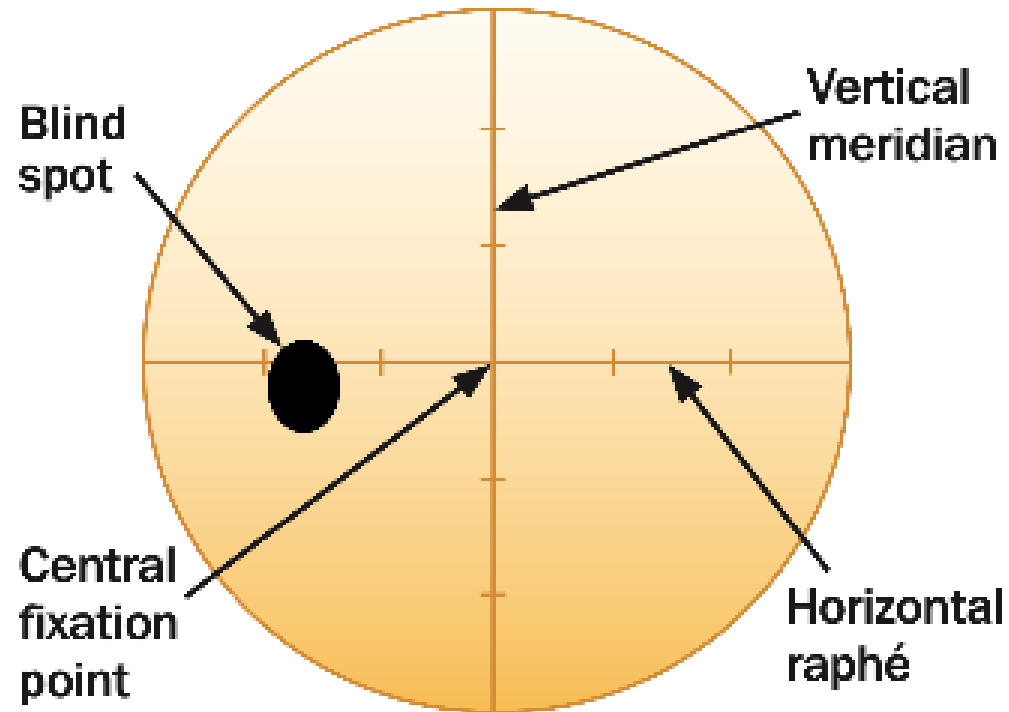
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Abbreviation

Normal Visual Field (Left Eye)

Figure 2. A normal visual field (left eye)



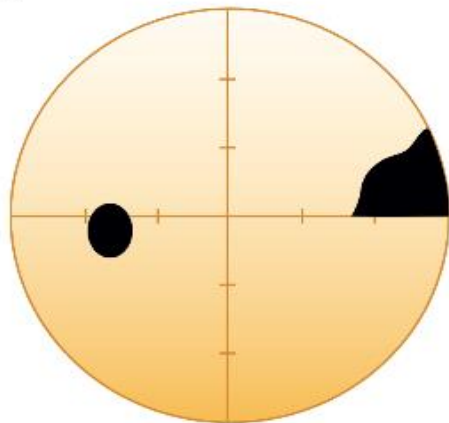
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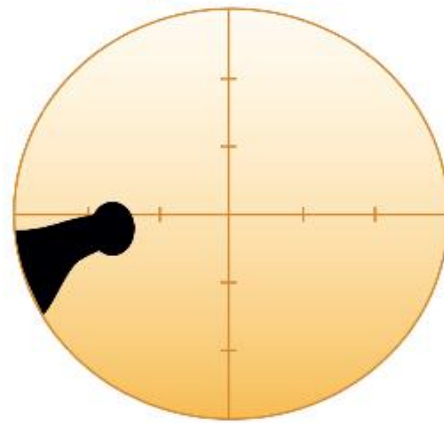
Abbreviation

Glaucomatous Field Defects in Left Eyes

Figure 3. Glaucomatous field defects in left eyes



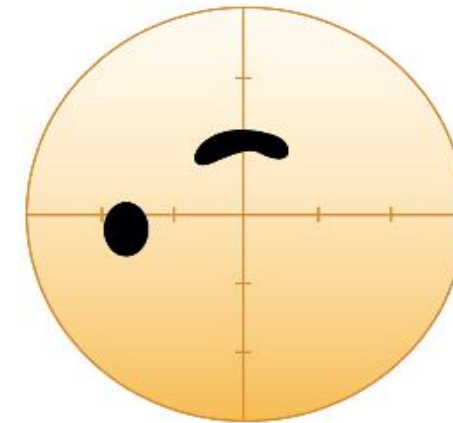
(a) nasal step



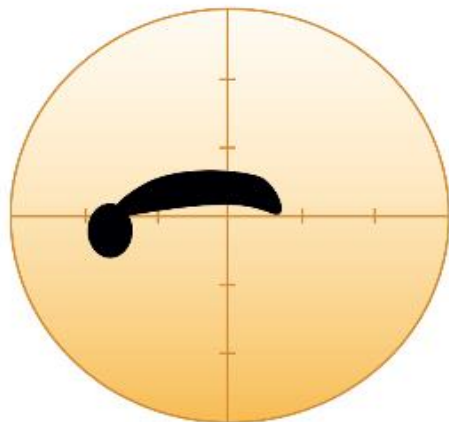
(b) temporal wedge



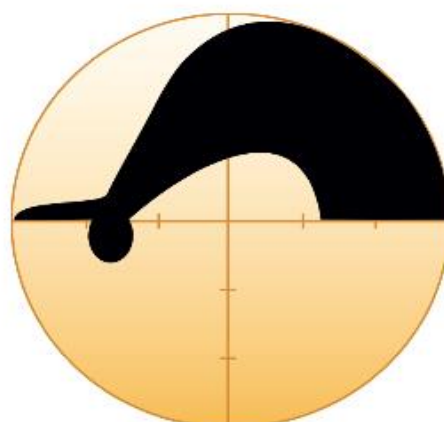
(c) established superior arcuate defect



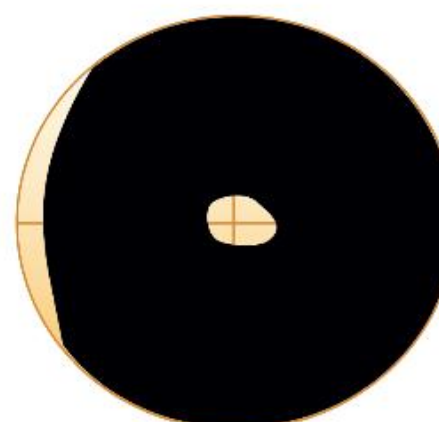
(d) early superior paracentral defect at 10°



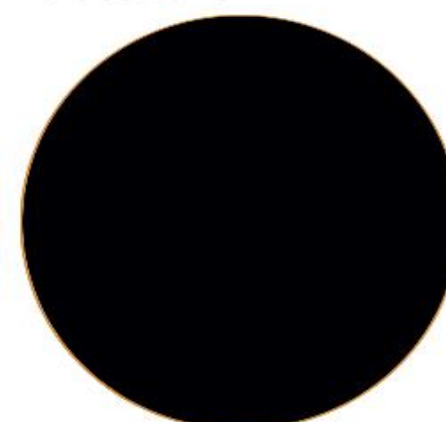
(e) superior, fixation-threatening paracentral defect



(f) superior arcuate with peripheral breakthrough and early inferior defect



(g) tunnel vision defect with temporal crescent sparing



(h) end stage, complete field loss

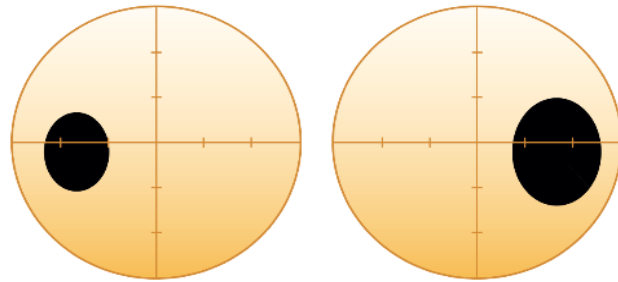
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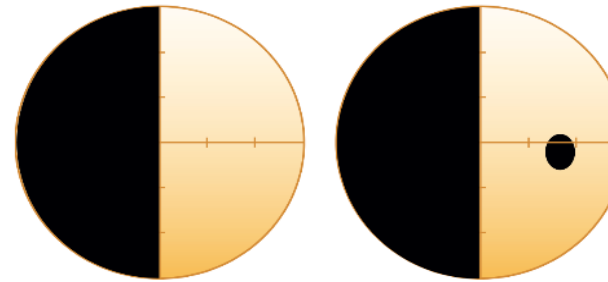
Abbreviation

Non-Glaucomatous Bilateral Visual Field Defects

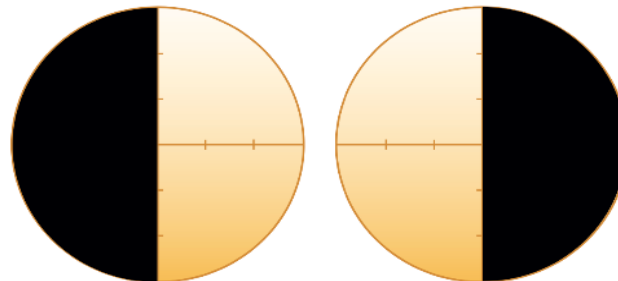
Figure 4. Non-glaucomatous bilateral visual field defects



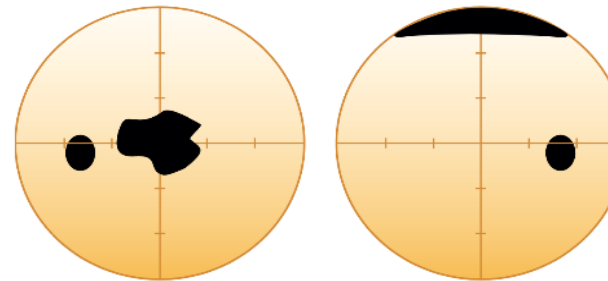
(a) Enlarged blind spots (e.g. papilloedema)



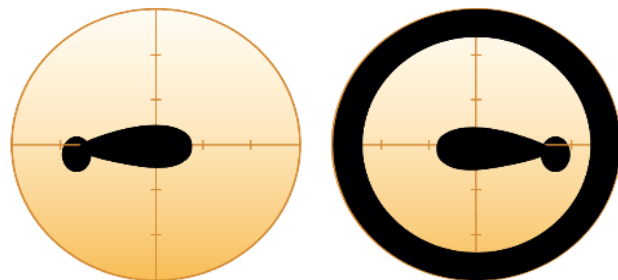
(b) Left homonymous hemianopia (e.g. stroke)



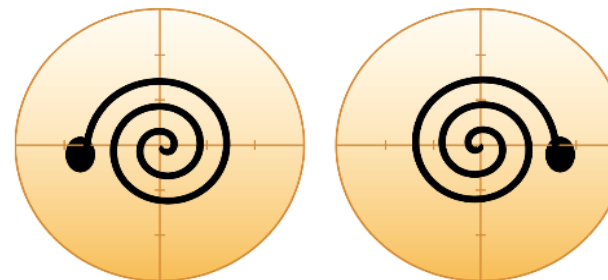
(c) Bitemporal hemianopia (e.g. pituitary tumour)



(d) Left central scotoma (e.g. central macular disease) and right upper eyelid artefact



(e) Centrocaecal scotomata (e.g. toxic optic neuropathy) and right lens rim artefact



(f) Spiral visual fields (the functional or hysterical visual loss associated with malingering)

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Exam



Surgery



Emergency



Drug

Abbreviation

眼球運動及眼位檢查 Ocular Motility and Alignment Testing

TOC



Abbreviation

Abbreviations		
∅	Orthophoric	雙眼正位
E	Esophoria	內隱斜視
X	Exophoria	外隱斜視
E(T)	Intermittent esotropia	間歇性內斜視
X(T)	Intermittent exotropia	間歇性外斜視
ET	Manifest esotropia	顯性內斜視
XT	Manifest exotropia	顯性外斜視
HT	Hypertropia (*preferred format)	上斜視
HypoT	Hypotropia (*only if restricted)	下斜視
ET'	Deviation at near	看近的時候內斜視

Ocular Motility

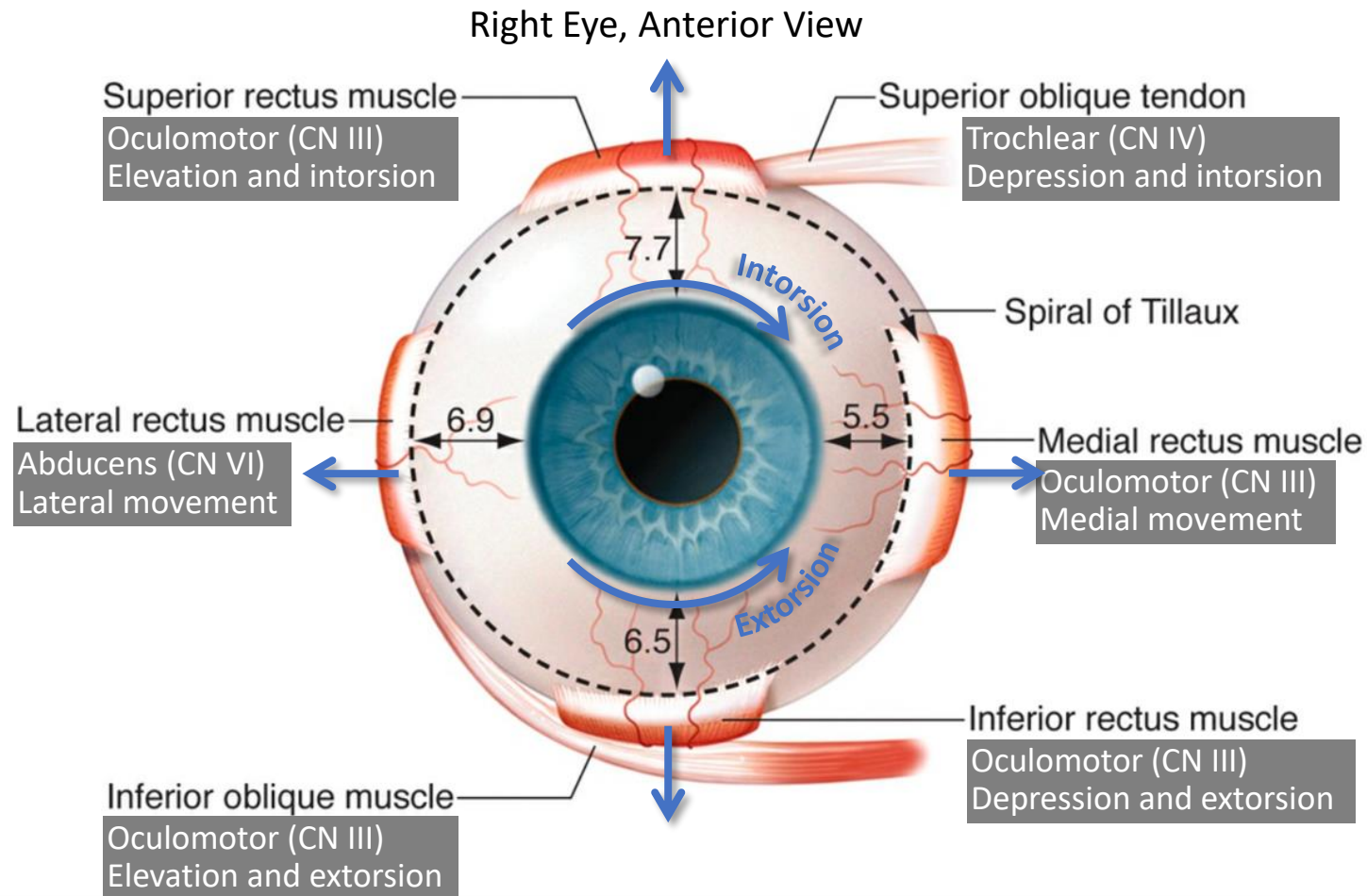


Figure 3-3 Spiral of Tillaux, right eye. *Note:* The insertion distances, given in millimeters, are maximum values. Insertion distances vary among individuals. (Illustration by Christine Gralapp.)

TOC



Abbreviation

Cardinal Position of Gaze and Yolk Muscles

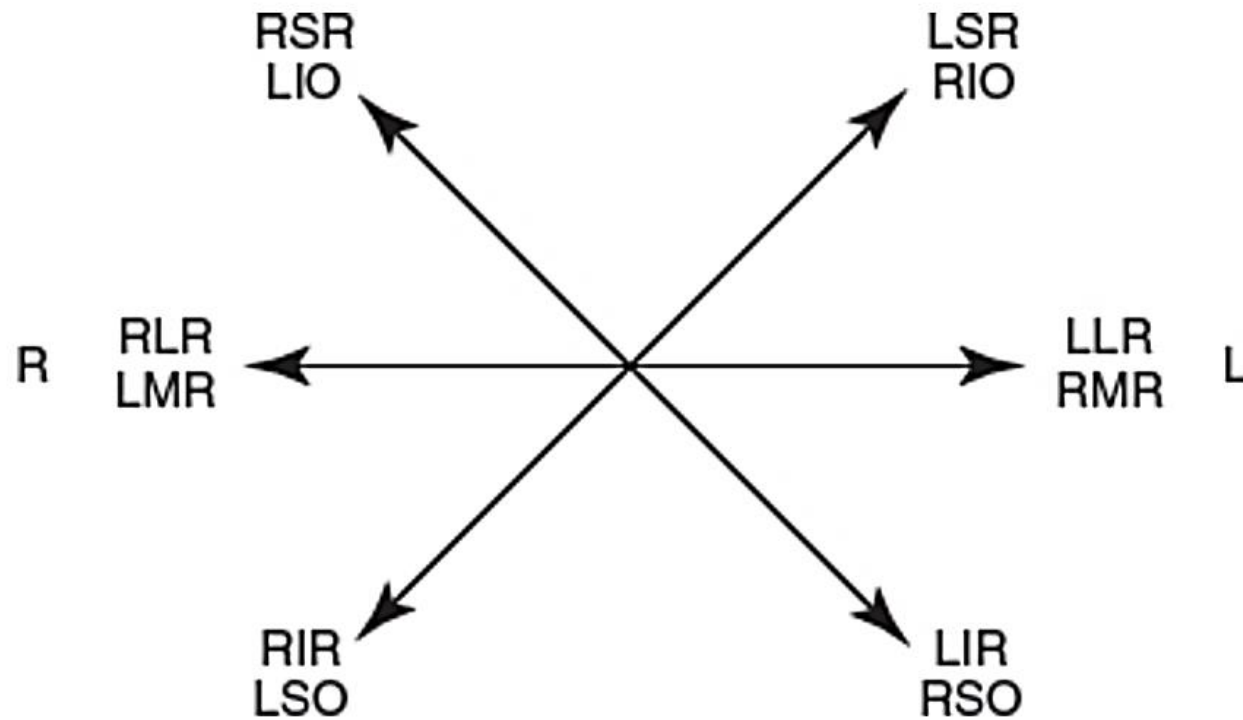


Figure 6.3 The cardinal positions of gaze and the yoke muscles acting in those positions.

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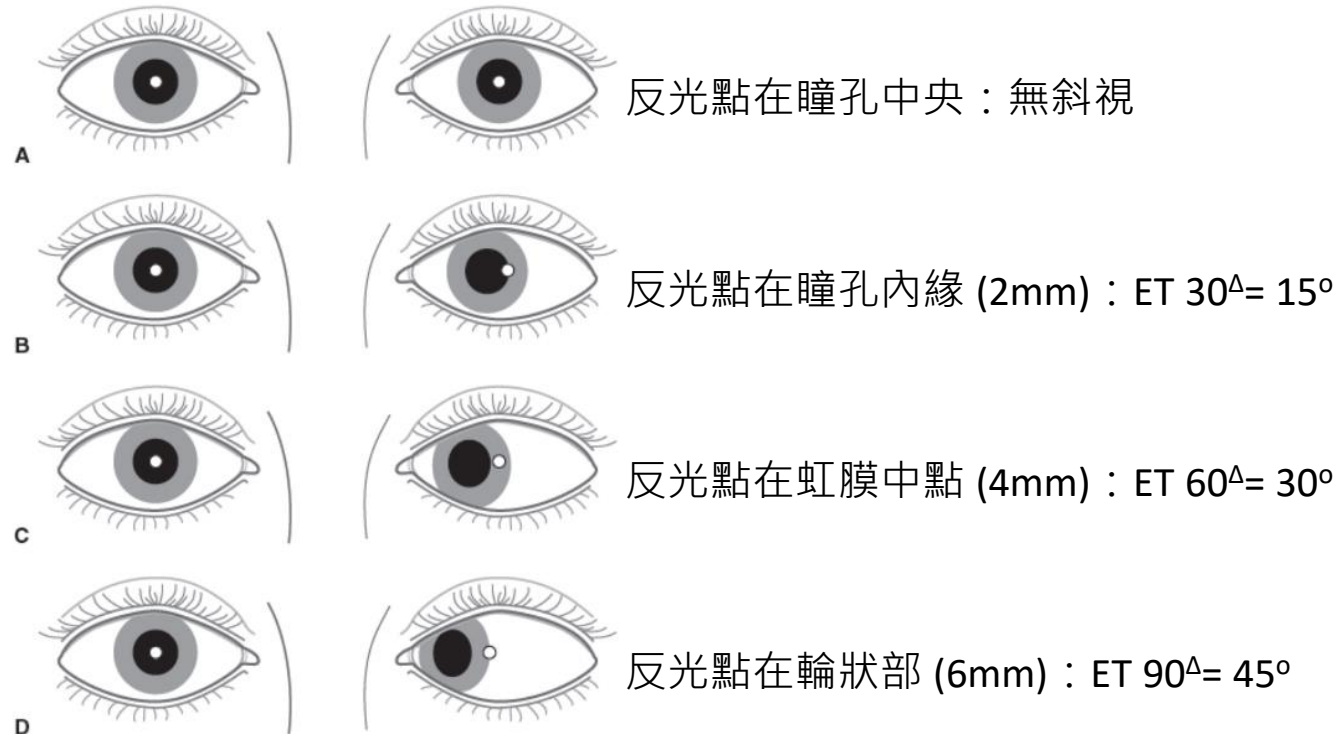


Abbreviation

Test of Alignment : (1) 角膜反光點測試 Corneal light reflection test

利用筆燈照射角膜產生的反光亮點在雙眼的相對位置來檢測眼位，正常狀況下應在雙眼對稱的位置 (若患者直視燈光，反光點會位在雙眼瞳孔正中央)。適合用於配合度較差的病患，例如嬰幼兒。測試的方法有Hirschberg 和 Krimsky test。

1. Hirschberg test：用角膜反射光點偏差的距離來估計斜視度數，每偏差1 mm = $7^\circ = 15^\Delta$ (prism diopters)。

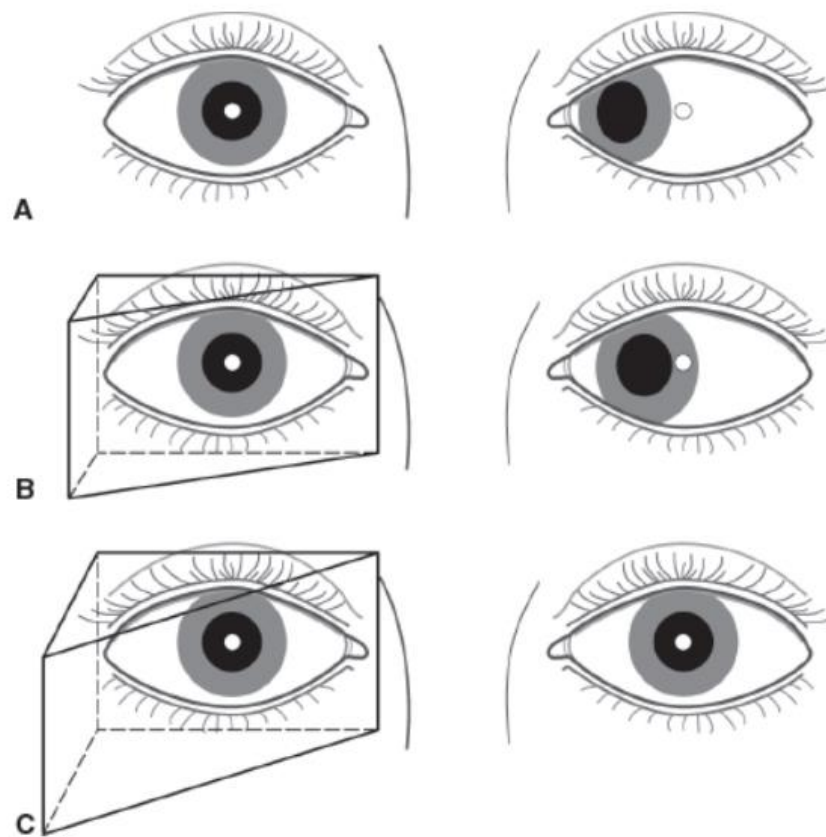


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Abbreviation

2. **Krimsky test**：將稜鏡置於固視眼 (fixating eye) 前，逐漸增加稜鏡度數直至偏斜眼 (deviating eye) 的角膜反射光點移至瞳孔中央，藉以測量斜視度數。



TOC



Abbreviation

Test of Alignment : (2) 遮蓋測試 Cover test

適合用於配合度好能夠固視 (maintain constant fixation) 的病患，例如學齡期兒童。有三種測試方法：monocular cover-uncover test、alternate cover test (或稱為cross cover test)、simultaneous prism and cover test。

- 1. Cover-uncover test**：執行時必須讓患者在適當的屈光矯正 (例如配戴常用的眼鏡) 之下，請患者正坐並目視正前方遠處的一個定點 (fixating point)，施測者在不遮擋患者固視點的狀況下坐在患者正對面。使用遮眼板或者施測者的手掌遮擋固視眼 (fixating eye)，觀察另一眼是否移動並記錄移動的方向；如果未遮蓋眼向外移動，代表患者有內斜視 (ET)，向內移動代表有外斜視 (XT)。把遮擋板移開之後等待至少三秒後在另外一眼執行相同動作，雙眼輪流完成後接著將固視點從遠處移至近處 (near point) 並依序重複以上動作。
- 2. Alternate cover test**：執行步驟大致與cover-uncover test相同，除了遮蓋雙眼之間的時間為連續交替；此時雙眼視覺 (binocular vision) 被干擾，因此在斜視 (tropia) 以及隱斜視 (phoria) 的患者都可觀察到未遮蓋眼的眼球移動。
- 3. Prism cover test**：用稜鏡置於一眼前並交替遮左右眼，藉由逐漸增加稜鏡度數來觀察何時兩眼的眼位不再因為交替遮眼而移動，進而量化斜視的嚴重程度。

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Abbreviation

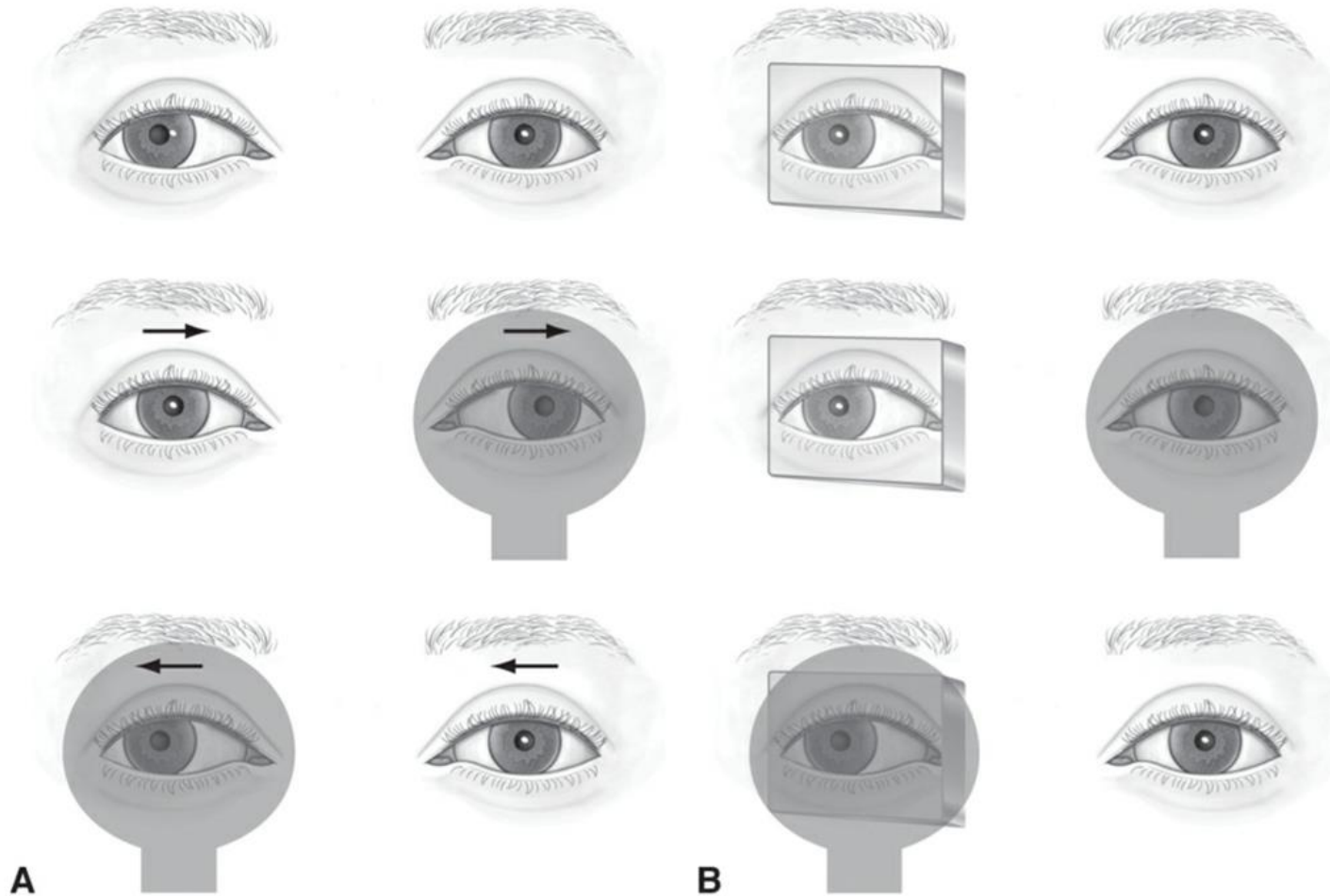


Figure 7-2 A, The alternate cover test. *Top*: Exotropia, left eye fixating. *Middle and bottom*: Both eyes move each time the cover alternates from one eye to the other. **B**, The prism alternate cover test. *Top*: The exotropia is neutralized with a prism of the correct power. *Middle and bottom*: The eyes do not move as the cover alternates from one eye to the other. (Illustration developed by Steven M. Archer, MD; original illustration by Mark Miller.)

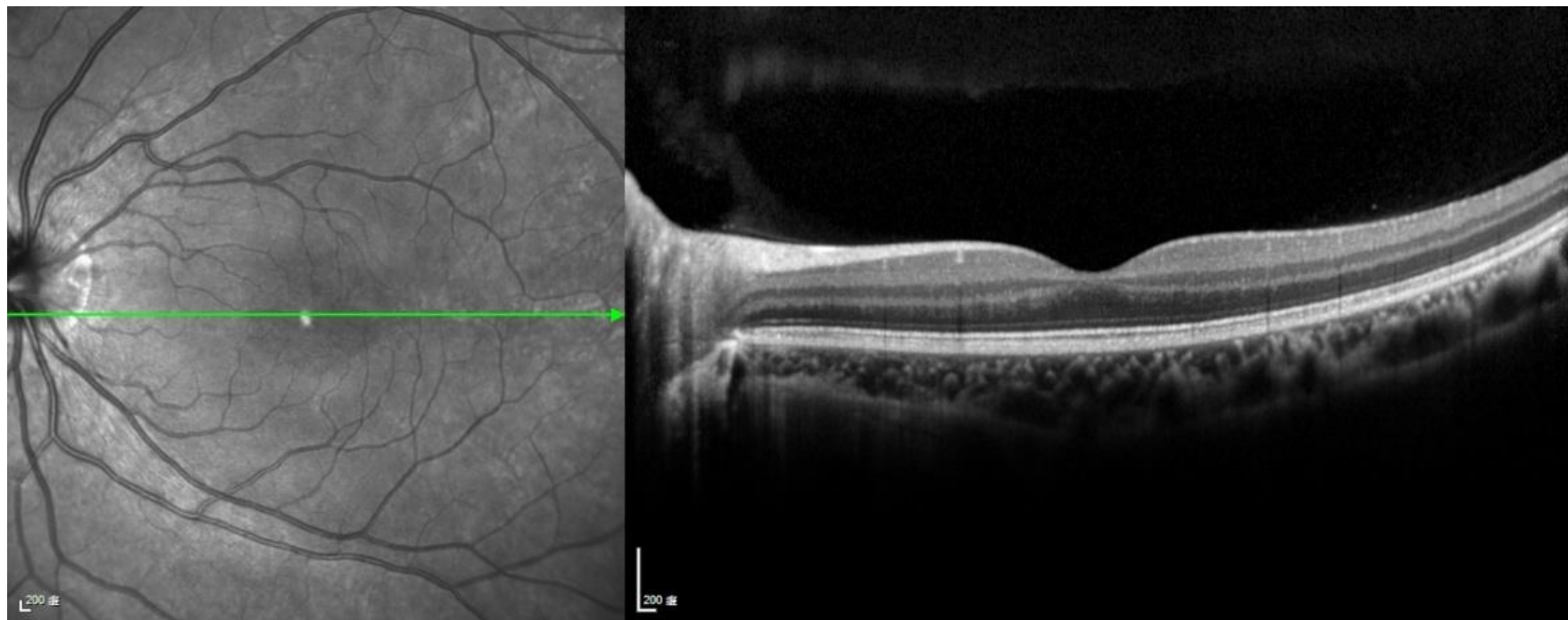
TOC



Abbreviation

光學同調斷層掃描Optical Coherence Tomography (OCT)

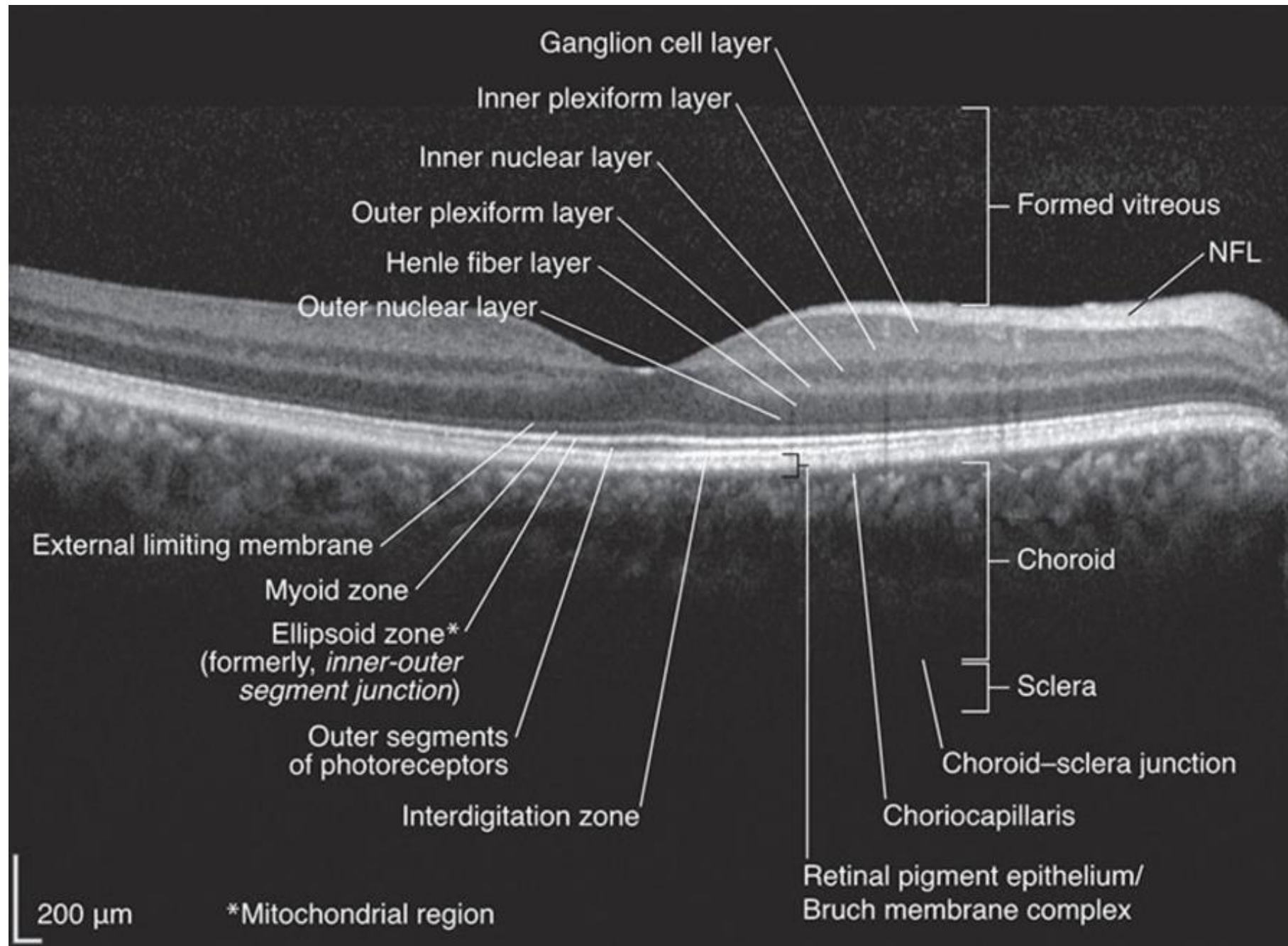
OCT就好比是視網膜的「電腦斷層」(請注意其英文不同)，可以提供眼底鏡無法看到的高解析度視網膜斷層切面影像，還能進一步使用電腦分析計算出視網膜各層厚度將其數據化，提供更豐富的診斷及追蹤資訊。因科技進步帶來攝影速度及影像解析度的不斷提升，最新的機型已可以利用此技術造影視網膜各層血管，稱之為OCTA (angiography)。OCT對視網膜特別是黃斑部疾病，以及慢性青光眼的診斷以及追蹤都有其重要性。



TOC



Abbreviation



TOC



Abbreviation

眼科常見手術

TOC



Exam



Surgery



Emergency



Drug

Abbreviation



眼科常見手術

白內障手術
Cataract Extraction



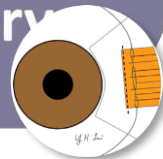
青光眼手術
Glaucoma Surgery



玻璃體視網膜手術
Vitreoretinal Surge



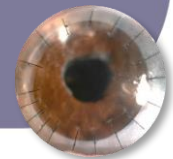
斜視手術
Strabismus Surgery



眼整形手術
Oculoplastic Surgery



角膜手術
Corneal Surgery



TOC



Abbreviation

白內障手術 Cataract Extraction

- 晶體摘除術 lens extraction

- ↳ 囊內摘除術 intracapsular cataract extraction (ICCE)
- ↳ 囊外摘除術 extracapsular cataract extraction (ECCE)
- ↳ 超音波晶體乳化術 phacoemulsification (phaco)

- 人工晶體植入術 intraocular lens implantation

- ↳ 前房人工晶體植入 anterior chamber intraocular lens implantation (ACIOL)
- ↳ 後房人工晶體植入 posterior chamber intraocular lens implantation (PCIOL)
- ↳ 二次晶體植入術 second IOL implant / suture IOL

- 前玻璃體切除術 anterior vitrectomy (anterior VT)

- 前房藥物注射術 intracameral injection (ICI)

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Abbreviation

白內障晶體乳化術步驟 Phacoemulsification Surgery Steps

麻醉 Anesthesia

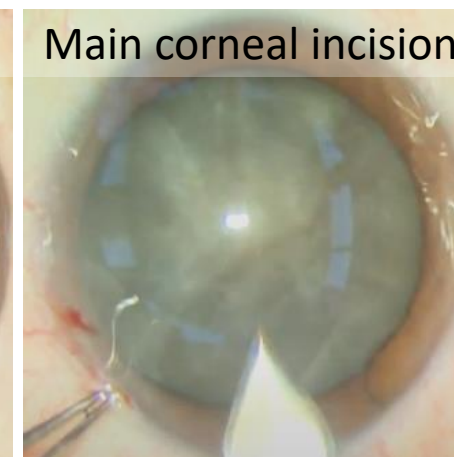
- 1 可以採球後麻醉 (retrobulbar anesthesia) 注射，或者用局部麻醉劑 (topical anesthesia，例如 alcaine) 點眼數次。

準備 Preparation

- 2 消毒並鋪單後使用開眼器 (lid speculum) 將眼皮撐開。

切口 Incisions

- 3 在術者輔助手 (若右撇子就是左手) 處於輪狀部穿刺約1 mm大小的傷口 (side port incision) 進入前房後，在前房注入透明粘彈劑 (viscoelastic，OVD)。在距離 side port incision約45-90度之間的輪狀部用角膜刀 (keratome slit knife) 進入前房 (main corneal incision)，此切口大小約2.2~3mm。



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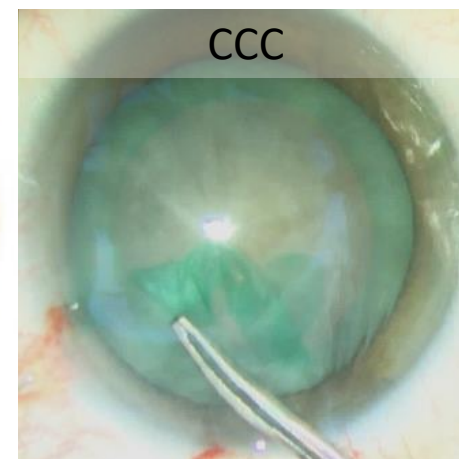
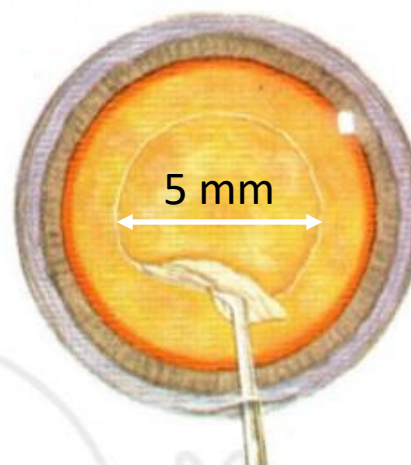


Abbreviation

4

前囊撕開術 Continuous curvilinear capsulorhexis

用鑷子或者尖端折過的針頭在前囊撕開一個直徑約5mm的完整圓形開口。



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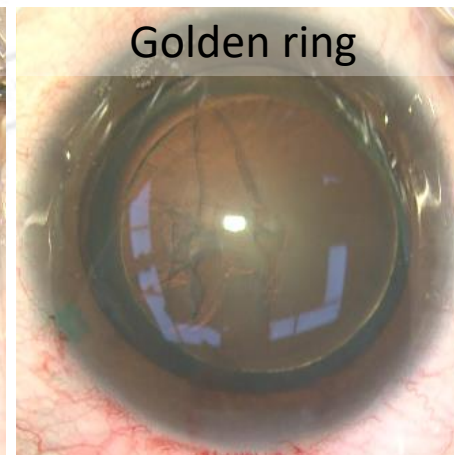
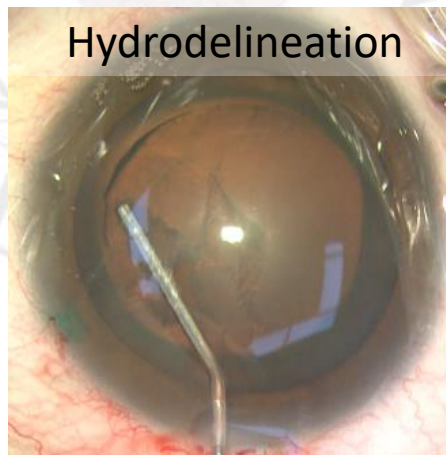
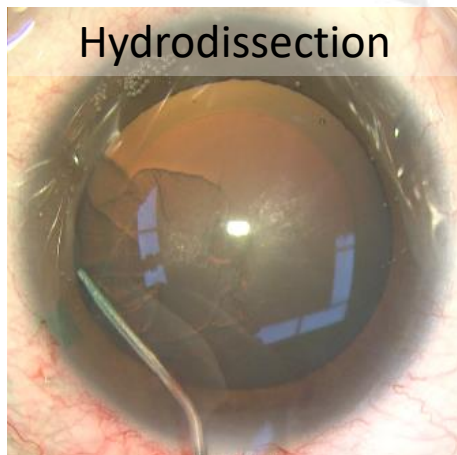
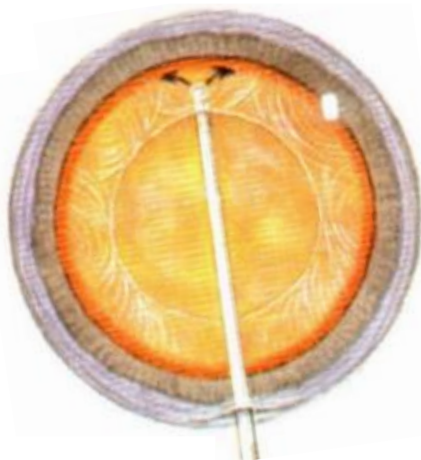


Exam

5

囊皮質分離術 Hydrodissection / hydrodelineation

利用注射液體的衝力將水晶體囊袋與皮質核分離，並嘗試將水晶體核旋轉以確認之。



Surgery



Emergency



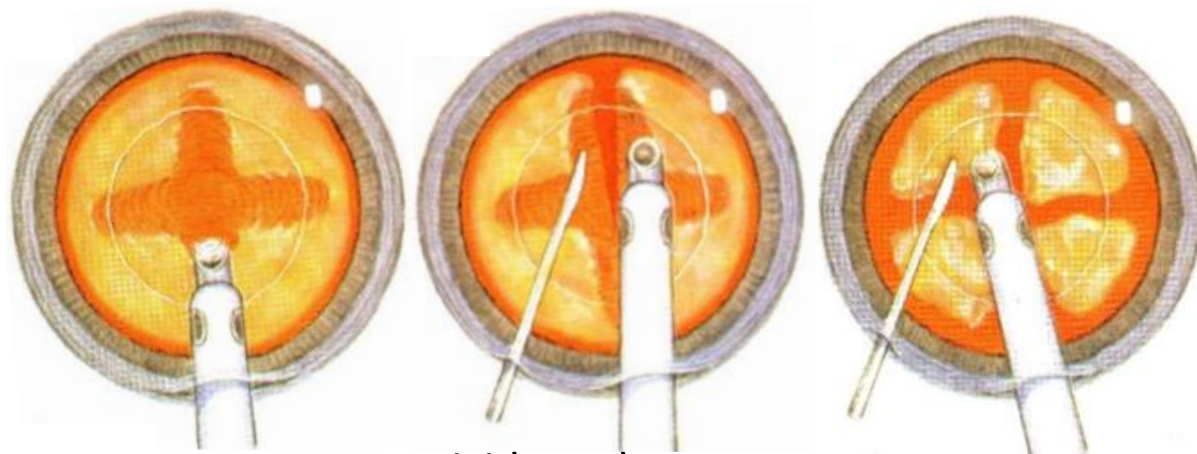
Drug

Abbreviation

6

超音波核乳化術 Phaco removal of nucleus

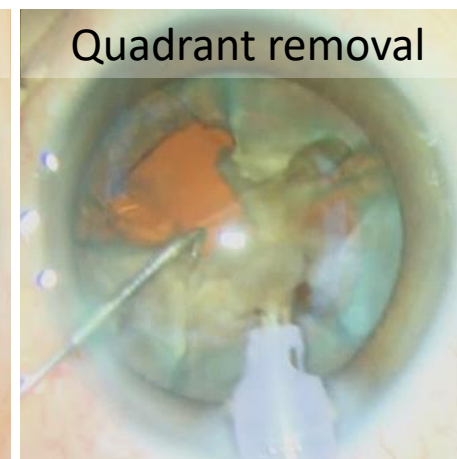
將超音波探頭 (phaco tip) 從corneal main incision伸入前房，side port incision伸入輔助工具 (second instrument，例如chopper)，雙手併用利用超音波乳化能量將水晶體核分成數塊逐步吸除。分塊的方式常見的有divide and conquer、phaco chop、stop and chop。



Divide and conquer



Phaco chop



Quadrant removal



Cortex aspiration



Capsule polish



皮質吸引術 Cortex aspiration

將殘餘黏附在囊袋上的晶體皮質用沖吸探頭 (I/A tip) 逐一剝除並將囊袋拋光 (capsule polish)。

7

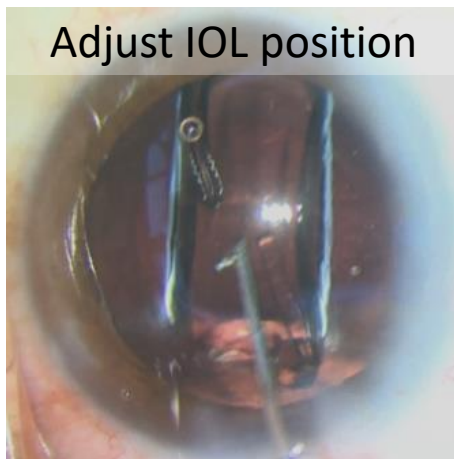
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Abbreviation

人工水晶體植入 IOL insertion

再次注入OVD至囊袋鼓脹之後，用注射器將折疊式人工水晶體 (foldable IOL) 打進囊袋，待人工水晶體完全展開後，用I/A tip將OVD移除乾淨。若是散光型人工晶體 (toric IOL) 須注意角度定位。



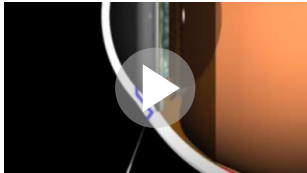
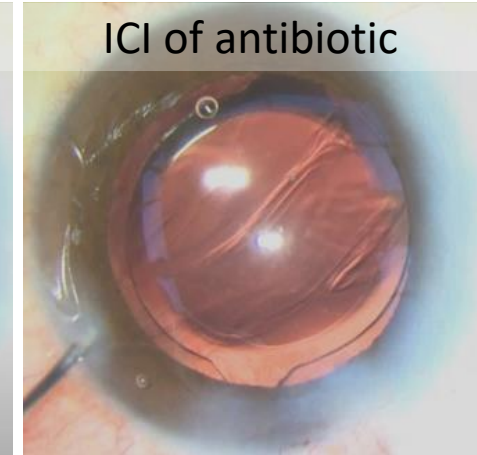
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Abbreviation

完成 Completion

檢查切口是否漏水，可在角膜基質注入手術用平衡鹽液 (BSS) 來封住傷口 (hydrosealing) 或者用Nylon 10-0線縫合。此外可以在前房注入縮瞳劑，也可以在前房內或結膜下注射預防性的抗生素溶液，或者類固醇。手術完成後覆上眼蓋 (protective eye shield) 保護。



Cataract Surgery 3D Animation

Oliver Findl

<https://www.youtube.com/watch?v=jxhX0nOoJLE>



Phacoemulsification with PCIOL implant

HY Tseng



Phacoemulsification with toric IOL implant

HY Tseng

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Abbreviation

青光眼手術 Glaucoma Surgery

- 青光眼濾過性手術 **glaucoma filtering surgery**

- ↳  小樑切除術 trabeculectomy (TRBC; or TRAB) 
- ↳ 周邊虹膜切除術 peripheral iridectomy (PI)
- ↳ 可拆式縫線 releasable suture (RS)
- ↳ 濾泡重建術 bleb revision
- ↳ 小樑切開術 trabeculotomy (TRBL)

- 青光眼雷射手術 **glaucoma laser procedures**

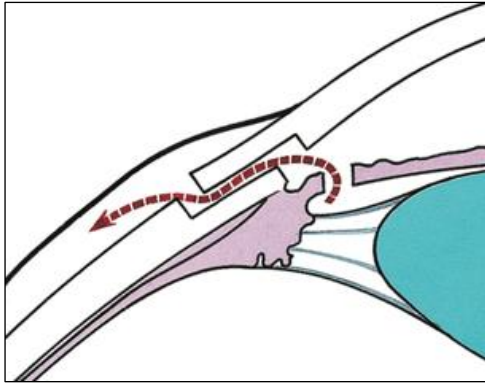
- ↳ 雷射 (周邊) 虹膜穿孔術 laser iridotomy (LI) / laser peripheral iridotomy (LPI)
- ↳ 雷射小樑整形術 laser trabeculoplasty
- ↳ 雷射虹膜整形術 laser iridoplasty
- ↳ 睫狀體二極體雷射破壞術 diode laser cyclodestruction / trans-scleral cyclophotocoagulation (TSCP)

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青光眼小樑切除術步驟 Trabeculectomy Surgery Steps

Principles

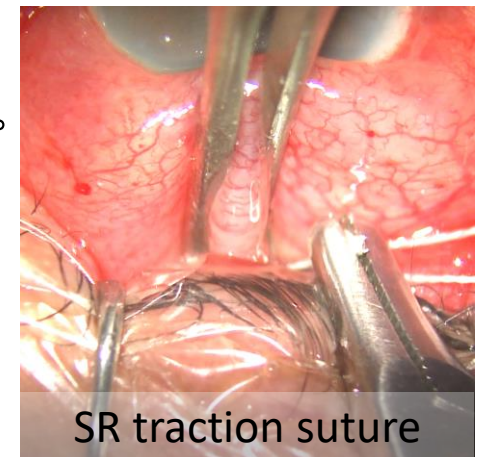


麻醉 Anesthesia

- 1 球後麻醉 (retrobulbar anesthesia) 注射，或者在結膜下注射局部麻醉劑 (local anesthesia)。

準備 Preparation

- 2 與其他大多數眼部手術不同，術前須點縮瞳劑 (miotics) 數次將瞳孔縮小。消毒鋪單使用開眼器 (lid speculum) 將眼皮撐開後，將12點鐘部位的輪狀部盡量露出；可以採用的方式包括上直肌牽扯固定 (superior rectus muscle traction suture) 或者角膜牽扯固定 (corneal traction suture)。



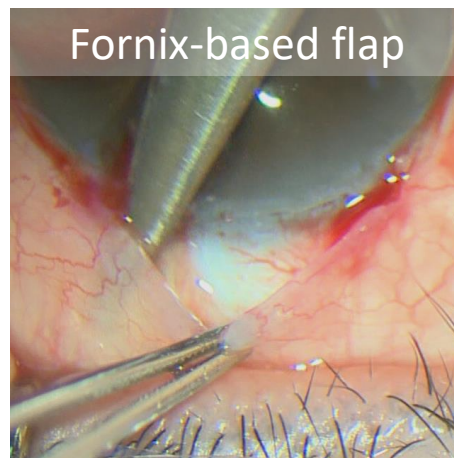
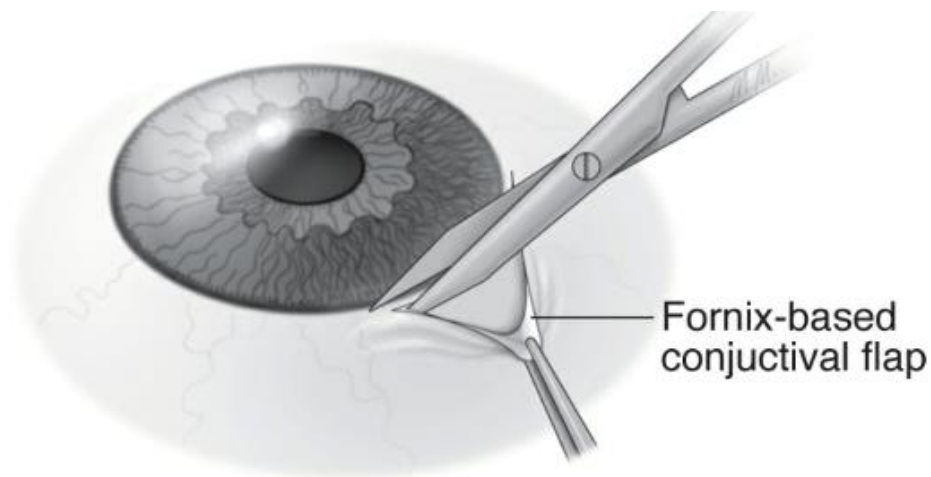
TOC



Abbreviation

結膜皮瓣 Conjunctival flap

在眼球上方10點鐘至2點鐘方向沿著輪狀部將結膜剪開約90~120度並將結膜與鞏膜分開，注意結膜瓣須保持完整不能有小裂孔 (buttonhole)，此種貼著輪狀部剪開的結膜瓣稱為 fornix-based flap；如果剪開的位置高於輪狀部7~10mm則稱之為limbal-based flap，兩種方法可擇一採用。



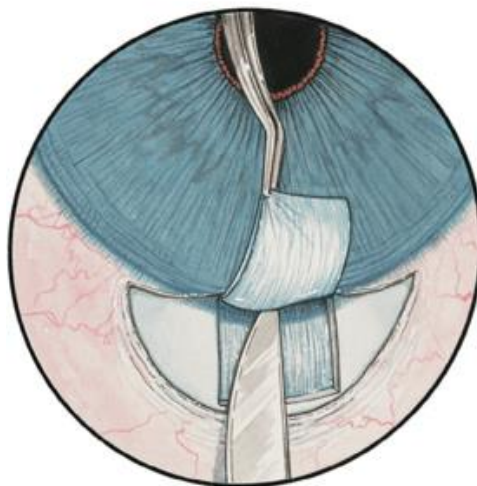
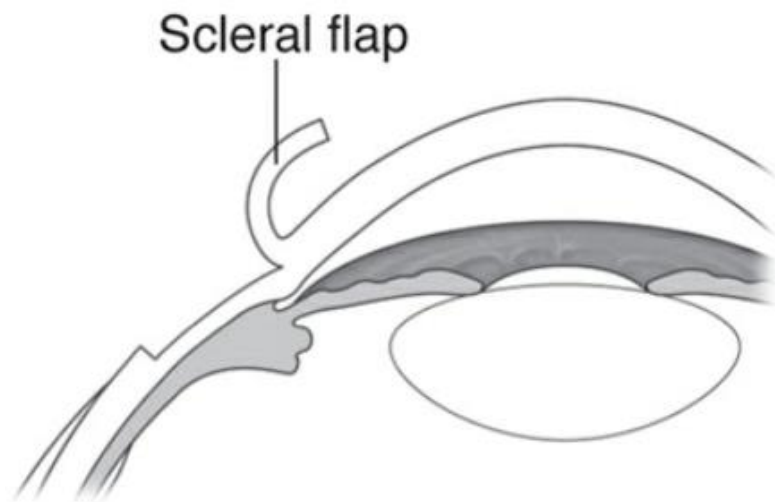
TOC



Abbreviation

鞏膜瓣 Scleral flap

在12點鐘方向距離輪狀部2.5mm~3mm處將鞏膜用新月刀 (crescent knife) 往角膜方向層狀切開，形成一個約50%鞏膜厚度的鞏膜瓣 (lamellar scleral flap, "trapdoor")。此鞏膜瓣的形狀可選擇三角形 (triangular)，長方形 (rectangular)，或者梯形 (trapezoidal)。



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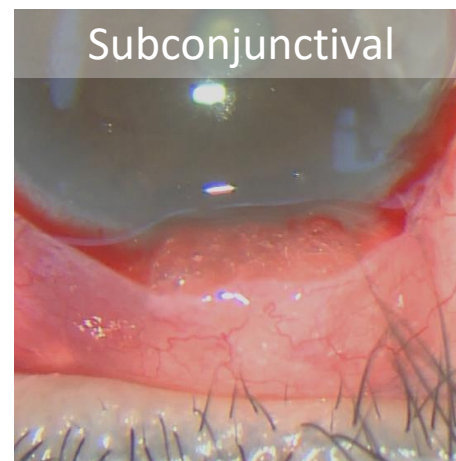


Abbreviation

5

浸泡抗纖維化藥劑 Soaking of antifibrotic agent

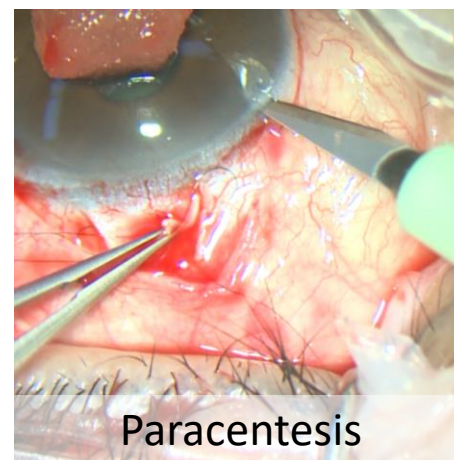
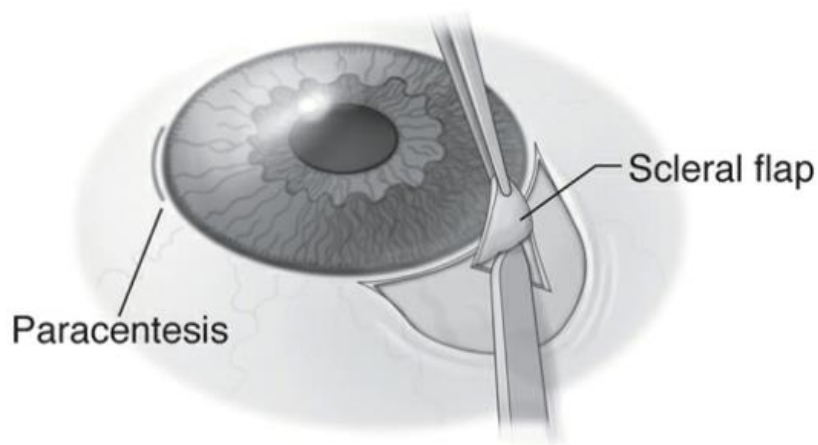
將浸泡過抗纖維化藥劑的海綿片放置結膜瓣以及鞏膜瓣下方數分鐘之後用大量平衡鹽液 (BSS) 沖洗。常用的抗纖維化藥劑有：mitomycin C 0.2~0.5mg/ml 或者 5-fluorouracil 50mg/ml，放置時間為1~5分鐘不等。



6

房水穿刺 Paracentesis

在鞏膜瓣附近的輪狀部角膜側穿刺一個進入前房的小傷口。此傷口可以在手術後期透過注射平衡鹽液 (BSS) 至前房來調整眼壓。



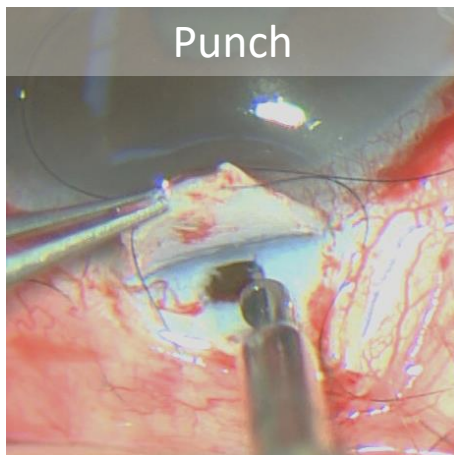
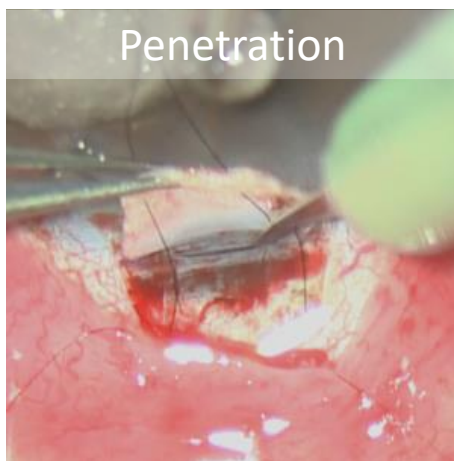
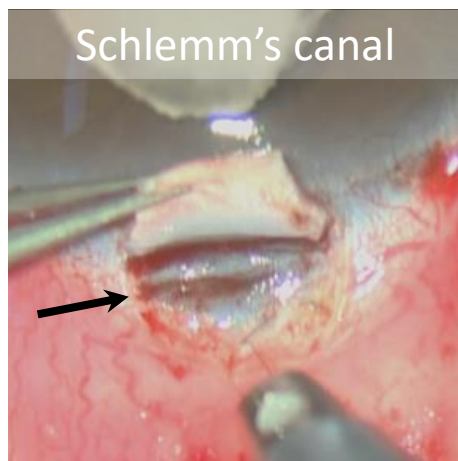
TOC



Abbreviation

深層鞏膜穿孔術 Inner sclerostomy

在鞏膜瓣的下層辨認Schlemm氏管之後在根部穿刺進入前房，再用打洞器 (trabeculectomy punch) 切除1x2mm的深層鞏膜。



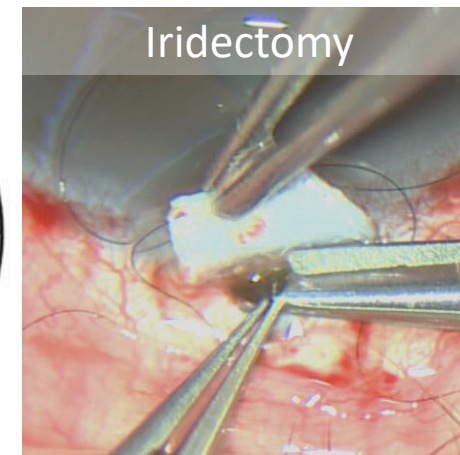
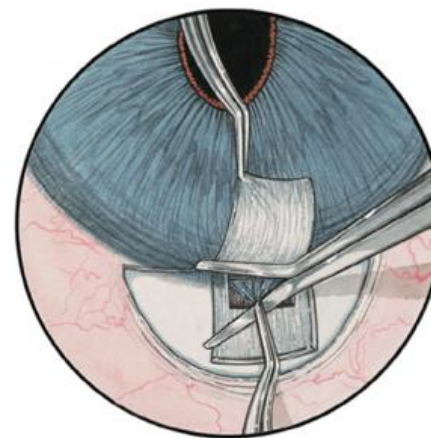
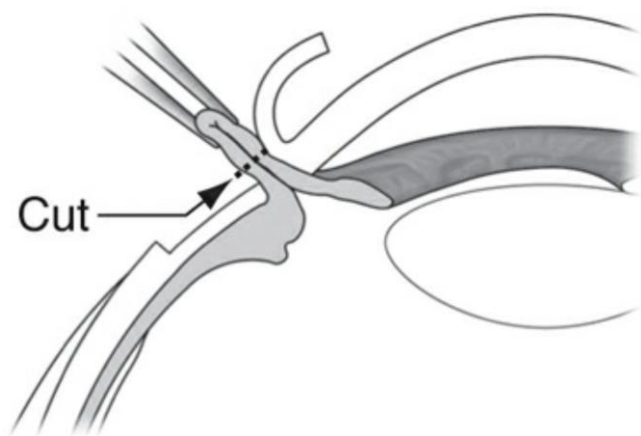
TOC



Abbreviation

虹膜切除術 Iridectomy

將鞏膜孔正下方對應處的虹膜用剪刀剪除以避免阻塞孔道。



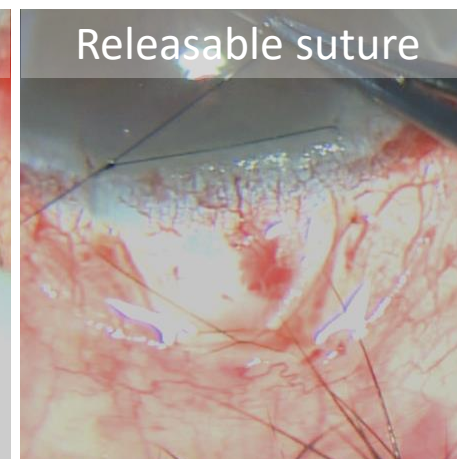
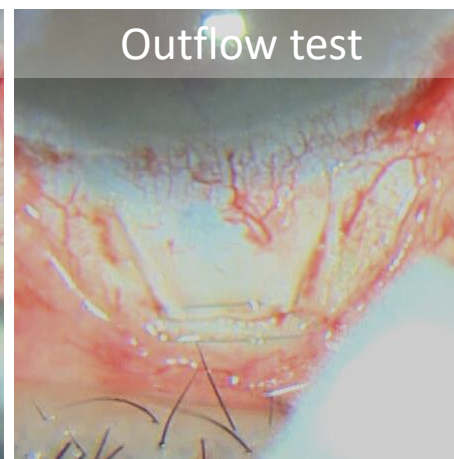
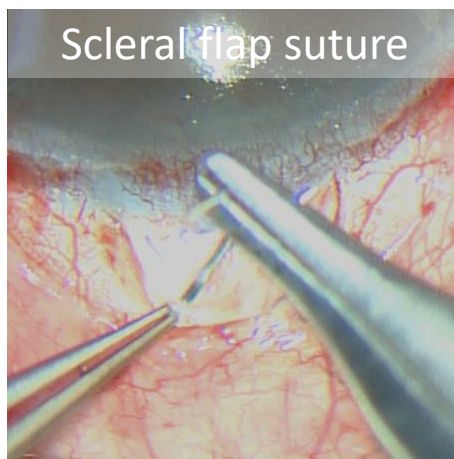
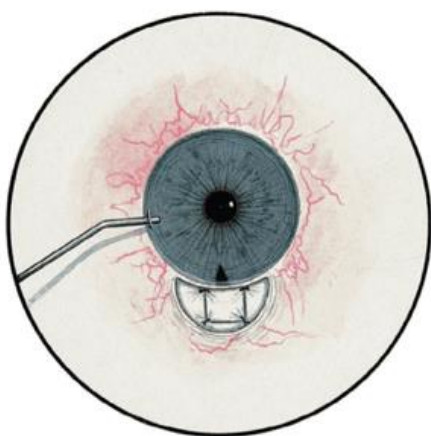
TOC



Abbreviation

鞏膜瓣縫合 Scleral flap closure

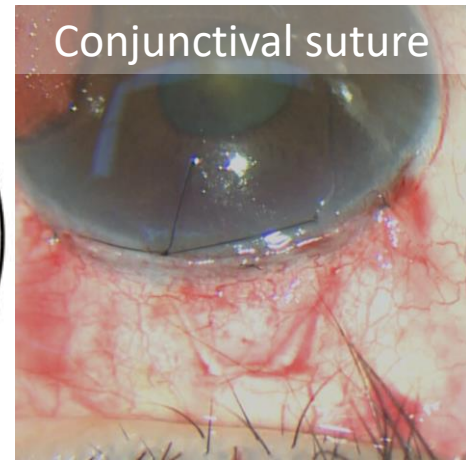
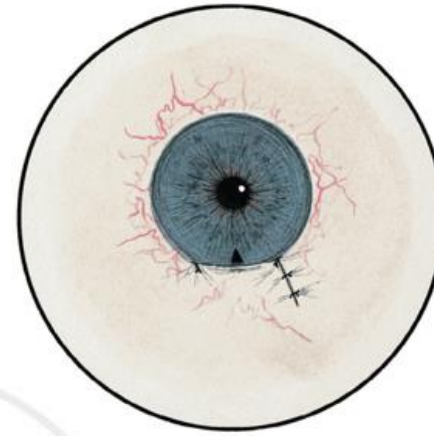
將鞏膜瓣縫合復位，此時可以藉由從房水穿刺傷口 (paracentesis) 注射BSS將眼壓提升後，再觀察房水從鞏膜瓣下排出的速度，進而調整縫線的鬆緊度。為避免術後低眼壓的早期併發症，宜採用可拆除 (releasable)、可調整 (adjustable) 或者可裂解 (lysable) 等不同的縫合手法。



10

結膜縫合 Conjunctival closure

將結膜瓣復位縫合，需縫合至完全不漏水的程度。

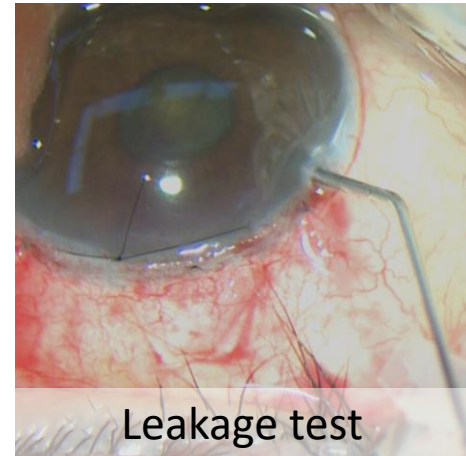


Conjunctival suture

11

完成 Completion

檢查結膜是否漏水，並可在房水穿刺傷口注入平衡鹽液 (BSS)、抗生素溶液或者類固醇溶液。結束時覆上眼蓋 (protective eye shield)。



Leakage test

TOC



Exam



Surgery



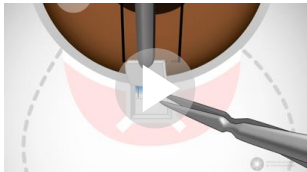
Emergency



Drug

Abbreviation

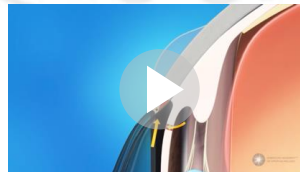
Online Videos



Trabeculectomy Fundamentals

Joseph Caprioli, MD, FACS

<https://www.aaof.org/basic-skills/trabeculectomy-fundamentals>



What happens during trabeculectomy surgery?

EyeSmart — American Academy of Ophthalmology

<https://www.youtube.com/watch?v=D7SUINefezg>



Trabeculectomy

HY Tseng

玻璃體視網膜手術 Vitreoretinal Surgery

-  **玻璃體切除手術 vitrectomy (VT) / pars plana vitrectomy (PPVT; or PPV)** 
 - ↳ 視網膜前膜剝除術 epiretinal membrane peeling (ERM peeling) / membrane peeling (MP)
internal limiting membrane peeling (ILM peeling)
 - ↳ 液氣體交換術 air-fluid exchange / fluid-air exchange (FAX)
 - ↳ 矽油 / 氣體填塞術 silicone oil (SiO₂; or SO) tamponade / air, SF₆, C₃F₈ tamponade
 - ↳ 眼內雷射 endolaser (EL) / photocoagulation (PC)
- **鞏膜扣壓術 scleral buckling (SB)**
- **玻璃體藥物注射術 intravitreal injection (IVI)**
 - ↳ 抗血管內皮生長因子 anti-vascular endothelial growth factor (anti-VEGF)

TOC



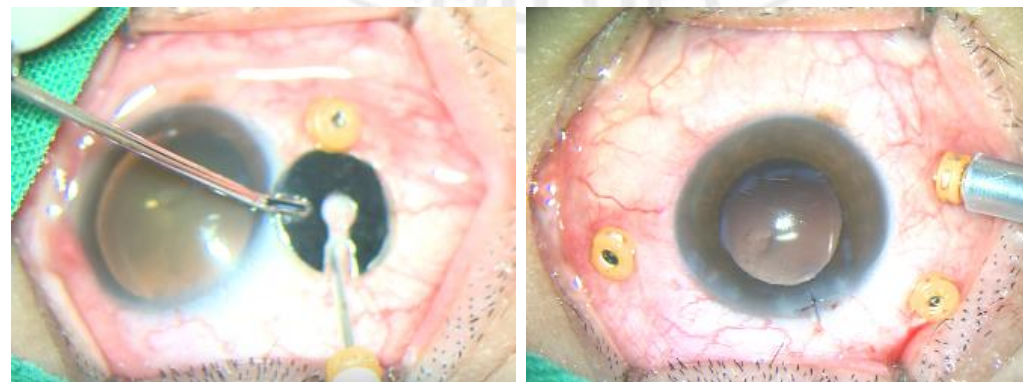
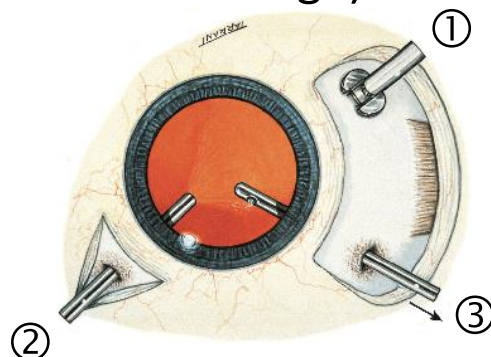
Abbreviation

經平坦部玻璃體切除手術步驟 Pars Plana Vitrectomy Surgery Steps

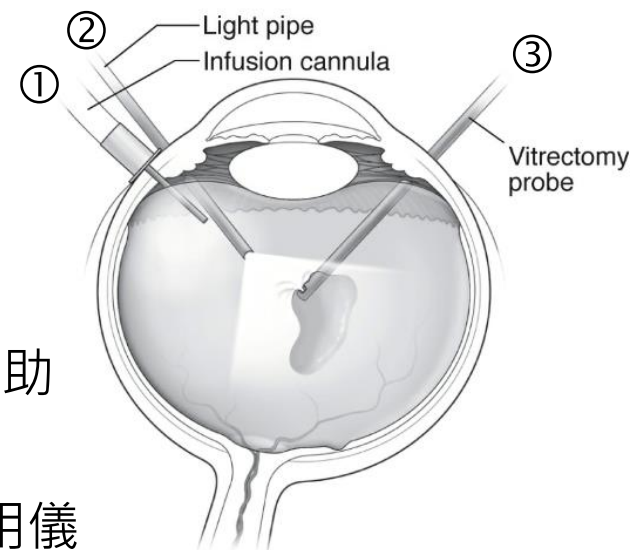
基本設置 Basic setup

在平坦部 (pars plana) (距離輪狀部3.5~4mm處) 經鞏膜穿刺三個25G (gauge) 或23G的切口 (sclerotomies)，並依序置入：

- ① 液體灌注管 (infusion cannula)：在眼外側下方，維持眼內壓力。
- ② 光源管 (endoillumination probe)：大多是主刀者非慣用手側，幫助眼內照明。
- ③ 玻璃體切除探頭 (vitrectomy cutter)：大多是主刀者慣用手側，用儀器的切吸功能施行玻璃體切除，例如移除玻璃體出血 (vitreous hemorrhage)。



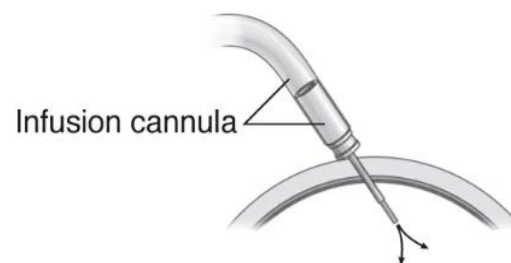
Right eye, Right-handed (Surgeon's view)



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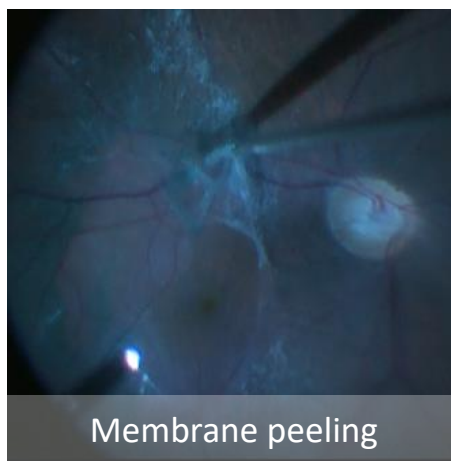
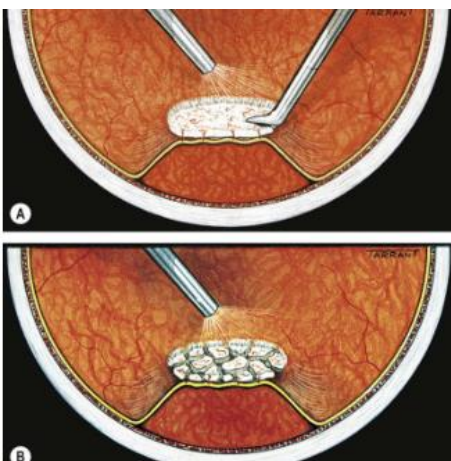
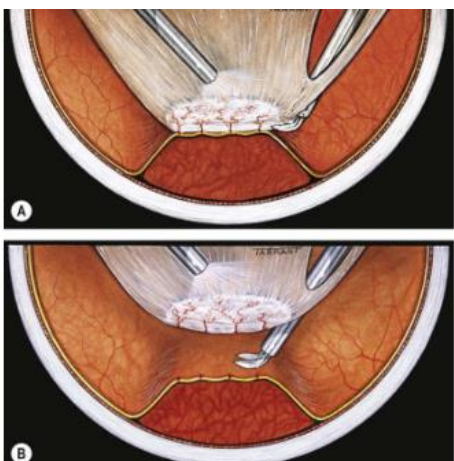
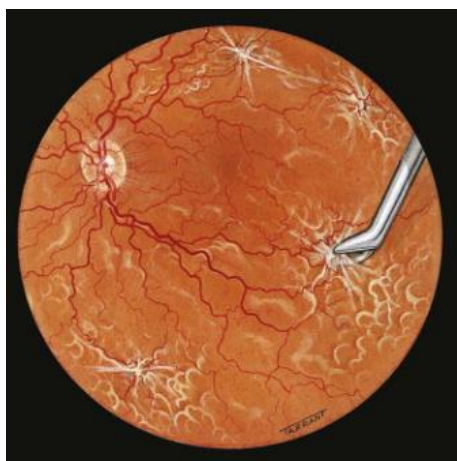
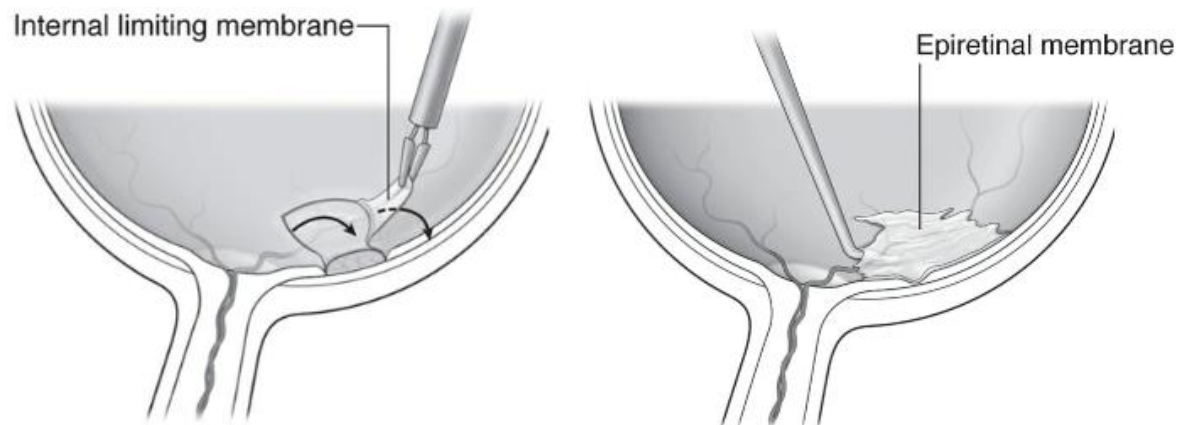


Abbreviation



膜剝除術 Membrane peeling

若是黃斑部裂孔 (macular hole) 或黃斑部皺褶 (macular pucker) 患者，施行玻璃體切除術後用眼內鑷或者眼內剪剝除內限膜 (internal limiting membrane) 或視網膜前膜 (epiretinal membrane)。若是增殖性糖尿病視網膜病變導致的牽扯性視網膜剝離 (tractional retinal detachment) 或者增殖性玻璃體視網膜病變 (proliferative vitreoretinopathy)，也可以用此方法切或剝除視網膜的增生膜。



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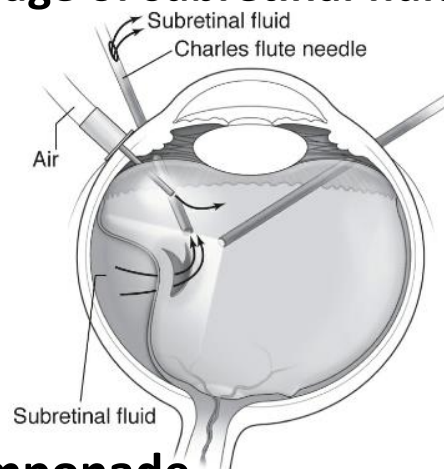


Abbreviation

3

視網膜造孔及視網膜下積液引流 Retinotomy and drainage of subretinal fluid

若是視網膜剝離 (retinal detachment) 患者，將玻璃體切除乾淨後，視情況由原裂孔處，或另在視網膜剝離的適當位置製造一個人工的視網膜裂孔 (retinotomy)，以便視網膜下積液可以被回沖針 (back flush needle) 吸出。



4

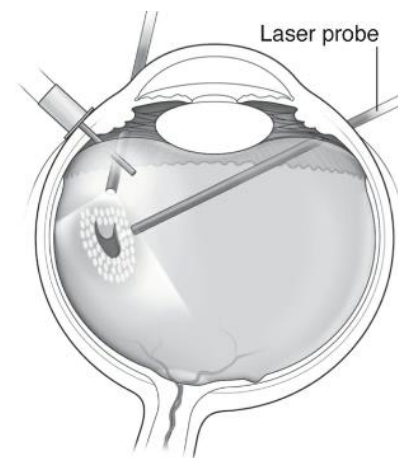
液氣交換術 Fluid gas exchange / 內部填塞 Internal tamponade

利用氣體 (C3F8或SF6) 或矽油來置換玻璃體液並填充玻璃體腔。利用其浮力及表面張力將裂孔封閉，使視網膜下積液可進一步排出幫助視網膜回貼。

5

眼內雷射 Endolaser

利用雷射使裂孔四周網膜與其下之組織黏合固定，防止玻璃體腔液體再灌入視網膜下腔。



Retinotomy



TOC

Silicone oil tamponade



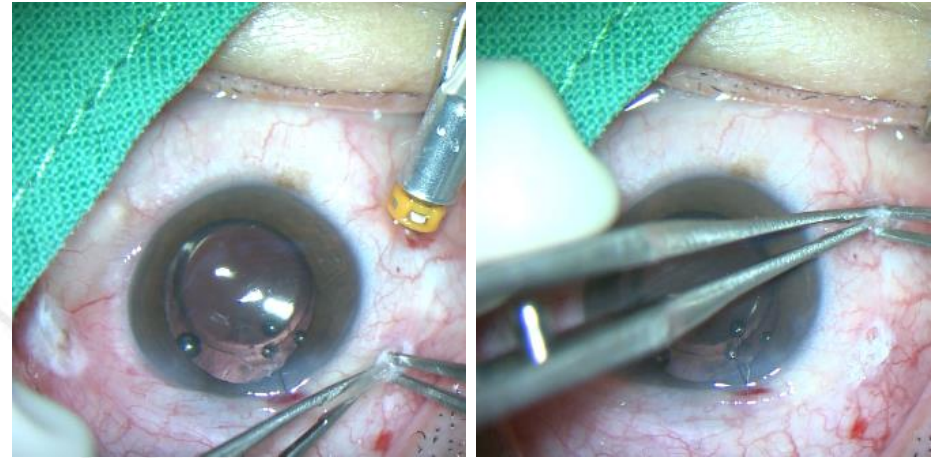
Endolaser



Abbreviation

完成 Completion

依序拔除相關器械組，首先移除光源管和玻璃體切除探頭，液體灌注管最後拔出。通常25G或23G的傷口以電燒密合傷口或棉棒加壓即可；特殊狀況下 (例如：移除玻璃體內矽油者)，可能需要使用Vicryl 7-0 做鞏膜或鞏結膜縫合，以避免傷口滲漏造成術後低眼壓。結束前也可在結膜下注射類固醇溶液抗發炎，或抗生素溶液預防感染。



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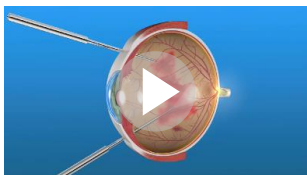


Abbreviation



Vitrectomy Surgery for Macular Holes
EyeSmart — American Academy of Ophthalmology

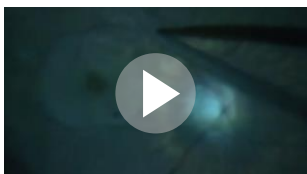
<https://www.youtube.com/watch?v=89VeAs9pMQM>



Vitrectomy Surgery for Diabetic Retinopathy

EyeSmart - American Academy of Ophthalmology

<https://www.youtube.com/watch?v=FeSb9ve9lwQ>



Pars plana vitrectomy and membrane peeling for macular hole

HJ Wu



Vitrectomy Surgery for Detached Retina
EyeSmart — American Academy of Ophthalmology

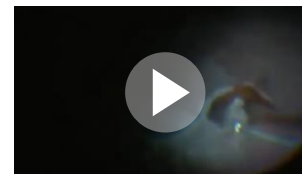
<https://www.youtube.com/watch?v=yDACxM-yM1w>



Vitrectomy in 60 seconds

Andres Rousselot

<https://www.youtube.com/watch?v=VGs7pGDnbqA>



Pars plana vitrectomy for retinal detachment

HJ Wu

斜視手術 Strabismus Surgery

- 眼外肌後徙 / 截短術 muscle recession & resection (MR&R)



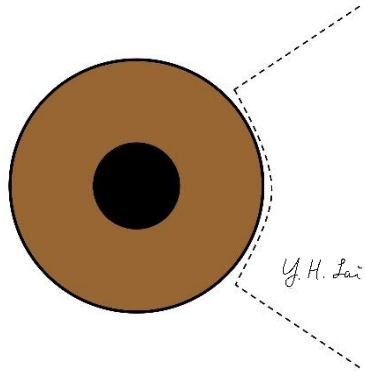
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Abbreviation

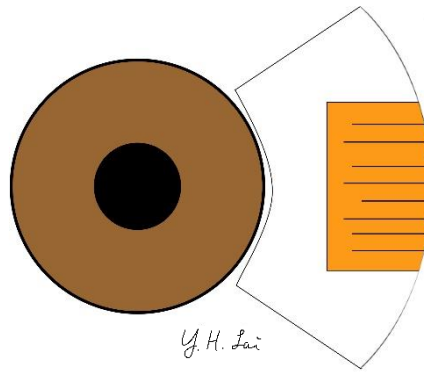
眼外肌後徙 / 截短術步驟 MR&R Surgery Steps : 眼外肌後徙 Recession

1



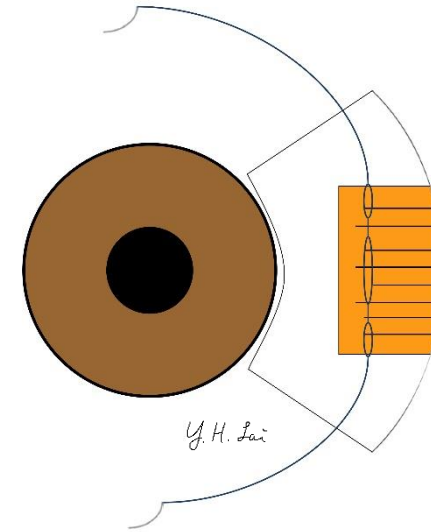
沿著虛線剪開結膜 (此為
limbal incision method)

2



小心剪開Tenon capsule以
顯露出眼外肌

3



以6-0 Vicryl在靠近肌肉附
著處 (insertion) 綁緊眼外
肌 (此法稱為Whiplock)

TOC



Exam



Surgery



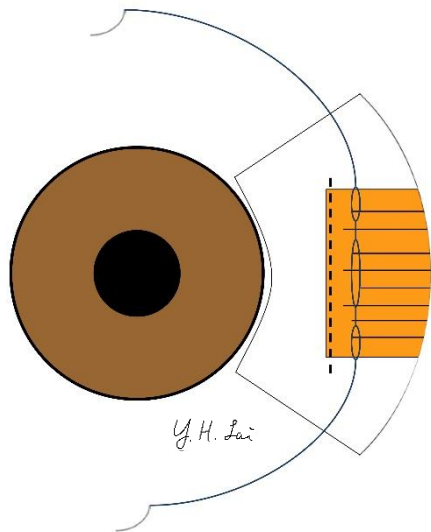
Emergency



Drug

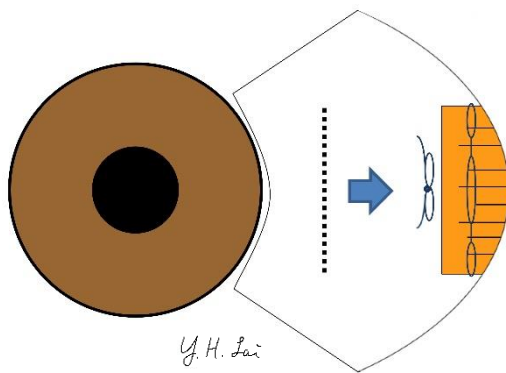
Abbreviation

4



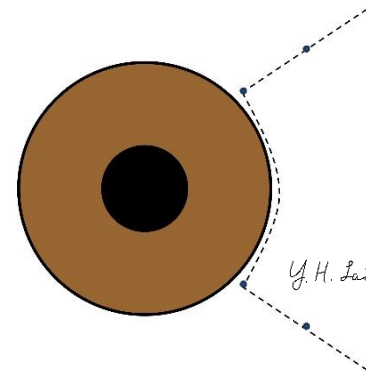
沿著虛線剪開眼外肌，使
眼外肌與鞏膜分離

5



將眼外肌原來的附著處往
後移，並且縫合在新的位
置，達到放鬆肌肉的效果

6



以8-0 Vicryl縫合結膜

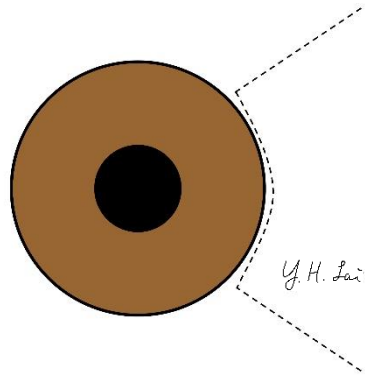
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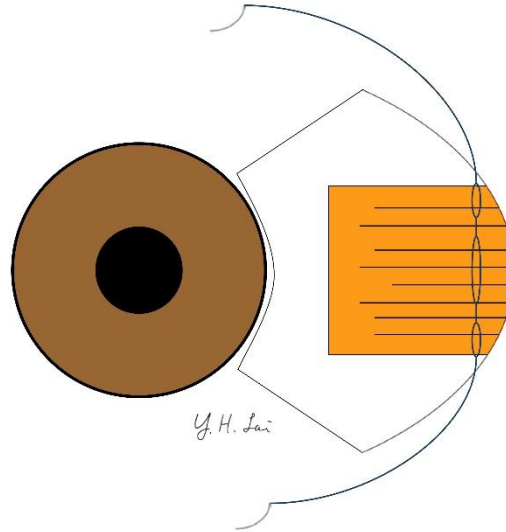
眼外肌後徙 / 截短術步驟 MR&R Surgery Steps : 眼外肌截短 Resection

1



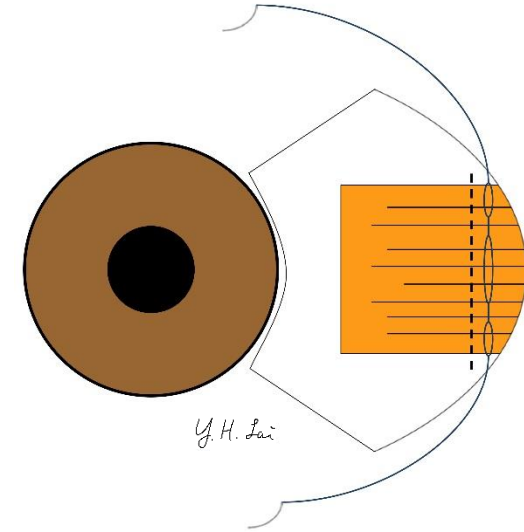
沿著虛線剪開結膜 (此為
limbal incision method)

2



以6-0 Vicryl在所欲截斷的
位置處綁緊眼外肌

3



沿著虛線剪開眼外肌，使
眼外肌與鞏膜分離

TOC



Exam



Surgery



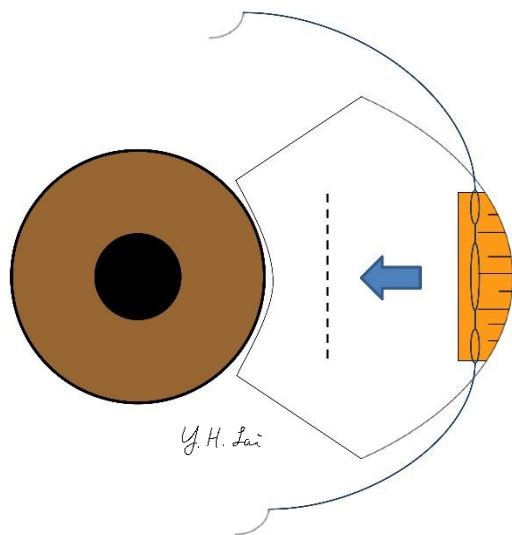
Emergency



Drug

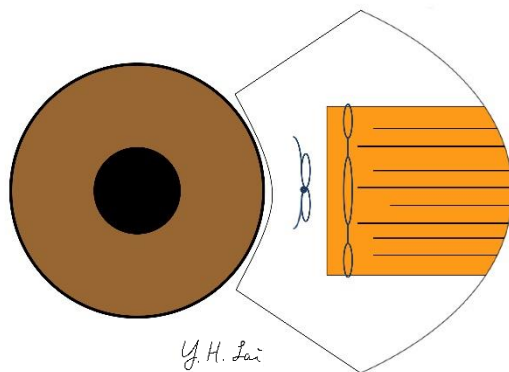
Abbreviation

4



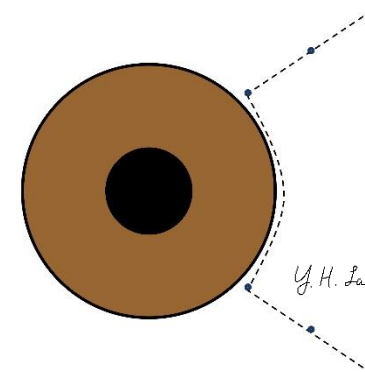
剪掉多餘的眼外肌(原來在
附著處端的殘餘眼外肌)，
並且將眼外肌往前拉

5



將眼外肌縫合在原來的附
著處

6



以8-0 Vicryl縫合結膜

Online Videos



Simulated Surgery: Strabismus Surgery: Resection Techniques
Cybersight

<https://www.youtube.com/watch?v=2VwCpkdpxWc>

TOC



Abbreviation

眼整形手術 Oculoplastic Surgery

- 眼瞼整形術 **blepharoplasty**

- ↳ 提上瞼肌筋膜復位術 levator aponeurosis advancement
- ↳ 眼瞼內翻 / 外翻矯正手術 entropion / ectropion correction

- 淚管手術 **lacrimal surgery**

- ↳ 淚囊鼻腔吻合術 dacryocystorhinostomy (DCR)
- ↳ 結膜淚囊鼻腔吻合術 conjunctivodacryocystorhinostomy (CDCR)

- 眼窩手術 **orbital surgery**

- ↳ 眼窩減壓術 orbital decompression
- ↳ 眼球內容物剝除術 evisceration
- ↳ 全眼球摘除術 enucleation

TOC

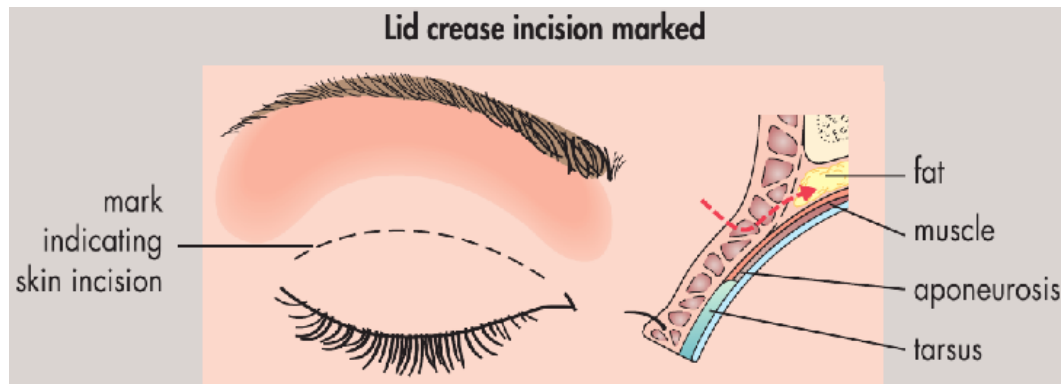


Abbreviation

提上瞼肌筋膜復位術步驟 Levator Aponeurosis Advancement Steps

1 標記 Marking

先在上眼瞼找出適合的皺摺處劃出標記線以呈現需要切除的皮膚範圍。



2 麻醉 Anesthesia

在標記處局部皮下注射麻醉劑 (local anesthesia)。



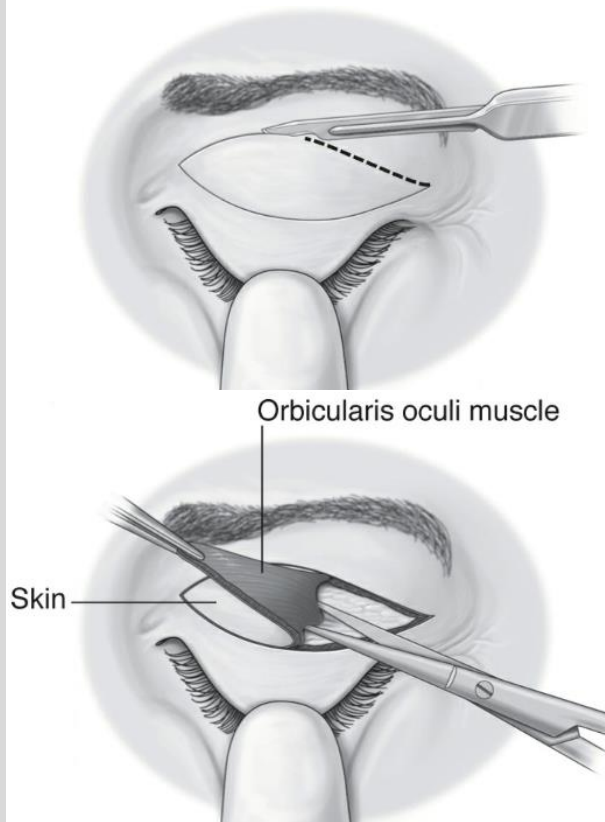
TOC



Abbreviation

切開 Incision

按照標記線利用15號刀片劃開，並用剪刀將皮膚剪下，露出下方的眼輪匝肌 (orbicularis muscle)，接著用剪刀將眼輪匝肌水平剪下，並用電燒 (electrocautery) 仔細止血 (hemostasis)。



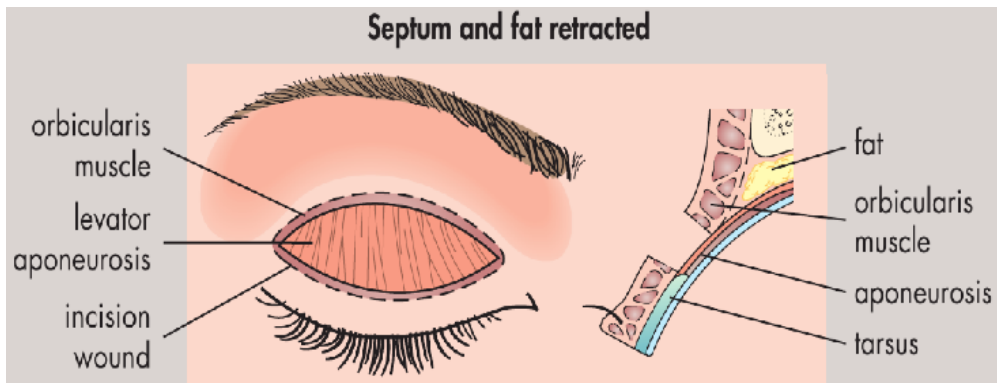
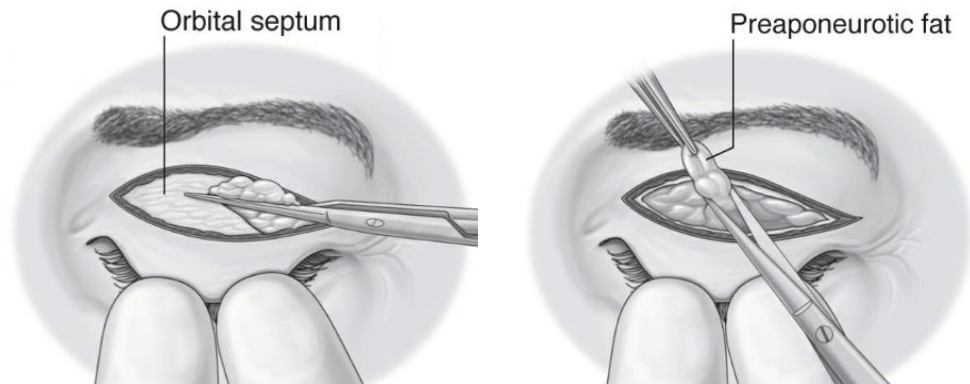
TOC



Abbreviation

確認提上瞼肌筋膜 Identification of the levator aponeurosis

眼輪匝肌剪下後會露出下方的眼眶膈 (orbital septum)，使用電燒止血後利用剪刀將眼眶膈從中間剪開露出下方的脂肪 (preaponeurotic fat)。視病人的情況裁減多餘的脂肪並使用電燒止血後，將剩餘的脂肪撥開就會露出下方白色的提上瞼肌筋膜 (levator aponeurosis)。



TOC



Exam



Surgery



Emergency

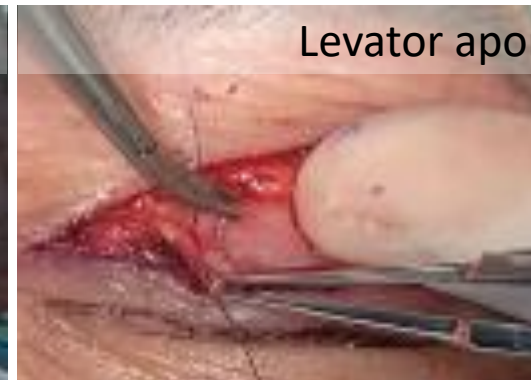
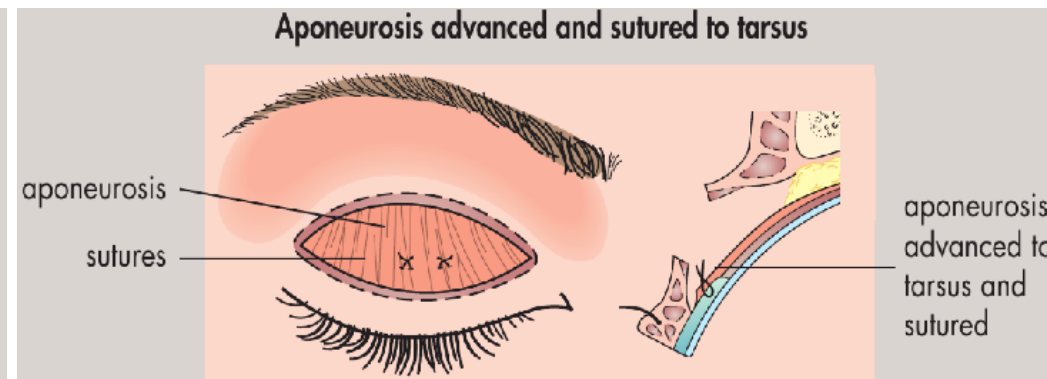
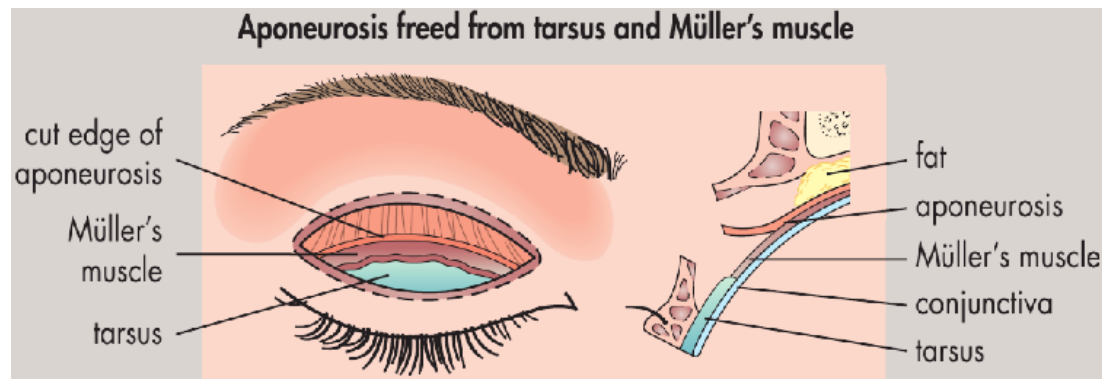


Drug

Abbreviation

提上瞼肌筋膜復位 Levator aponeurosis advancement

利用剪刀將提上瞼肌筋膜與眼瞼板 (tarsus) 分離之後，再使用縫線 (例如Vicryl 5-0雙頭針) 將提上瞼肌筋膜縮短縫回眼瞼板。



TOC



Abbreviation

6

術中調整 Intraoperative adjustments

讓病人睜開眼睛，觀察雙眼眼瞼高度以微調縫線位置，此程序可以反覆數次直到高度適中且對稱為止。

縫合 Closure

使用縫線 (例如Nylon 6-0) 將皮膚縫合後覆蓋抗生素眼藥膏 (ophthalmic antibiotic ointment)。術後宜局部冰敷 (ice packing) 以減少瘀血與腫脹。

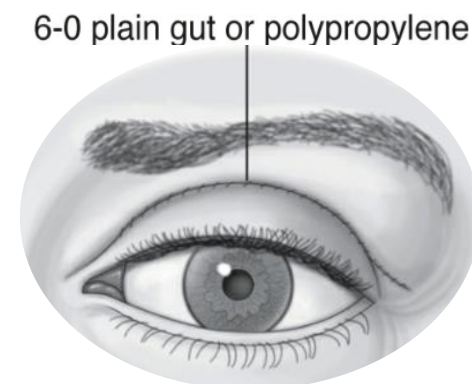
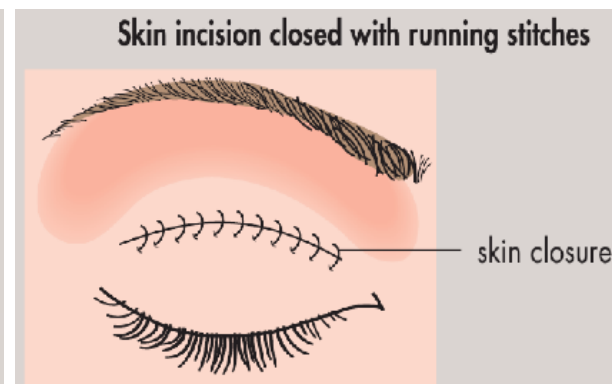
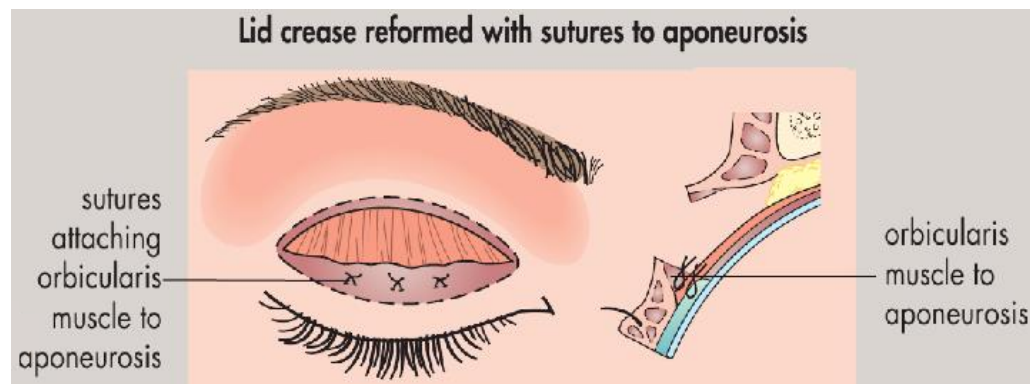


TOC



Abbreviation

7



Online Videos



Bloodless Ptosis Repair: Levator Resection Surgery

Dr Milind Naik, MD

<https://www.youtube.com/watch?v=CA52ZutRawM&feature=youtu.be>

角膜手術 Corneal Surgery

-  **全層角膜移植術 penetrating keratoplasty (PK, or PKP)** 
- **板層角膜移植術 lamellar keratoplasty (LK, or LKP)**
 - ↳ 角膜內皮細胞移植術 Descemet stripping automated endothelial keratoplasty (DSAEK)
 - ↳ 深層前角膜移植術 deep anterior lamellar keratoplasty (DALK)
- **眼翳刮除手術 pterygium excision, or bare sclera**
- **羊膜移植手術 amniotic membrane transplantation (AMT)**
- **屈光手術 refractive surgery**
 - ↳ 放射狀角膜切開術 radial keratotomy (RK)
 - ↳ 雷射屈光角膜切除術 photorefractive keratectomy (PRK)
 - ↳ 雷射層狀角膜塑型術 laser assisted in situ keratomileusis (LASIK)

TOC



Abbreviation

全層角膜移植術步驟 Penetrating Keratoplasty Surgery Steps

1

麻醉 Anesthesia

可以採全身麻醉 (general anesthesia) 或者球後麻醉 (retrobulbar anesthesia)。若採取球後麻醉，須注意避免眼球壓力上升，可以在注射麻醉藥後從外部施加壓迫 (external compression)。

2

事前準備 Preparation

檢視捐贈角膜 (donor cornea) 的基本資料包括保存期限、狀態及是否已送達刀房；各種移植手術所需器械及角膜環刀 (trephine、punch) 尺寸是否齊全。

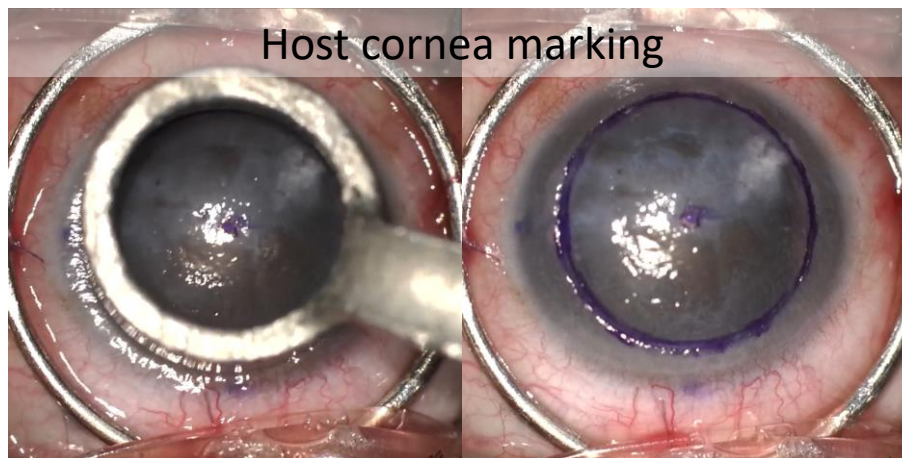
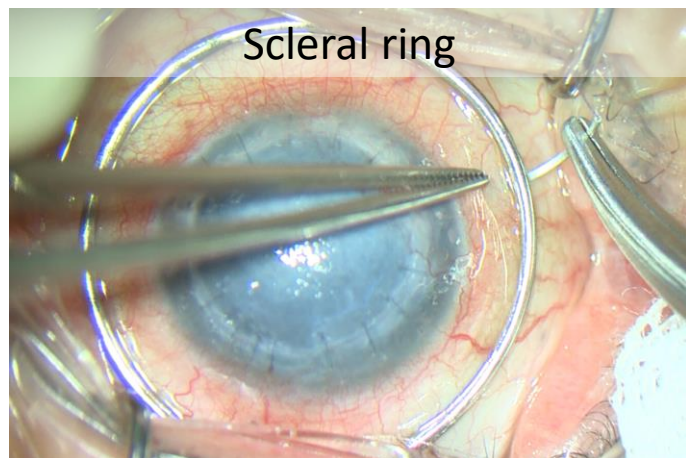
TOC



Abbreviation

量測及定位 Sizing and centration

照一般程序消毒並鋪單後使用開眼器 (lid speculum) 將眼皮撐開。可以選擇性使用Vicryl 7-0將鞏膜支撐環 (scleral support ring, Flieringa ring) 縫在上鞏膜 (episclera) 以加強眼球支撐。用尺規 (caliper) 量測受贈角膜 (host cornea) 大小，以決定適當移植角膜 (graft cornea) 尺寸；並用標記筆 (marking pen) 或者角膜標記器 (corneal marker) 先做好中心定位記號。常見的移植角膜大小約為7.5mm左右，但仍需視狀況 (受贈角膜大小、病變位置) 調整。



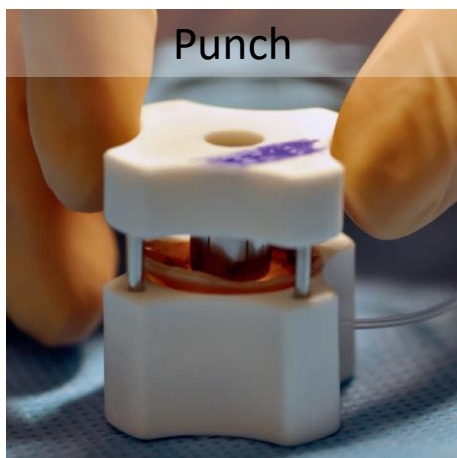
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Abbreviation

準備移植角膜 Graft preparation

將捐贈角膜自保存液中取出，依照剛才選好的尺寸，利用disposable punch block apparatus (拋棄式角膜打洞器) 以內皮細胞面朝上的方向放置，切下適當大小的中央角膜。切下的graft尺寸通常需比host大0.25 - 0.5mm左右。切好之後可以用少許OVD或者角膜保存液覆蓋在內皮細胞側以保護之。



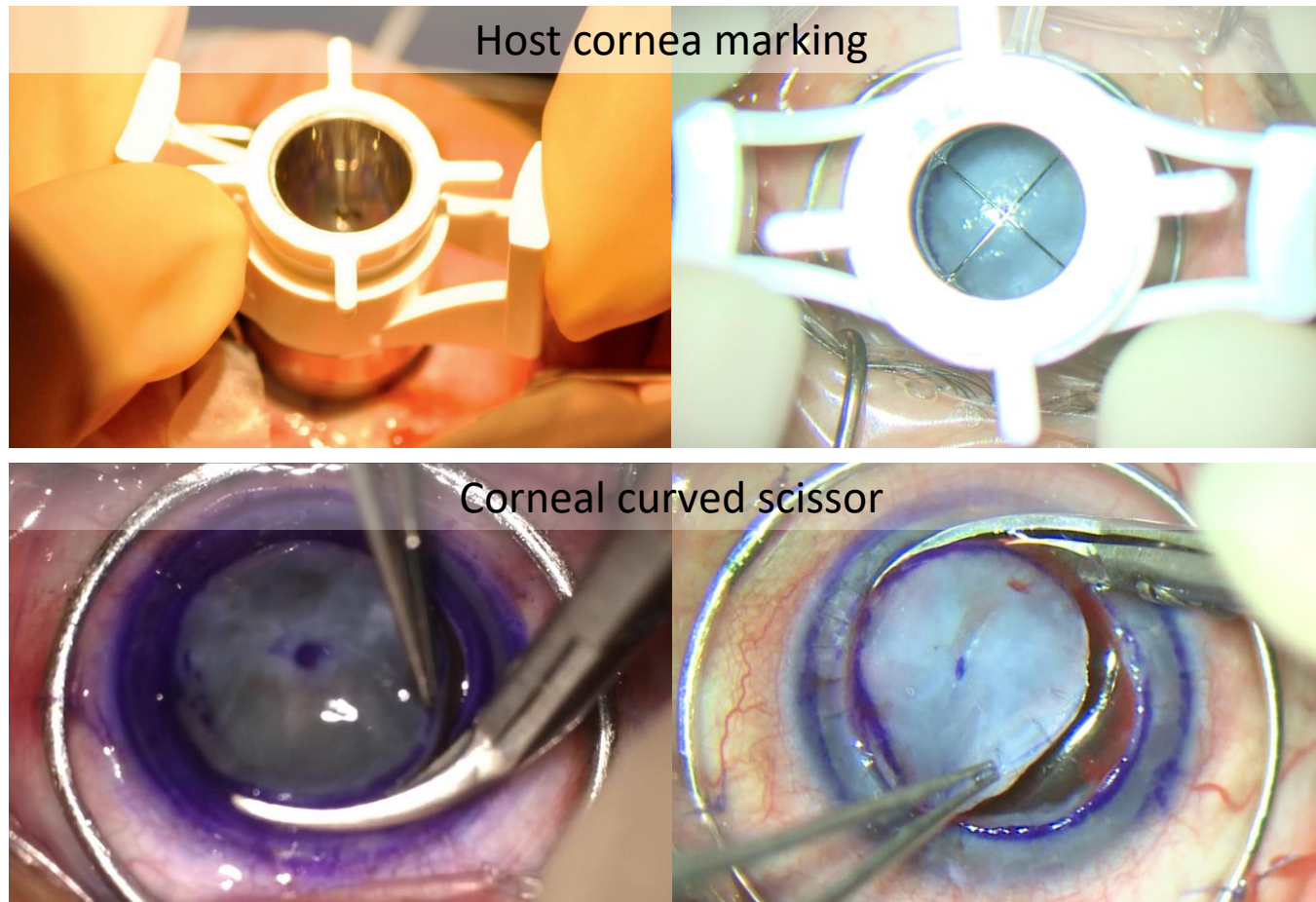
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Abbreviation

受贈角膜環切Host trephination

使用角膜環刀 (corneal trephine, manual or disposable vacuum trephine) 將host cornea 的中央病變位置切除，須注意對齊先前作的標記，以及抓準中心部位避免偏斜。切到深層角膜之時需移開環刀，改用角膜彎剪 (corneal curved scissor) 將整個病變角膜360度垂直切下，以避免傷到虹膜或者水晶體。



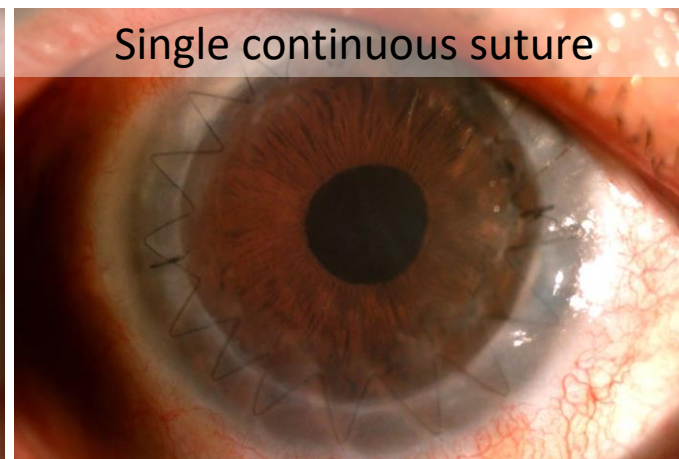
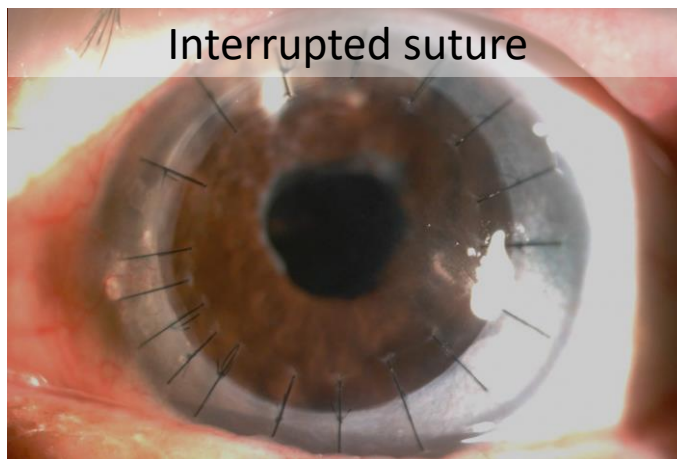
TOC



Abbreviation

縫合Suturing

將縮瞳劑Miostat滴在虹膜表面並以OVD適度填充之後，將備好的graft內皮細胞側朝下放置在host上，並用Nylon 10-0依序在12點、6點、3點、9點鐘方位以縫線固定。此4處主要縫線 (cardinal suture) 完成之後，可以採取多種不同方式將graft進一步縫合，包括 all interrupted sutures, single or double continuous suture, single continuous with interrupted sutures等。



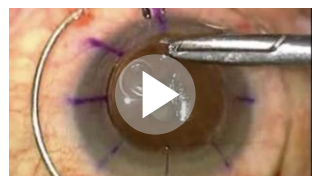
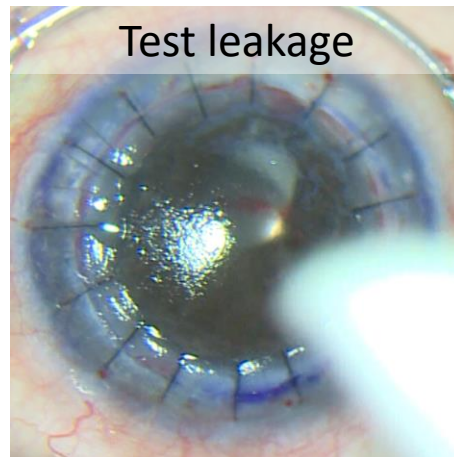
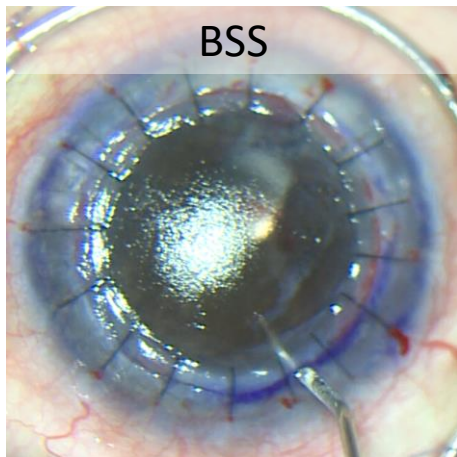
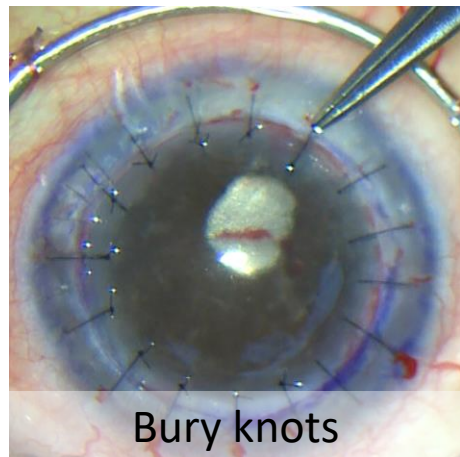
TOC



Abbreviation

完成Completion

可以使用沖洗針頭將OVD移除並注入BSS置換，檢查角膜縫合處是否漏水，若需要可以適度增加縫針，並將線頭埋入 (bury knots) 勿使外露。手術完成後覆上眼蓋 (protective eye shield)



Penetrating Keratoplasty Technique

Irving M. Raber

<https://eyetube.net/videos/penetrating-keratoplasty-technique>

眼科急症

TOC



Exam



Surgery



Emergency



Drug

Abbreviation



眼科急症

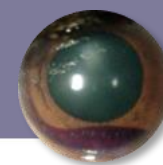
急診患者眼部檢查
Eye Examination in the
Emergency Room



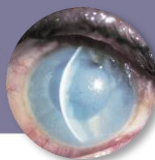
眼球撕裂傷或眼球破裂
Laceration and Rupture
of the Globe



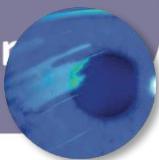
眼球鈍傷
Blunt Trauma



化學性灼傷
Chemical Burns



角膜擦傷
Corneal Abrasion



中央視網膜動脈阻塞
Central Retinal Artery
Occlusion



急性隅角閉鎖型青光眼
Acute Angle Closure
Glaucoma



急性結膜炎
Acute Conjunctivitis



近期突發性視力減退
Decreased Vision of
Recent Onset



TOC



Abbreviation

急診患者眼部檢查 Eye Examination in the Emergency Room

基本上急診的病人仍應該要接受完整的眼部檢查，除非情況特殊無法進行。如果病人狀況允許檢查設備也充足，建議依序進行下列檢查：

1. **History taking**：詳盡尤佳，但chemical burn、acute angle closure glaucoma、central retinal artery occlusion等應盡速進行治療的急症除外。
2. **Visual acuity (with near card)**：使用pinhole或+2.5D的鏡片來修正refractive error或老花眼。另外很重要的是記得要詢問病人過去的視力如何，以排除弱視的病史。
3. **External examination**：看看有無hematoma, laceration, foreign body, fracture。
4. **Motility**：觀察眼球運動是否順暢，必要時可使用forced duction test。
5. **Pupil**：觀察病人的瞳孔反射非常重要，在尚未確定有無眼神經或瞳孔病變時，不要隨便散瞳。
6. **Visual field**：使用confrontation test。
7. **Anterior segment**：若病人無法坐到細隙燈前，請使用portable slit-lamp來檢查。
8. **Tension**：在確定無眼球破裂之後檢查。
9. **Fundus**：用direct or indirect ophthalmoscope來檢查。

以下介紹常見的眼科急症：

TOC



Abbreviation

眼球撕裂傷或眼球破裂 Laceration and Rupture of the Globe

- 臨床表現包括decreased vision, soft globe, hyphema, vitreous hemorrhage, excessive chemosis, unusually deep or flat anterior chamber, uvea prolapse。
- 在檢查的時候必須小心輕柔，以免進一步擠壓眼球造成組織損傷，在檢查完畢後記得幫病人蓋上保護蓋 (eye shield)。
- 安排眼部CT攝影以排除眼內異物 (IOFB) 的可能。
- 對於需要開刀縫合的患者，請病人禁食並記錄最後進食時間，聯絡開刀事宜，並給予 prophylactic IV antibiotics。
- 最重要的是當你不確定是否有眼球破裂，記得explore it surgically best under GA。

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Abbreviation

眼球鈍傷 Blunt Trauma

重點在於 (1) 有無globe rupture，(2) 有無hyphema，(3) 有無intraocular foreign body。

處理原則如下：

- A set of X-ray：包括 Waters view, skull AP and lateral view。
- Orbital CT including coronal view。
- Check pupil：看看有無traumatic mydriasis or miosis, Marcus Gunn Pupil。
- Slit lamp exam：corneal laceration, hyphema, cells, iridodialysis, lens subluxation, vitreous cells。
- Fundus exam：commotio retinae, retinal hemorrhage, choroidal rupture, vitreous hemorrhage, retinal detachment。
- 對於uncomplicated case的處理主要針對消腫、止痛，並請病人回診追蹤。
- 對於有骨折的case，通常都是等消腫後(約一個禮拜)再仔細評估是否需要手術矯正。
- 有hyphema的病人，可以視嚴重度請病人住院或門診治療，而不論住院與否都必須臥床休息且密切追蹤眼壓。可以給予病人類固醇藥水及散瞳劑，若有眼壓高的現象，則視情況需要給予口服或局部的降眼壓藥物。

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Abbreviation

化學性灼傷 Chemical Burns

- 診斷眼部化學灼傷主要依據病史history，而非signs and symptoms。
- 主要症狀包括：紅眼、疼痛、畏光、視力不等程度的減退。
- 要注意的是鹼性溶液灼傷 (alkaline burns) 比酸性溶液灼傷 (acid burns) 更嚴重，因為鹼性溶液會穿透組織，破壞角膜上皮細胞、基質、內皮細胞，甚至在灼傷發生後持續穿透至前房。
- 最重要的處理原則是立刻以大量清水沖洗患眼 (prompt, copious fluid irrigation of the affected eye)，沖洗一個階段後反覆以石蕊試紙檢測，直到pH值恢復接近中性 (6-8) 為止。
- 沖洗的時候必須檢查眼瞼下有無化學物質殘留 (常見如水泥粉) 以徹底清除，若有結膜或角膜的組織壞死，可以用沾濕的無菌棉棒輕輕去除。
- 治療藥物包括：topical steroids (if significant corneal or AC inflammation) , topical antibiotic ointment, topical cycloplegics, preservative-free artificial tears, antiglaucoma medications if elevated IOP, oral analgesics。

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Abbreviation

角膜擦傷 Corneal Abrasion

病人會抱怨有異物感、流淚、畏光、眼皮水腫，處理原則如下：

- 可以先給予病人局部麻醉劑緩解不適感以利於檢查。
- 可以使用螢光染色輔助在slit-lamp下檢查。
- 如果有線狀的刮痕，記得要看看有無異物卡在tarsal conjunctiva。
- 治療可以給予短效型睫狀肌麻痺藥水 (以緩解 ciliary spasm)、預防性抗生素藥水或藥膏，並緊密包紮患眼 (compress patching)。

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Abbreviation

中央視網膜動脈阻塞 Central Retinal Artery Occlusion

- 病人通常自述突然視力模糊，常常降至count fingers or hand motion level。
- 患眼瞳孔反射呈現Marcus Gunn pupil。
- 病人眼底可見pale nerve head, severe attenuated arteries with / without segmentation of blood columns, retinal diffuse edema, fovea showing cherry-red spot。
- 需要立即治療，讓病人吸carbogen (5%的CO₂及95%的O₂)，降眼壓 (使用diamox or mannitol)。
- 如果懷疑是因為vasculitis所引起，可考慮使用systemic steroids。
- 通常需要在發病兩小時以內開始治療，否則預後很不好。
- 住院後應幫病人安排EKG, cardiac echo, doppler of carotid artery及FAG等檢查。

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Abbreviation

急性隅角閉鎖型青光眼 Acute Angle Closure Glaucoma

- 病人常會抱怨頭痛或眼痛，單或雙側視力減退，畏光、光暈 (colored haloes around lights)，有時伴隨噁心 / 嘔吐。
- 眼科檢查會觀察到眼壓明顯升高，眼紅 (尤其是輪狀部)，瞳孔中度擴張並喪失光反射 (fixed, mid-dilated pupil)，角膜水腫，前房狹窄。
- 必須儘快幫病人降眼壓：靜脈注射mannitol，口服CAI (diamox)，局部給予縮瞳劑 (2% pilocarpine, q15mins x 2, then qid) 及減少房水生成的降眼壓藥物 (如beta-blocker, alpha-agonist, CAI等)，同時對側眼睛也應給予縮瞳劑預防急性發作，待眼壓降低後，適時施行laser iridotomy。

TOC



Abbreviation

急性結膜炎 Acute Conjunctivitis

- 主要是由濾過性病毒所引起，如腺病毒、腸病毒、以及科沙奇病毒，少數是由非濾過性病毒所造成，如披衣菌，流行的途徑主要是公共場所或家庭中接觸傳染。
- 主要症狀包括：conjunctival injection, tearing, serous discharge, edematous eyelids, subconjunctival hemorrhages, pseudomembrane formation and palpable preauricular lymph nodes.
- 需鑑別診斷的疾病：uveitis, corneal abrasion, acute glaucoma, scleritis, episcleritis, blepharitis, corneal ulcer等等。
- 處理方式：cold compresses, artificial tears, topical vasoconstrictors, preventive topical antibiotics and steroids。如果表面有膜 (membrane)，可使用沾濕的無菌棉棒或鑷子輕輕剝除。

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Abbreviation

近期突發性視力減退 Decreased Vision of Recent Onset

對於突然發生視力模糊的鑑別診斷有下列幾種：

Table. Differential Diagnosis for Decreased Vision of Recent Onset

Refracting surface abnormality Cornea: edema, laceration, ulcer Lens: cataract, swelling, subluxation	Optic nerve abnormality Avulsion Optic neuritis Contusion or laceration
Media abnormality Tear film: conjunctivitis, tearing Aqueous: uveitis, hyphema, hypopyon Vitreous: uveitis, vitreous hemorrhage	CNS abnormality Cortical blindness Optic chiasm / tract / radiation insult
Retina abnormality Edema: traumatic or inflammatory Hemorrhage: traumatic, inflammatory, or secondary to intracranial phenomenon Central retinal artery occlusion Detachment: traumatic or nontraumatic	Functional Hysterical Malingering

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Abbreviation

眼科常用藥物

TOC



Exam



Surgery



Emergency



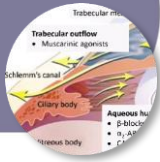
Drug

Abbreviation

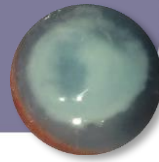


眼科常用藥物

Drugs for
Glaucoma



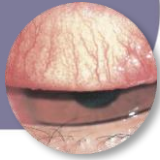
Antimicrobial
Drugs



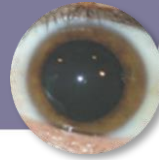
Anti-inflammatory
Drugs



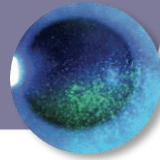
Antiallergic
Drugs



Mydriatics and
Cycloplegics



Drugs for
Dry Eye



Drugs for Intravitreal
Injection (IVI)



Miscellaneous



TOC



Exam



Surgery



Emergency



Drug

Abbreviation

Drugs for Glaucoma

Alphagan P	0.15 %	Brimonidine	5 ml/bt	α_2 -agonist	BID / TID
Timoptol-XE	0.5 %	Timolol	2.5 ml/bt	β -blocker	QD
Azopt	1 %	Brinzolamide	5 ml/bt	CAI	BID / TID
Xalatan	0.005 %	Latanoprost	2.5 ml/bt		
Travatan	0.004 %	Travoprost	2.5 ml/bt	PGF _{2α} analogue	
Taflotan-S	0.0015 %	Tafluprost	0.3 ml/UD		HS
Vyzulta	0.024 %	Latanoprostene bunod	5 ml/bt	NO-donating PGF _{2α} analogue	
Eybelis	0.002 %	Omidenepag	2.5 ml/bt	PGE ₂ agonist	
Isoptocarpine	2 %	Pilocarpine	15 ml/bt	Cholinergic agonist	QID*

* q15mins x 3 then qid in acute angle closure attack caused by phakic pupillary block
 CAI = carbonic anhydrase inhibitor; NO = nitric oxide; PG = prostaglandin; UD = unit dose

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Abbreviation

Combigan	0.5 % Timolol 0.2 % Brimonidine	5 ml/bt	β -blocker α_2 -agonist	BID
Azarga	0.5 % Timolol 1 % Brinzolamide	5 ml/bt	β -blocker CAI	
Cosopt	0.5 % Timolol 2 % Dorzolamide	5 ml/bt		
Simbrinza	0.2 % Brimonidine 1 % Brinzolamide	5 ml/bt	α_2 -agonist CAI	BID / TID
Duotrav	0.5 % Timolol 0.004 % Travoprost	2.5 ml/bt	β -blocker PGF _{2α} analogue	HS

CAI = carbonic anhydrase inhibitor; PG = prostaglandin

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Abbreviation

Side Effects

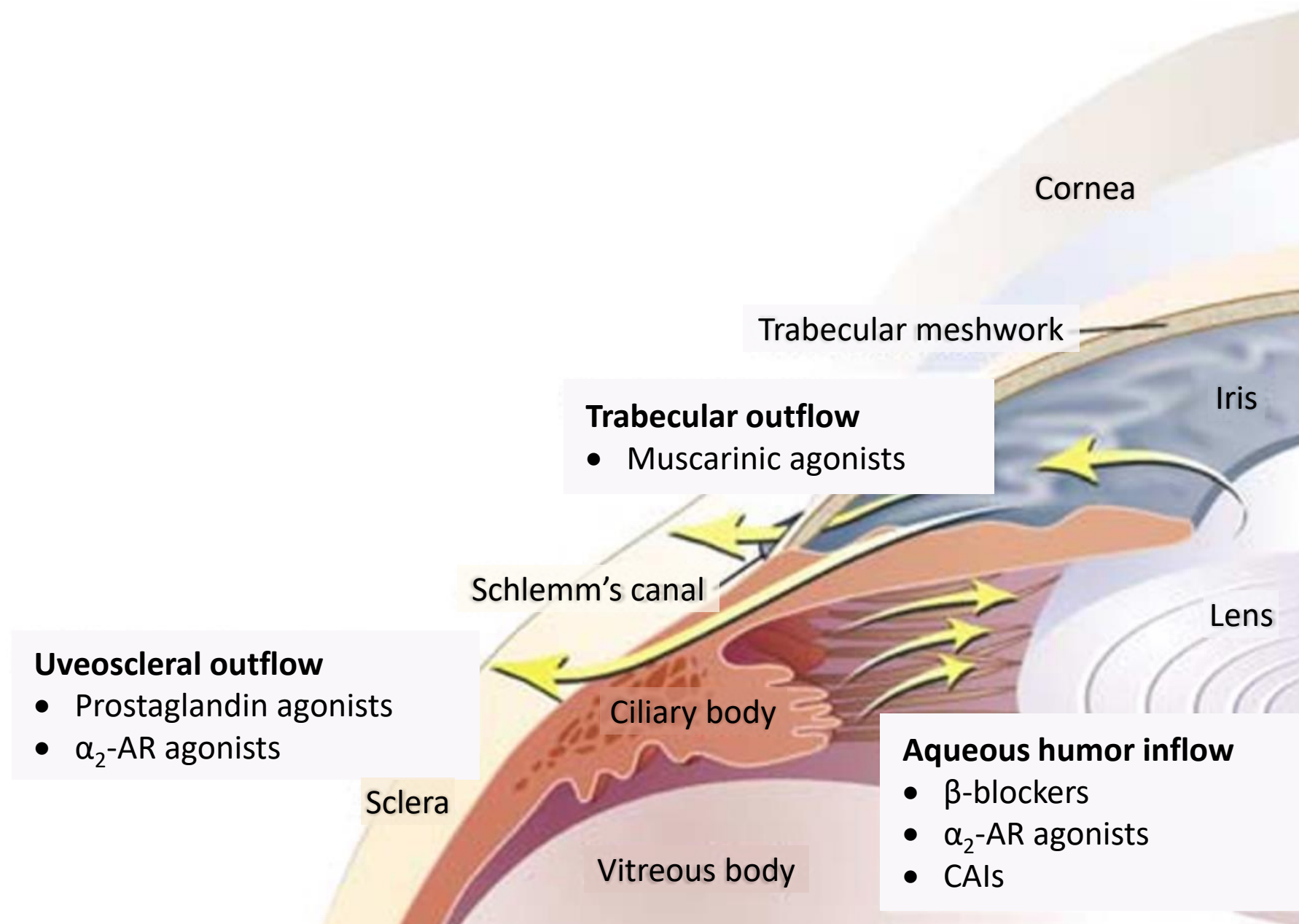
α_2-agonist	allergic conjunctivitis, fatigue, dry mouth
β-blocker	punctate keratitis, bronchospasm, bradycardia, heart block, hypotension, syncope, lethargy, depression, decreased libido, dyslipidemia
CAI	stinging, a transient bitter taste following administration, allergic blepharoconjunctivitis, may precipitate corneal decompensation in patients with corneal endothelial dysfunction
PG analogue	conjunctival hyperemia, iris hyperpigmentation, hyperpigmentation of periocular skin, eyelash lengthening and thickening, periocular atrophy, anterior uveitis, cystoid macular edema
Cholinergic agonist	miosis, brow ache, myopic shift, exacerbation of the symptoms of cataract

CAI = carbonic anhydrase inhibitor; PG = prostaglandin

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Abbreviation



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Abbreviation

TRABECULAR MESHWORK

(70%-80% of outflow)

- Adenosine A1 agonist (trabodendoson)
- Inhibitor of Rho kinase and norepinephrine (Rhopressa, Roclatan)
- M3 agonist (pilocarpine)

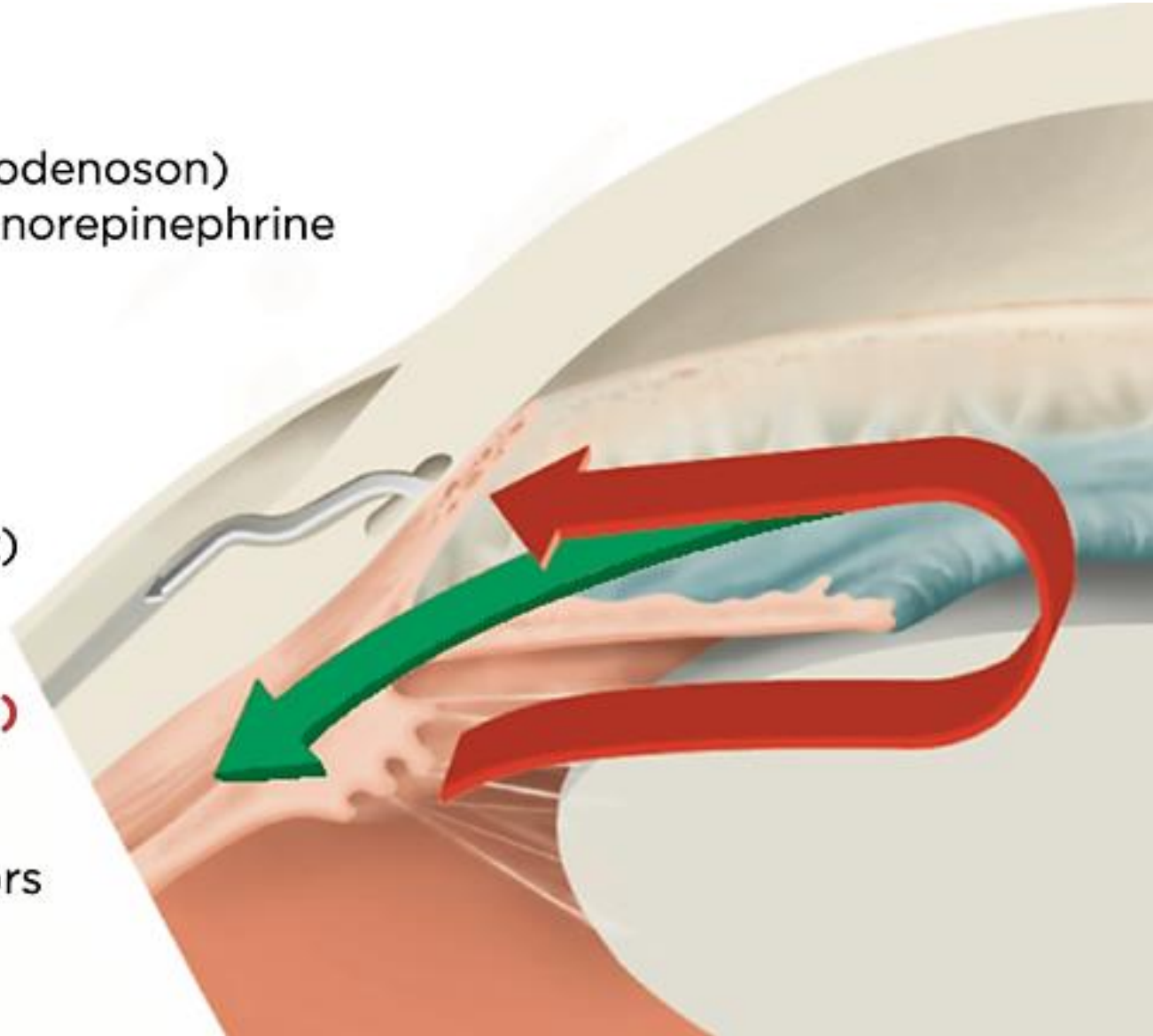
UVEOSCLERAL PATHWAY

(20%-30% of outflow)

- Prostaglandins (latanoprost)
- α -2 agonists (brimonidine)

AQUEOUS HUMOR (INFLOW)

- β -blockers (timolol)
- α -2 agonists (brimonidine)
- Carbonic anhydrase inhibitors
- Rhopressa, Roclatan
- Trabodendoson



TOC



Abbreviation

Antimicrobial Drugs

Chloramphenicol	0.25 %	Chloramphenicol	5 ml/bt	Chloramphenicol	
Sinomin	4 %	Sulfamethoxazole	15 ml/bt	Sulfonamide	
Gentax	0.3 %	Gentamicin	0.5ml/UD		
Tobrex sol'n	0.3 %	Tobramycin	5 ml/bt	Aminoglycoside	Frequency based on clinical condition
Tobrex oint	0.3 %	Tobramycin	3.5 g/tube		
Cravit	0.5 %	Levofloxacin	5 ml/bt	Fluoroquinolone	
Erythromycin	0.5 %	Erythromycin	3.5 g/tube	Macrolide	
Natacyn	5 %	Natamycin	15 ml/bt	Antifungal agent	
Devirus oint	3 %	Acyclovir	4.5 g/tube	Antiviral agent	5x/day

UD = unit dose

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Abbreviation

Antibacterial Spectra	Gram positive										Gram negative											
	E. faecalis (S) ¹	E. faecalis (VRE) ²	S. aureus (MSSA) ³	S. aureus (MRSA) ⁴	Staph coag-neg (S) ⁵	Staph coag-neg (R) ⁶	S. epidermidis (S) ⁵	S. epidermidis (R) ⁶	Strep. pyogenes	Strep. pneumoniae	Anaerobe	P. acnes	B. fragilis	E. coli	K. pneumoniae	Salmonella sp.	H. influenzae	M. catarrhalis	N. gonorrhoeae	P. aeruginosa	C. trachomatis	M. pneumoniae
Pip-Tazo																						
Cefazolin																						
Ceftazidime																						
Amikacin																						
Gentamicin																						
Tobramycin																						
Levofloxacin																						
Vancomycin																						
Chloramphenicol																						
Erythromycin																						

++	Recommended	Agent is a first line therapy : reliably active in vitro, clinically effective, guideline recommended, recommended as a first-line agent or acceptable alternative agent in the Sanford Guide.
+	Active	Agent is a potential alternative agent (active in vitro, possesses class activity comparable to known effective agents or a therapeutically interchangeable agents and hence likely to be clinically effective, but second line due to overly broad spectrum, toxicity, limited clinical experience, or paucity of direct evidence of effectiveness).
±	Variable	Variable activity such that the agent, although clinically effective in some settings or types of infections is not reliably effective in others, or should be used in combination with another agent, and/or its efficacy is limited by resistance which has been associated with treatment failure.
0	Not Recommended	Agent is a poor alternative to other agents because resistance to likely to be present or occur, due to poor drug penetration to site of infection or an unfavorable toxicity profile, or there is insufficient clinical data to support effectiveness.

NOTE: The data provided are intended to serve as a **general guide** to antibacterial usefulness based on treatment guidelines and recommendations, in vitro activity, predominant patterns of susceptibility or resistance and/or demonstrated clinical effectiveness. **Variability in resistance patterns due to regional differences or as a consequent of clinical setting (e.g., community-onset vs. ICU-acquired infection) should be taken into account** when using this table because activities of certain agents can differ significantly from what is shown in the table, which are by necessity based on aggregate information.

TOC



Abbreviation

Anti-inflammatory Drugs

0.02% Flucason	0.02 %	Fluorometholone	5 ml/bt		
0.1% Flucason	0.1 %	Fluorometholone	5 ml/bt		
Fusone	0.1 %	Betamethasone	0.5ml/UD	Corticosteroid	Frequency based on clinical condition
Econopred plus	1 %	Prednisolone	5 ml/bt		
Dexagel	0.1 %	Dexamethasone	5 g/tube		
Tobradex	0.3 %	Tobramycin	5 ml/bt	Aminoglycoside	
	0.1 %	Dexamethasone		Corticosteroid	
Acular	0.5 %	Ketorolac	5 ml/bt	NSAID	QID

NSAID = non-steroidal anti-inflammatory drug; UD = unit dose

TOC



Abbreviation



Table. Anti-Inflammatory Efficacy in the Cornea of Topically Administered Corticosteroids*

Generic Name	Trade Name	Anti-Inflammatory Effect
0.1% Dexamethasone acetate	N/A	55%
1% Prednisolone acetate	Econopred Plus, Pred Forte	51%
0.1% Dexamethasone alcohol	Maxidex	40%
0.1% Fluorometholone alcohol	FML Liquifilm	31%
1% Prednisolone sodium phosphate	Inflamase Forte	28%
0.1% Dexamethasone sodium phosphate	Decadron Phosphate	19%

N/A = not available

*Corneal epithelium intact.

Table 16-15 Comparison of Anti-inflammatory^a and IOP-Elevating^b Potencies

Glucocorticoid	Relative Potency	Rise in IOP, mm Hg
Dexamethasone, 0.1% (equivalent to betamethasone, 0.1%, less than or equivalent to difluprednate, 0.05%)	24.0	22
Fluorometholone, 0.1% ^c	21.0	6
Prednisolone acetate, 1%	2.3	10
Medrysone, 1% ^d	1.7	1
Tetrahydrotriamcinolone, 0.25%	1.4	2
Hydrocortisone, 0.5%	1.0	3

^a **Anti-inflammatory potency** determined by **in vitro assay** of inhibition of lymphocyte transformation. Anti-inflammatory potency of difluprednate, 0.05%, is equal to or stronger than betamethasone, 0.1%, which has a 6-fold anti-inflammatory potency compared with that of prednisolone or equivalent to that of dexamethasone. In clinical trials on uveitis, a significant increase in IOP occurred in 6% of patients treated with difluprednate, 0.05%, emulsion compared with 5% of those treated with prednisolone acetate, 1%.

^b **IOP effects** determined in **topical dexamethasone responders** .

^c Rapid metabolism of fluorometholone in the eye compared with dexamethasone.

^d Relatively poor ocular penetration.

TOC**Abbreviation**

Antiallergic Drugs

Zaditen

0.025 %

Ketotifen

5 ml/bt

H1-antagonist,
Mast cell stabilizer

BID

TOC



Abbreviation

Mydriatics and Cycloplegics

Antol	1 %		5 ml/bt		
Tropine	0.125 %	Atropine	5 ml/bt	Anticholinergic	HS / BID / TID
0.01% Atropine	0.01 %		0.5 ml/UD		
1% Mydriacyl	1 %	Tropicamide	5 ml/bt		
Mydrin-P	0.5 %	Tropicamide	10 ml/bt	Anticholinergic	q10mins x 3
	0.5 %	Phenylephrine		Sympathomimetic	

UD = unit dose

TOC



Abbreviation

Table 16-5 Mydriatics and Cycloplegics

Generic Name	Trade Name	Strengths	Onset of Action	Duration of Action
Phenylephrine HCl	AK-Dilate	Solution, 2.5%, 10%	30–60 min	3–5 h
	Altafrin	Solution, 2.5%, 10%		
	Mydfrin	Solution, 2.5%		
	Neofrin	Solution, 2.5%		
	Neo-Synephrine	Solution, 2.5%		
	Available generically	Solution, 2.5%, 10%		
Hydroxyamphetamine hydrobromide, 1%		Available as powder for compounding	30–60 min	3–5 h
Atropine sulfate	Atropine-Care	Solution, 1%	45–120 min	7–14 d
	Isopto Atropine	Solution, 1%		
	Available generically	Solution, 1%		
		Ointment, 1%		
Cyclopentolate HCl	AK-Pentolate	Solution, 1%	30–60 min	1–2 d
	Cyclogyl	Solution, 0.5%–2%		
	Cylate	Solution, 1%		
	Available generically	Solution, 1%, 2%		
Homatropine hydrobromide	Isopto	Solution, 2%, 5%	30–60 min	3 d
	Homatropine			
	Homatropaire	Solution, 5%		
Scopolamine hydrobromide	Isopto Hyoscine	Solution, 0.25%	30–60 min	4–7 d
Tropicamide	Mydral	Solution, 0.5%, 1%	20–40 min	4–6 h
	Mydriacyl	Solution, 1%		
	Tropicacyl	Solution, 0.5%, 1%		
	Available generically	Solution, 0.5%, 1%		
Cyclopentolate HCl/phenylephrine HCl ^a	Cyclomydril	Solution, 0.2%/1%	30–60 min	1–2 d
Hydroxyamphetamine hydrobromide/tropicamide ^b	Paremyd	Solution, 1%/0.25%	20–40 min	4–6 h

^a A dilute combination agent for infant examinations.

^b Used for dilating the pupil; cannot be used to test for Horner syndrome.

TOC



Abbreviation

Drugs for Dry Eye

Tears naturale	0.4 %	Duasorb	15 ml/bt		
Ginpol gel	0.2 %	Carbomer	10 g/tube		TID / QID
Liposic gel	0.2 % 1 %	Carbomer MCT	10 g/tube		
Vitamin A oint	5000 U/g	Vitamin A	3.5 g/tube	Artificial tear and ocular lubricant	HS
Systane ultra	0.4 % 0.3 %	PEG 400 PG	0.5 ml/UD		
Optive fusion	0.5 % 1 %	CMC Glycerin	0.4 ml/UD		PRN
Cationorm	0.5 % 0.5 %	Mineral oil Light mineral oil	0.4 ml/UD		
Diquas	3 %	Diquafosol	5 ml/bt	Stimulates the secretion of tear fluid and mucin	6x/day
Restasis	0.05 %	Cyclosporine	0.4 ml/UD	Immunomodulator, Anti-inflammatory	BID

CMC = carboxymethyl cellulose; MCT = medium-chain triglyceride; PEG = polyethylene glycol; PG = propylene glycol;
UD = unit dose

TOC



Abbreviation

The concept of TFOT (Tear Film Oriented Therapy)

Target for Therapy		Topical eye therapy
Lipid layer		Warm compress and lid hygiene Low-dose ophthalmic ointment Certain types of OTC *Diquafosol sodium
Aqueous / Mucous layer	Aqueous component	Artificial tears Sodium hyaluronate Diquafosol sodium Punctal plug
	Secretory mucin	Diquafosol sodium Rebamipide
Epithelium	Membrane associated mucins	Diquafosol sodium Rebamipide
	Epithelial cells (Goblet cell)	Autologous serum EGF (Rebamipide)
Ocular surface inflammation		Cyclosporin Steroids **Rebamipide

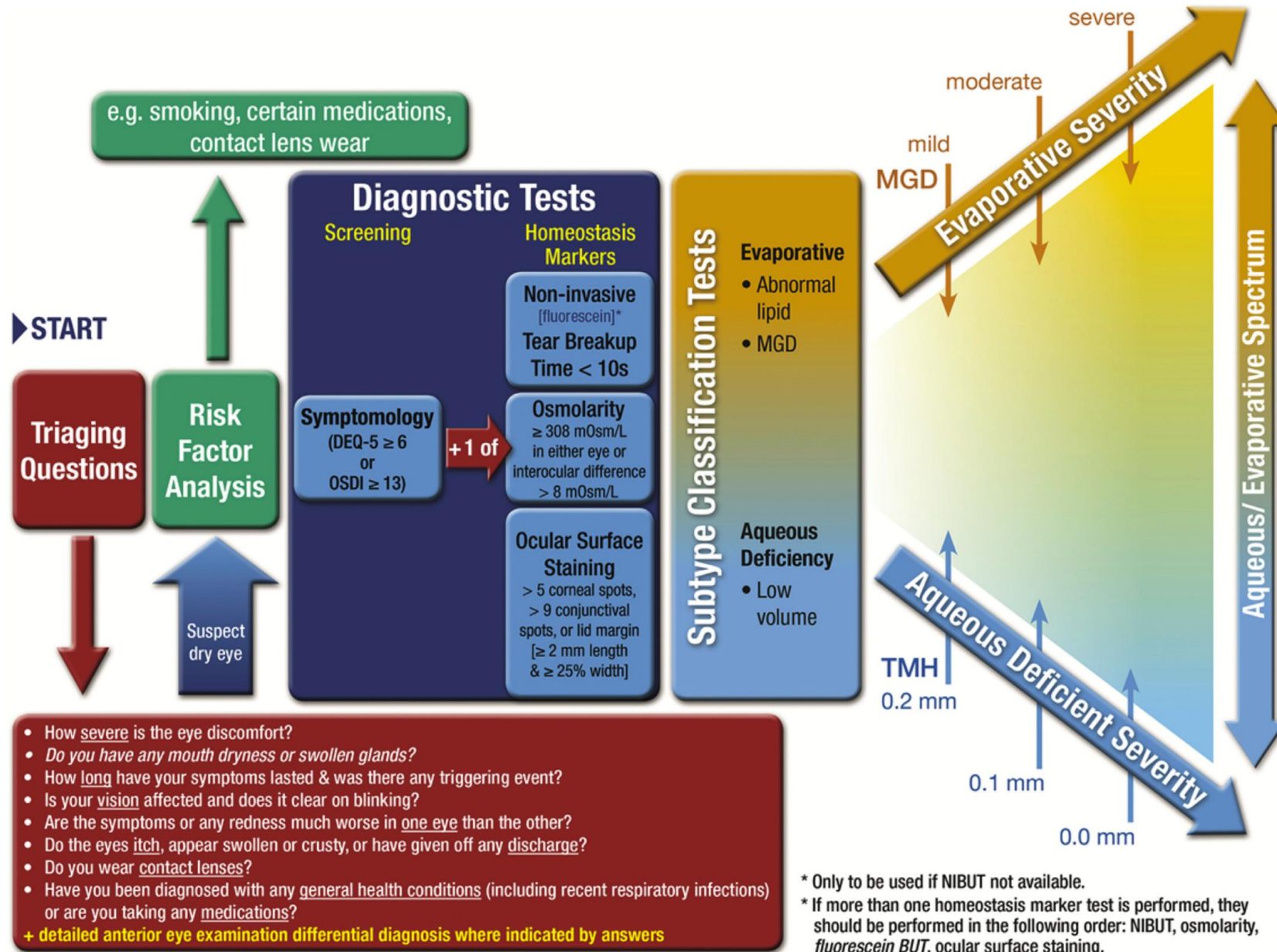
Figure 9. The concept of TFOT (Tear Film Oriented Therapy). *Diquafosol sodium may increase the function of the tear film lipid layer by promoting spreading of the lipid layer through lipid and tear fluid secretion. **Rebamipide may suppress the inflammation of the ocular surface in dry eye by its anti-inflammatory action. Figure provided by Dry Eye Society Japan.

TOC



Abbreviation

Recommended Diagnostic Approach for Dry Eye Disease



TOC



Abbreviation

Drugs for Intravitreal Injection (IVI)

TOC

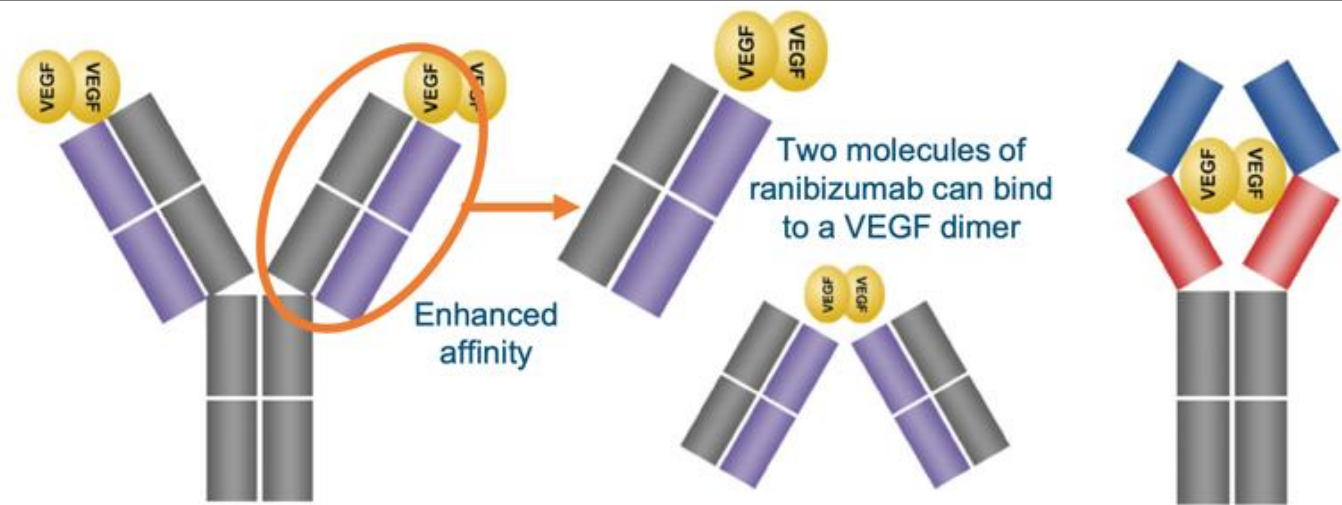
Avastin	Bevacizumab	100 mg / 4 ml / vial	Anti-VEGF
Lucentis	Ranibizumab	1.65 mg / 0.165 ml / PFS	
Eylea	Aflibercept	4 mg / 0.1 ml / vial	
Ozurdex	Dexamethasone	0.7 mg / applicator	Corticosteroid

PFS = pre-filled syringe; VEGF = vascular endothelial growth factor



Abbreviation

Structure and Characteristics of Anti-VEGF Agents



Bevacizumab

Ranibizumab

Aflibercept

Structure	Humanized IgG monoclonal antibody	Fab fragment of humanized IgG monoclonal antibody	2nd Ig domain of VEGFR1 and 3rd Ig domain of VEGFR2 fused to the Fc portion of human IgG1
Size (kDa)	149	48	115
Target	VEGF-A	VEGF-A	VEGF-A, VEGF-B, PlGF, Galectin-1
K _D for VEGF-A ₁₆₅ (pM)	58	46	0.490
Intravitreal half-life (days)	6.7-10	7.1	9 (estimate)

TOC



Abbreviation

Miscellaneous

TOC

ONSD	0.01 %	Neostigmine	10 ml/bt	Cholinesterase inhibitor	
Sancoba	0.02 %	Cyanocobalamin	5 ml/bt	Vitamin B ₁₂	BID - QID
Kary uni	0.005 %	Pirenoxine	5 ml/bt	Anti-cataract agent	
Alcaine	0.5 %	Proparacaine	15 ml/bt	Anesthetic	STAT
Benoxil	0.4 %	Oxybuprocaine	5 ml/bt		
Visudyne		Verteporfin	15mg/vial	Photosensitizer in PDT	For PDT

PDT = photodynamic therapy



Abbreviation

眼科常用縮寫

TOC



Exam



Surgery



Emergency



Drug

Abbreviation

眼科常用縮寫

A B

C D

E F

G H I

K L M

N O

P

R S

T-Y

TOC



Abbreviation

AB

AACG	acute angle-closure glaucoma
AAION	arteritic anterior ischemic optic neuropathy
AAU	acute anterior uveitis
ABK	aphakic bullous keratopathy
A/C	anterior chamber depth to corneal thickness ratio
AC	anterior chamber
ACD	anterior chamber depth
ACIOL	anterior chamber intraocular lens
AHP	abnormal head position
AION	anterior ischemic optic neuropathy
AL	axial length
ALT	argon laser trabeculoplasty
AMD	age-related macular degeneration (=ARMD, SMD)
APCT	alternate prism cover test
APD	afferent pupillary defect
App	applanation tonometry

ARC	abnormal retinal correspondence
ARMD	age-related macular degeneration (=AMD, SMD)
ARN	acute retinal necrosis
AS	anterior synechiae
ASC	anterior subcapsular cataract
Auto-P	automated perimetry
Auto-R	autorefractor
BCVA	best-corrected visual acuity
BDR	background diabetic retinopathy
BRAO	branch retinal artery occlusion
BRVO	branch retinal vein occlusion
BSCL	bandage soft contact lens
BSS	balanced salt solution
BUT	tear breakup time

TOC



Abbreviation

CD

C₃F₈	perfluoropropane (gas)
CACG	chronic angle-closure glaucoma
CB	ciliary body
CCT	central corneal thickness
C/D	cup to disc ratio
CDCR	conjunctivodacryocystorhinostomy
CF	count fingers (=ND)
CIS	canalicular intubation set
CL	contact lens
CME	cystoid macular edema
CMT	central macular thickness
CNV	choroidal neovascularization
CNVM	choroidal neovascular membrane
CR	cycloplegic refraction
CRA	chorioretinal atrophy
CRAO	central retinal artery occlusion
CRVO	central retinal vein occlusion
CSC	central serous chorioretinopathy (=CSCR)
CSCR	central serous chorioretinopathy (=CSC)

CSM	central, steady, maintained
CSNB	congenital stationary night blindness
CSR	central serous retinopathy
Cyl	cylinder
D	diopter
DALK	deep anterior lamellar keratoplasty
DCR	dacryocystorhinostomy
DHD	dissociated horizontal deviation
DME	diabetic macular edema
DMEK	Descemet membrane endothelial keratoplasty
DMRP	diabetic retinopathy
DSAEK	Descemet stripping automated endothelial keratoplasty
DTS	dysfunctional tear syndrome
DVD	dissociated vertical deviation

TOC



Abbreviation

EF

E	esophoria	FA	fluorescein angiography (=FAG, FFA)
ECCE	extracapsular cataract extraction	FAF	fundus autofluorescence
EKC	epidemic keratoconjunctivitis	FAG	fluorescein angiography (=FA, FFA)
EL	endolaser	FAX	fluid-air exchange
ELM	external limiting membrane	FBS	foreign body sensation
EOG	electrooculogram	F'd	fundus
EOM	extraocular movement; extraocular muscle	FDT	frequency doubling technology
ERD	exudative retinal detachment	FFA	fundus fluorescein angiography (=FA, FAG)
ERG	electroretinogram	ffERG	full-field electroretinogram
ERM	epiretinal membrane	F/R	foveal reflex
E(T)	intermittent esotropia / AE(T)*, alignment intermittent esotropia		
ET	esotropia / AET*		
EUA	examination under anaesthetic		

TOC



Abbreviation

GHI

GCC	ganglion cell complex
GON	glaucomatous optic neuropathy
GPC	giant papillary conjunctivitis
HM	hand motion (=MM)
Ho	hypophoria
Ho(T)	intermittent hypotropia / AHo(T)*
HoT	hypotropia / AHoT*
HSK	herpetic simplex keratitis
H(T)	Intermittent hypertropia / AH(T)*
HT	hypertropia / AHT*
HZO	herpes zoster ophthalmicus

ICCE	intracapsular cataract extraction
ICE	iridocorneal endothelial syndrome
ICG	indocyanine green
ICGA	indocyanine green angiography
ICI	intracameral injection
IFIS	intraoperative floppy iris syndrome
ILM	internal limiting membrane
IO	inferior oblique / EOMIO*, extraocular muscle inferior oblique
IOFB	intraocular foreign body
IOL	intraocular lens
IOOA	inferior oblique overaction
IOP	intraocular pressure
IR	inferior rectus / EOMIR*
IRMA	intraretinal microvascular abnormality
IVI	intravitreal injection

TOC



Abbreviation

KCS	keratoconjunctivitis sicca
KP	keratic precipitates
KSD	keratitis superficialis diffusa; diffuse superficial keratitis (=SPK)
LASIK	laser (or laser-assisted) in situ keratomileusis
LI	laser iridotomy (=LPI)
LK	lamellar keratoplasty (=LKP)
LKP	lamellar keratoplasty (=LK)
LMF	levator muscle function
LP	light perception (=SL)
LPI	laser peripheral iridotomy (=LI)
LR	light reflex; lateral rectus / EOMLR*
LT	lens thickness
LTG	low tension glaucoma (=NTG)
LTP	laser trabeculoplasty
LWE	lid wiper epitheliopathy

MA	microaneurysm
mfERG	multifocal electroretinogram
MGD	meibomian gland dysfunction; meibomian gland disease
MH	macular hole
MHRD	macular hole with retinal detachment
MIGS	minimally invasive glaucoma surgery
MM	motus manus; hand motion (=HM)
MMC	mitomycin C
MP	membrane peeling
MR	medial rectus / EOMMR*
MRD	margin-reflex distance



NO

NAION	nonarteritic anterior ischemic optic neuropathy	OCT	optical coherence tomography
NC	non-correctable	OD	oculus dexter (right eye)
NCNSNM	not-central, not-steady, not-maintained	OE	ophthalmic examination
NCT	non-contact tonometry	OHT	ocular hypertension
ND	numerous digitorum; number of digits (=CF)	OIS	ocular ischemic syndrome
Nd-YAG	neodymium-doped yttrium aluminum garnet (laser)	OKN	optokinetic nystagmus
NLD	nasolacrimal duct	Ortho-K	orthokeratology
NLDO	nasolacrimal duct obstruction	OS	oculus sinister (left eye)
NLP	no light perception	OU	oculus uterque (both eyes)
NPDR	nonproliferative diabetic retinopathy	OVD	ophthalmic viscosurgical device
NRC	normal retinal correspondence		
NS	nuclear sclerosis / NSc*		
NTG	normal tension glaucoma (=LTG)		
NVA	neovascularization of the angle		
NVD	neovascularization of the disc		
NVE	neovascularization elsewhere		
NVG	neovascular glaucoma		
NVI	neovascularization of the iris (rubeosis iridis)		

TOC



Abbreviation

P

PACG	primary angle-closure glaucoma	POAG	primary open angle glaucoma
PAN	periodic alternate nystagmus	POHS	presumed ocular histoplasmosis syndrome
PAS	peripheral anterior synechiae	PP	posterior pole
PBK	pseudophakic bullous keratopathy	PPA	peripapillary atrophy
PC	photocoagulation; posterior chamber	PPDR	preproliferative diabetic retinopathy
PCF	pharyngoconjunctival fever	PPV	pars plana vitrectomy (=PPVT)
PCIOL	posterior chamber intraocular lens	PPVT	pars plana vitrectomy (=PPV)
PCO	posterior capsular opacity	Prism IN	prism diopter base in
PCV	polypoidal choroidal vasculopathy	Prism OUT	prism diopter base out
PD	pupillary distance; prism diopters	PRK	photorefractive keratectomy
PDR	proliferative diabetic retinopathy	PRP	panretinal photocoagulation
PDT	photodynamic therapy	PS	posterior synechiae
PED	pigment epithelial detachment	PSC	posterior subcapsular cataract (=PSCO)
PEE	punctate epithelial erosions	PSCO	posterior subcapsular opacity (=PSC)
PH	pinhole	PTK	phototherapeutic keratectomy
Phaco	phacoemulsification	PUK	peripheral ulcerative keratitis
PI	peripheral iridectomy	PVD	posterior vitreous detachment
PION	posterior ischemic optic neuropathy	PVR	proliferative vitreoretinopathy
PK	penetrating keratoplasty (=PKP)	PXF	pseudoexfoliation
PKP	penetrating keratoplasty (=PK)	PXG	pseudoexfoliation glaucoma

TOC



Abbreviation

RS

RAPD	relative afferent pupillary defect	SF₆	sulfur hexafluoride (gas)
RC	recession / Rc*	SiO	silicone oil (=SO)
RD	retinal detachment	SL	sensus luminis; sense of light (=LP)
RGP	rigid gas permeable lenses	SLK	superior limbic keratopathy
RICS	relaxing incision and compression suture	SLT	selective laser trabeculoplasty
RK	radial keratotomy	SMD	senile macular degeneration (=AMD, ARMD)
RNFL	retinal nerve fiber layer	SO	sympathetic ophthalmia; silicone oil (=SiO); superior oblique / EOMSO*
ROP	retinopathy of prematurity	SOOA	superior oblique overaction
RP	retinitis pigmentosa	Sph	sphere
RPE	retinal pigment epithelium	SPK	superficial punctate keratitis (=KSD)
RRD	rhegmatogenous retinal detachment	SR	superior rectus / EOMSR*
RS	releasable suture; resection / Rs*	SRF	subretinal fluid
RVO	retinal vein occlusion	SRNV	subretinal neovascularization
SB	scleral buckle	SRNVM	subretinal neovascular membrane
SCH	subconjunctival hemorrhage	SS	scleral spur
SCI	subconjunctival injection		
SCL	soft contact lens		
SE	spherical equivalent		

TOC



Abbreviation

TUVXY

TM	trabecular meshwork	VMA	vitreomacular adhesion
TO	thyroid orbitopathy (=TRO)	VMT	vitreomacular traction
TON	traumatic optic neuropathy	VOD	visual acuity of right eye
TRAB	trabeculectomy (=TRBC)	VOS	visual acuity of left eye
TRBC	trabeculectomy (=TRAB)	VSR	venous stasis retinopathy
TRBL	trabeculotomy	VT	vitrectomy
TRD	tractional retinal detachment	X	exophoria
TRO	thyroid-related orbitopathy (=TO)	X(T)	intermittent exotropia / AX(T)*
UBM	ultrasound biomicroscopy	XT	exotropia / AXT*
VA	visual acuity	YAG	yttrium-aluminum-garnet laser
VEGF	vascular endothelial growth factor		
VEGFR	vascular endothelial growth factor receptor		
VEP	visual evoked potential		
VF	visual field		
VH	vitreous hemorrhage		
VKC	vernal keratoconjunctivitis		
VKH	Vogt-Koyanagi-Harada disease		

TOC



Abbreviation

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Your feedback is welcome and much appreciated!